

Math 120 The Fundamentals of Mathematics: Precalculus

Numbered Transcript

MATH 120 · Dr. Robert Snellman

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Lecture 1: Understanding Functions

At a glance — Understanding functions.

- 1.1 All right. Welcome to the course, Fundamentals of Mathematics for Precalculus. The idea for today is to motivate what precalculus is, what it's about, what are we going to be doing, and what are some of the topics that lead into this? Well, a precalculus course ultimately is meant to give you the foundations necessary to transition into calculus, as the name would suggest. It's a precalculus class, which begs the question, well, what is calculus?
- 1.2 Calculus has its roots ultimately in physics. Newton is associated with the creation of calculus. I think Newton put it best when he has this quote where he says, "If I've seen further than anybody else, it's because I've stood on the shoulders of giants." That's a really humble statement to make in the sense that people attribute the creation of calculus to him. What he's saying is, he's like, "Look, if I've done more than other people, it's because the groundwork was already laid for me. There were mathematicians that preceded me which did all of these amazing things, and I just put everything together and put my creative genius to it." I think that's a really beautiful quote.
- 1.3 But what is calculus? Again, it's all about putting a mathematical structure to physics. What is physics? We're trying to understand the world around us. We're trying to understand, if I were to take this marker and throw it across the room, could I somehow predict its trajectory, where it's going to land, how it's going to hit the ground, what angle it's going to hit the ground, and so forth?
- 1.4 But at the end of the day, calculus is really about rates of change. How fast is a particular quantity changing? If you get in your car or you're on a bicycle or whatever you're doing, and you look down and it says you're traveling in your car at 75 miles per hour, it's like, that miles per hour, it's a rate of change. It's your distance miles per time hour. It's saying, in one hour period of time, you will go 75 miles if you sustain that 75 miles per hour rate of change.
- 1.5 Precalculus is going to start with the fundamentals we need to be able to talk about calculus. By the end of this course, we'll be able to get into the beginnings of calculus with what are called limits. A limit is simply just an idea of trying to take some particular quantity and bringing it into a very small window and understanding what's happening in this little, small amount of time. Those limits will allow us to talk about derivatives, which are then rates of change in an instantaneous level, like what your speedometer is reading with your car.
- 1.6 But in order to get there, we have to talk about things like functions. Those are going to be the objects of interest. That's what we're talking about today. We're going to talk about the fundamentals of functions and what we need. Now, a function, the way you

can think about a function is a rule which simply just takes something from A to B. It takes something from a given input to a given output. That's the idea.

- 1.7 But I want to introduce the official notation for this because, unfortunately, this isn't really covered all that much when it comes to introducing functions. You may have seen in a previous course, like an algebra course or some other math course, the notation, things like this, where we say, I'm going to give you something like F of X is equal to, say, X squared. You may have seen something like this.
- 1.8 You may even say, "Oh, I'm comfortable with that. I associate that with a function. It's a function that takes X and it gives me an X squared." That's fine. That's completely fine. That's completely correct. But what I want to shed light on is, what is the actual notation for the function? Because technically speaking, this is not the function. F of X is not the function, even though that's what we associate with it. We say, "Oh, F of X is the function." It's like, no, it's not. F of X is the output that we get by applying F , the function, to the input, X , and it gives me an output of X squared.
- 1.9 For example, if you think about functions as saying, take an input, give me an output, this function right here would say, take the input, which is, say, the number 3. That's X is 3. Then the function is going to do something to 3. What is the output going to be? The output is going to be 3 squared, which is 9. It's weird. It's like you have on this side, you have the input. It goes through this black box, which is the function, F . Then on the output side of it, I get 9. But we conflate the fact that the middle black box thing, which is the function, is actually F , and the output is F of X .
- 1.10 The notation for functions is the following. You notate it by saying, you give it a name. Mathematicians are not very unique in how we name things. So function starts with F , so we'll call it little F . That's fine. You take little F , and then you simply define what are things that you're going to plug into the function. That's going to go here. In this case, if I'm taking real numbers, real numbers is denoted with this bold-faced $\mathbb{R}$. What do I mean by bold-faced $\mathbb{R}$?
- 1.11 You can think about this in a lot of different ways. Some people like to think about it as the infinite line that you think of as any number whatsoever that you can think of lives there. You could think of the number 1, the number 2, the number 0, or you could say the number one-half in here somewhere. You could say somewhere over here is the square root of 2. Any number that you've ever encountered in your life, you can think of as just living on this line, which we call the real line. They're real numbers.
- 1.12 You could also think about this as minus infinity to infinity. You just say anything from negative infinity off to positive infinity, that's a possibility. Or you can think about it as just its name, which is given as this bold-faced $\mathbb{R}$. These are called the real numbers. These are the real numbers. Again, this is going to be something that's— real numbers is a misnomer in terms of our actual English of saying these things. Real numbers, we're

like, "What do you mean real? They exist? Like, it's a real entity?" It's like, no, no, no, no, no. I mean, mathematically, there's a very precise meaning as to what real means.

- 1.13** We can visualize them in these three different camps. Whatever I'm allowed to plug into my function goes here, and then whatever the output of the function is goes here. In this case, if I say I'm going to take a real number, F is going to do something to it and output a real number, then this is my corresponding notation. This notation, unfortunately, is very, let's say, not emphasized enough, in my opinion.
- 1.14** I'll out myself here in terms of my own education. When I first took a calculus class, my professor wrote this on the board all the time, every single day. Every single lecture, this statement was on the board. I was too scared to ask questions because I didn't realize the value of actually asking questions and failing and then learning from those failures. I didn't say anything for the 15 weeks of a 16-week semester, until the 15th week. I finally went into the professor's office hour and I just sat down and I said, "Look, can you explain this to me?"
- 1.15** And this was one of the most profound lessons I ever learned. The professor looked at me and goes, "You don't know what this means?" And I was like, "No, I'm sorry, I don't. I have no clue what this means." And I fully expected to be made fun of. I fully expected to just be made to feel like I was a complete moron, all of this stuff. The professor looked at me and goes, "It's the 15th week of the semester." I said, "Yes." She said, "Why didn't you ask? Why didn't you just ask?" I said, "Well, I was scared to."
- 1.16** And she was like, "I could have answered this day one, and now you would have known exactly what we were talking about the entire semester." She said, "Just ask. What do you have to lose?" It was a very profound lesson because I was too scared. I was too scared to ask questions. But truly, the way to learn is to ask questions and be OK with feeling like a failure and feeling like you're silly or you don't understand something. Because arguably, I'm sure I had students in my class who probably felt the same way that I did at that time.
- 1.17** So this notation literally just says, this is where my inputs are going to come from. These are my inputs, and these are where my outputs are going to live. In terms of this notation, for this function, F of X is equal to X squared, for this function, F of X is equal to X squared, it's literally saying, I take an X that lives over here. What do I do? I send that to F of X , which is going to live in here. Then I have to give a definition as to what F is actually doing. I'm going to say that it's going to take it and square it. That's the notation for it.
- 1.18** What can we do with these things? What can we do with functions? How do we add them? What's the arithmetic that I can actually do? The arithmetic of functions. This is just a general mathematical principle. If you study mathematics long enough, you realize that there are patterns that show up in terms of procedure overall. What I mean by that is, you study, say, algebra, and you see all of these different techniques that are

laid out before you. Then you study a trigonometry course and you study an optimization course, and you study calculus and precalculus and all of that.

- 1.19** You start realizing, if you step back a little bit, you start realizing these patterns that start emerging. You're like, oh, in algebra, we define some objects, and then we ask, what can we do with those objects? Can I add them, subtract them, multiply, divide? Can I do arithmetic with them? OK, well, then in a calculus course or a precalculus course, we define some objects, we call them functions, and we asked, can I add them, subtract them, multiply, and divide?
- 1.20** Then in a linear algebra course, we define objects like vectors and matrices and yada-yada, and we ask the same questions, what operations can I do? You realize that this pattern is really just universal in mathematics. I'm going to define a new class of objects, and then I'm going to ask, what can I do with those objects? Can I add, subtract, multiply, divide, and so on? The arithmetic of functions is going to go as follows.
- 1.21** If I give you two functions, all right? Number one, if I give you two functions, F and G , and I say, can I take the functions F and G , which are labeled as F and G , create a new function, which is the sum of the two, F plus G , and evaluate it at X ? The answer is yes. You can guess what the answer is going to be.
- 1.22** What would I do if I gave you a function like this? If I gave you G of X is equal to $3X$ minus, let's say, E to the X , something like that. I give you some function that looks like that. I can say, well, what would F plus G be? I think a natural thing to do would be to say, well, just add X squared to G . So take X squared plus $3X$ minus E to the X . And the answer is, that's what you do.
- 1.23** So the definition of this is, you take F of X and you add G of X . I'll work out the example over here. So this will be the example side of things. Because it's important, I mean, it's one thing to push symbols on a board. It's another to actually have a practical understanding of how you actually apply these things. You want to be able to, I mean, you could sit there all day long and study, I don't know, study architecture, for example.
- 1.24** You could study architecture all day long and you have this Gothic architecture or the Tudor style of housing and things like that. It's another entirely to have a theoretical understanding of that, pushing symbols on the board, than it is to actually get your hands dirty and build the dang thing. Those two things are very different. I'm going to try and juxtapose this with the pushing symbols on the board, the definitional side, with the example saying, if these are my two functions, F and G , how do I illustrate this property with those two functions? Does that make sense?
- 1.25** For number one, I would say F plus G of X is equal to what? It's F of X here, X squared, plus G of X , $3X$ minus C to the X . There it is. You just add the two pieces. Well, if I have addition, then I have subtraction, presumably. Addition and subtraction are basically one and the same, more or less. If I gave you the numbers 5 minus 3 , that's really the

same thing as 5 plus negative 3. It's plus and minus, you can think about as the same object.

- 1.26** Number two, you have F minus G of X . Well, that's what you expect to do. Well, if addition just said just add F and G , subtraction is just going to say just subtract the two. OK, so what is number two? If I take F minus G of X , so this will be F minus G . What is that? X squared minus $3X$ minus E to the X . Now, I have to be careful here because the minus sign is going to have to distribute to both terms, and that's going to make that one a positive. You have to be careful when you're doing subtraction.
- 1.27** It matters, like, if I take 5 plus 3, it doesn't matter if I take 3 plus 5 or anything like that. The order is irrelevant. I'm still going to get 8 at the end of the day. But if I take 5 minus 3, it's like, the order matters. 5 minus 3 is 2, 3 minus 5 is negative 2. So the order matters.
- 1.28** The third one is, how do I take F times G ? I'm just going to denote that as F , and then you can put a dot in between the two if you want, F times G . Again, it's what your intuition says it should be. I'm just going to take the definition for the output, F of X , and the definition for the output, G of X , and multiply the two quantities together. All right. So that's going to give me F G of X . It should be what? F is X squared, and G is $3X$ minus E to the X . OK.
- 1.29** And then I would have to distribute via the distributive property that we cover in algebra. Those two things have to distribute over. That's great. Division, same thing. I'm going to take F divided by G at X . This one's equal to F of X , divided by G of X , as we expect.
- 1.30** Now, this one, I have to be very careful because, look, when I'm dividing by something, I'm potentially introducing zeros in the denominator. Remember, you're not allowed to divide by zero. You have to be very careful about, if I divide by a function, being aware of, how do my inputs, my domain values, how do those change accordingly?
- 1.31** For number four, if I take F divided by G , add X , this should be what? F was X squared, and G is $3X$ minus E to the X . The last one that we have, and it's not obvious that there is another one. It's not obvious because we've really just gone through the arithmetic that we're used to, like addition, subtraction, multiplication, and division. It's not too surprising that those two things are actually still valid in the function theoretic sense. But now we have one which is called function composition.
- 1.32** This last one, number five, it's denoted with F and then a circle G . It looks like FOG , the word fog, but it's just F , circle, G , evaluated at X . What is it really saying here? It's saying, well, look, if functions are just rules that you take an input and you plug it into a function, then why can't I iterate that? If you think about a function as moving from one point to another, so let's say I want to move from this side of the room to this side of the room. Well, I can do that in a lot of different ways.

- 1.33** I can start at that point, and then I can just walk directly to this point, and that could be a function that takes me from that point to this point. Or I could start from there. I could walk half the way, and then I could walk the rest of the way. It's like a two-step iterative process. That's how you think about function composition. Function composition says this is equal to F of G of X . Let's walk through what that means.
- 1.34** It says, look, start with the input of X . Take that X and plug it into the function of G . That's going to give me an output, whatever it is. Whatever that output is becomes the input for the outermost function, which is F . It's this iterative process. Start over here, that's X . Plug it into G , that took me here; that's G of X . Take G of X , plug it into F , that's going to take me over there. That's the idea I want you to keep in your head.
- 1.35** This one's a little tricky to do. How do I do that? This is F of G of X . This is F of, what is G of X , though? I have to go and say, do I have a definition for it? G of X is equal to this guy, $3X$ minus E to the X . I'm going to have a $3X$ minus E to the X . Then I have to say, what does F do to this input? Well, what does F do? You give me an input and I just square it. That's it. What is F going to do to this? It's going to take the input, which is $3X$ minus E to the X , and just square it. That's all it's going to do. Those are the operations that I can do with functions.
- 1.36** Now, function composition is probably, arguably, the most difficult to understand because it's a two-step process. These ones, it's nice. It's like, oh, you gave me F , you gave me G . I do one thing, which is add them. You gave me F , you gave me G , I do one thing, I subtract, one thing, multiply, one thing, divide. This one's saying, you give me F and G , and I have to do two things. I have to plug G into—I have to first take X , plug it into G , and then I have to take that and plug that into F . It's this two-step process.
- 1.37** Moreover, it's a new operation that we haven't been doing since forever, basically, grade-school thing. When you first start learning arithmetic, it's like, you start doing addition, subtraction, multiplication from a very young age. It's like, I'm willing to bet that you probably weren't doing function composition when you were learning your ABCs and 1, 2, 3s kinds of things. This is a relatively new concept for us in terms of what we're learning. That makes it a little more difficult as well.
- 1.38** But this function composition is really key, especially when you get into calculus in general. When you first start talking about these rates of change, which are in calculus, how fast is something changing? How quick is something decreasing or increasing or along those lines? What is the shape of its graph and so forth? It's really the notion of, what will be involved in terms of evaluating that concept is a derivative, and derivatives always, always, always use the chain rule, and the chain rule uses function composition.
- 1.39** We're never going to get away from function composition. This isn't one of those gimmicky math things. It's like, you learn it once and then you never see it again. Function composition is probably the most universal thing you're going to see going

forward. It's important to have a thorough understanding of it. Let's talk about functions from a function composition standpoint. Let's talk about compositions.

- 1.40** What's interesting about composition is that it opens up the door for us to be able to define inverse functions. It opens the door for inverse functions. Now, inverse functions may not seem useful on the surface. You might look at them and just be like, well, why are you trying to invert something? That seems lame. What's the point? If I give you the function F of X squared, why would I ever want to invert that? That's a valid question. Maybe you don't. Maybe you don't want to invert it. Maybe you do.
- 1.41** What would the value of that be? Well, let me motivate it in the following sense. If I gave you something like this, I want to—here's the motivation. I want you to say, let X be positive. Suppose X is a positive number, anything positive. I want you to solve for X in the equation given by X squared is equal to 16. You may look at this and be like, well, that's kind of ridiculous. It's easy. It's 4. 4 times 4 is 16, so I know the answer. It's just 4.
- 1.42** I would say, OK, that's fair. You know the answer, but I'm trying to illustrate a principle. I'm trying to illustrate a principle. The way I could view this is the following. I could say, look, let's view this in a function context. Let's view this in terms of functions and see if we can somehow understand why inversion of functions would be useful. What I'm going to do is, I'm going to take the F of X that we had initially. Again, my notation for this is like my S-O-L, upper N, is just my solution. What I write down here is the solution to the corresponding example.
- 1.43** I'm going to say, let F of X equal X squared. That's my function. My notation for it is, I'm going to have F takes in a real number and it outputs a real number. That's really what the function is, and its output is X squared. F of X is equal to X squared. Then this equation, so then X squared equals 16 becomes what? Well, F of X equals X squared, so I'm going to say F of X equals 16. That's what it becomes. I just replace my X squared with a name, F of X .
- 1.44** Now, if there exists an inverse to F , what do I mean by inverse? Well, you can think about it in a lot of different ways. One way to think about it, which personally, that's the way I think about it is, if I think about a function as taking me from A to B , like the example that we had earlier with function composition, is saying, look, if you give me an X on this side of the room and you apply it to a function and that function takes me to this point over here on this side of the room, what would the notion of inverse be?
- 1.45** Well, it would be something that would, I mean, it's at least plausible to say that the inverse of a function would be to undo what the function did. If I have a function which takes me from this side of the room to this side of the room, then the inverse function, plausibly, would be, take me from this side of the room back to where I started on that side of the room. Do we agree? It's just going to undo what I did.
- 1.46** The inverse function to F , if there exists an inverse function to F , call it, and this is its actual notation, F upper minus 1, then what is the property that it has to satisfy? It has

to satisfy the following thing. Look, what is F doing? It's taking an X and it's moving it to F of X . That's what it's doing. This is actually the appropriate mathematical notation for this. This little arrow that's pointing to the right with this little, I don't know, vertical hash mark to it or something on the left-hand side. This is actually what it's doing.

- 1.47** If F takes X to F of X , that's what F is doing. Well, then F inverse should do what? It should take F of X , back to X . That's what F inverse should do. If I start on the right side of the room, over here, and I apply F , now I'm on the left side of the room. If I'm on the right side of the room, and then I move over to the left side of the room, then what should the inverse function do? It should move me from the left side of the room back to the right side of the room, and I shouldn't have changed. That's what the inverse should do.
- 1.48** In particular, if F inverse exists, again, there's a big if here, there's a big if here. If that function exists, let's call it F inverse. F inverse has to satisfy the property of what? Well, look, if I start with this, if I take F of X and I feed it in as an input to F inverse, I have to get X . What does that mean function composition-wise? It means, if I take F inverse of F of X , I have to get X back. That's what it says.
- 1.49** X first moves from the right side of the room to the left side of the room via F of X , and then F inverse takes it from the left side of the room back to the right side of the room. You have to end up where you started. That's the property that it has to satisfy. OK, well, how do I use that? Well, let's go up to this equation right here. Just like in algebra, whatever we do to one side of the equation, we have to do the other side of the equation. Let's just apply F inverse on both sides and see what happens.
- 1.50** Well, I'd have F inverse of F of X is equal to F inverse of 16. By definition, this thing has to be X , which is nice because it gives me a systematic way to solve, quote, solve for my variable X , whatever that means. But then X is equal to F inverse of 16. Now it begs the question, well, what the heck is F inverse then? What is it? What is it actually doing?
- 1.51** Now, if you sit back for a little bit, you realize that, how do you undo a square? Well, you take its square root. If you give me the number 2 and you square it, that gives me 4. If I want to get back to 2, what do I need to do to 4? Well, if I take its square root of 4, that takes me back to 2. You start with 3, 3 goes to 9, square root of 9, that's back to 3. 16 comes from 4, so 4 goes to 16. 16 goes back to 4.
- 1.52** You quickly realize that the inverse function is the square root of X , which is why we needed to make sure that only X was positive. We need to make sure that X is positive because the inverse function doesn't exist otherwise. And so this becomes the square root of 16, which is equal to 4. It recovers what we know. It recovers what we already knew to some extent. You may have looked at this question and you're like, gosh, dang it, this is taking forever. The answer is 4, for goodness' sakes. X is equal to 4. Just take it. X is 4. That's the answer.

- 1.53** And it's like, that's fine. But that's not a systematic way. It's a fluke. It's a fluke because if I gave you something like this, I can easily— this is one thing with mathematics that people need to truly understand, is that the examples that we see, the examples that we can actually work out, we want to try and generalize them to what are their underlying principles, so that I can extract those principles and abstract them.
- 1.54** For example, I could very easily break this example to something that's suddenly like, I have no clue how to solve it. I have no clue how to solve it. For example, what if I said X squared is equal to 3? We knew the answer here because it was just 4. We said, well, 4 times 4 is 16, so we know that's the answer. But it's like, OK, well, what's the answer here? If you say, oh, the answer is the square root of 3, it's like, ah, but what are you using? You're using the inverse function, the square root of X . You're not just using your intuition of 4 times 4 is 16.
- 1.55** Now you're saying, oh, I'm actually using something else that I know. Its square root function satisfies it. Square root of 3 times square root of 3 is 3. OK, fine. Then you say, all right, fine. I'll cede that ground. You're allowed your square root function. That's fine. But now I'm going to give you this. X squared is equal to minus 6. You're like, oh, the answer is the square root of minus 6. Then I counter that by saying, well, what is that? What is the square root of minus 6? Can you draw it on the ground?
- 1.56** Like the ancient Greeks liked to do with all of their mathematics, is that a number didn't exist unless they could literally draw it on the ground with some ruler or straightedge and compass thing. Draw me the square root of negative 6. Draw me a triangle that gives me the square root of negative 6. Immediately you're like, OK, don't know what that is. But we'll get into that in a precalculus course.
- 1.57** We'll talk about what are complex numbers with these imaginary, quote, unquote, "imaginary" numbers and things like that. But it all stems from this idea. Look, how do I undo whatever this function is in order to extract the actual variable I'm interested in solving for? It comes from inverting functions. Let's see some other examples on how we can do this.
- 1.58** I think mathematics is really interesting in kind of a frustrating way, which is OK, in the following way. You quickly, at least this happened to me, so I'm projecting here a little bit, but for me, it was very interesting that once I felt I had some sort of grasp on mathematics and I actually understood things, somebody would ask me the smallest question, and suddenly I'd realize I don't know anything. Somebody would ask me—it's very similar to this.
- 1.59** It's like, if I ask you, what is an integer? You're like, oh, those are negative 3, negative 2, negative 1, 0, 1, 2, 3, 4. It's like things that you can count with or do accounting with, with whole numbers, kind of thing. You're like, OK, that's fine. Then you jump that up a little bit and you say, well, what are rational numbers? You say, well, those are just quotients of integers. So like, 1 over 3, 7 over 2, 27 over 14, things like that. Those are

rational numbers. You're like, OK, I have a grasp on those. That's fine. I've eaten a pizza kind of thing, so I could say, oh, half of a pizza. I can make sense of that geometrically a little bit. OK, that's fine.

- 1.60** Then you immediately get asked the question, what's a real number? And you're like, it's a thing. It's, what is it? And you fudge your way around a little bit. You say, oh, it's a line. You say it's any point on that line. OK, fine. Whatever. And then you say, well, it's the thing from negative infinity to infinity. OK, that's fine. I'm still asking you, what is a real number? Give me a definition for what is a real number. Mathematically, it's incredibly complicated. Mathematically, a real number is incredibly complicated.
- 1.61** Probably the best definition for a real number is something that you can write in decimal expansion, something you can write in terms of a decimal, like the number pi, 3.14159, blah, blah, blah. You can keep going. You're like, ah, it's a real number because I can write it as a decimal. That's probably the best explanatory definition as to what a real number actually is. But mathematically, real numbers are incredibly complicated.
- 1.62** You quickly learn that the amount of things you don't know mathematically far surpasses the things that you actually do know, which I think is very humbling on a lot of levels. Let's do another example. Let's talk about exponential and logarithmic functions. Exponential and logarithmic functions. What are they? Where do they come from? Why do we care?
- 1.63** Exponential functions simply look like this. They're functions that look like you have some number, we call it B because B is called the base. Again, mathematicians are not very unique in naming things. We're going to call it the base, and base starts with B, so we'll give it the letter B. OK, that's fine. I raise it to the X power, where we're going to assume that B is positive and B is not equal to 1. B is not equal to 1. Because if B is equal to 1, that's lame. I'm saying 1 to the X power. I'm literally just saying it's constant equal to 1. That's not really an exponential function. So, B is not equal to 1.
- 1.64** So, as examples, we can have things like F of X is equal to 2 to the X. So I'm just doubling every single time. That's the idea. We can look at their graphs associated to this. So here's X, here's Y. I know that if I take F of zero, I get 2 to the zero, which one of our exponent rules from algebra, that's equal to 1. I at least know, if I have an input of zero, my output is 1. I'm going to hit this point here. That's going to be 0, 1.
- 1.65** Now, I'm a big fan of free utility, free graphing utilities. When you have access to a free utility, you should take full advantage of that. There's a wonderful tool online called Desmos that you can use to actually graph these things. You type in literally just F of X equals 2 to the X, and this is what you get. You'll realize that its graph does something like that. It gets very, very big, very quickly, as you move off to the right-hand side. You can convince yourself of that just by plotting points.

- 1.66** If X is 1, I get 2 to the 1, which is 2. If X is 2, I get 2 squared, that's 4. If I get X is 3, that's 8. X is 4, that's 16. 32, 64, 128, 256, 512, 1,028. It gets big really fast. Really fast. You don't have to go too far out in the X direction to quickly realize you're suddenly 100,000, million, 100 million, billion, all of this. The values are getting big very quickly.
- 1.67** But you can also convince yourself that, going off to the left, you're going to get arbitrarily close to zero. How do you see that? Well, look, what is 2 to the negative 1? Remember, how do negative exponents work? We did this in algebra, right? This is 1 over 2. You take the number and you take 1 over that number. At negative 1, I'm equal to a half. At negative 2, I'm equal to 1 over 2 squared, which is a fourth, and so on. Negative 3, I'm at an eighth. Negative 4, I'm 116th, 132nd, 164th, 128th, or 1 over 128, so forth. I'm getting arbitrarily small.
- 1.68** That's how exponential functions look. That's if B is bigger than 1. This is the graph in general of B greater than 1. They all have a similar shape to this, depending on how big B is. When B is really close to 1, like 1.0001, which is typically the interest rate you're going to get at a brick-and-mortar bank, basically, this seems basically flat. It's very flat. It doesn't grow very quickly. But the larger B is away from 1, so B is 2, 3, 5, 20, things like that, this immediately spikes really quickly. But in general, they're going to have this upward-looking shape. All right.
- 1.69** What happens when B is in between zero and 1? If zero is less than B is less than 1, the shape just inverts on itself. It does this. You still go through 0, 1. That's fine. It just inverts on itself. It goes that way. Again, exponential and logarithmic functions. The one thing about these things is that, look, they satisfy the horizontal line test. The horizontal line test says what? It says if you draw the graph of the function, and I draw a whole bunch of family of horizontal lines throughout the entire thing, do these horizontal lines intersect in one and only one point?
- 1.70** If the answer is yes, it means that the function is invertible. It means the function is invertible. You imagine, how do you see that, how do you see that proves invertibility? If this is my input of X , I take that input, I plug it into the function, and I get this output, which is 2 to the X in this example, then what should the inverse function do? The inverse function is going to take this 2 to the X as my input. It's going to send it back down the chain and say, oh, it's equal to X .
- 1.71** If there is only one intersection point for this particular horizontal line, then there is only one X value that I can reasonably define for the output of my inverse function, which is why the horizontal line test has to be true, in order for a function to be invertible. If that's the case, then what's the inverse function? The inverse function is the logarithm. If F of X is B to the X , let's say, if F of X equals B to the X , then F inverse of X is equal to the logarithm base B of X . That's what this is called. This is the log logarithm base B .

- 1.72** What do I mean by that? I mean that it's its inverse function. It's its inverse function. What does that mean? Well, remember that inverse functions have to have this property, that when I take F inverse of F of X , I get X , and when I take F of F inverse of, let's say, Y , I get Y . They have to undo each other. They have to undo each other. In particular, what does that mean? Well, we get some properties based on that. This means that if I take B raised to the logarithm base B of X , I have to get X back. If I take logarithm base B of B to the Y , I have to get Y back. Those are two of my properties, simply just taking advantage of this inverse relationship.
- 1.73** Now, again, the motivation for inverse functions is to somehow undo a known function so that we can solve for the variable at hand. If that's the case, how do I take advantage of this when trying to solve particular equations? Let's walk through a couple of examples. Again, pushing symbols on the board is one thing, being able to solve for equations is another. Being able to put that into practice is completely different. Here's some examples.
- 1.74** Solve for X in the following. I'm going to give you two examples illustrating these two things. Number one, I'm going to say, let's say, log base 3 of, let's say, $X + 4$ is equal to 2. That's that one. And the second one is, I want to take, let's say, 5 raised to the $2X$ minus 6 is equal to 4. I want to solve for X in these two things.
- 1.75** Now, the age-old question of mathematics is like, well, when am I ever going to use this? Is this relevant? Is this that? I think that's a fair question, if I'm being honest. I think that's a fair question. I would arguably say that you're probably never going to roll up on this equation on your day-to-day. You're probably not going to wake up in the morning and be like, "Oh, this is the first thing I'm going to do this morning, is I'm going to solve this guy." You're probably not going to do that. I mean, I'm a mathematician, and I don't do that.
- 1.76** You're probably not going to do that. It begs the question, well, then what's the point? It begs the question, well, what's the point? I would argue that the point is the following. Well, in your day-to-day, you have a sequence of events that take place, and you need to somehow solve the problems associated with that. You're driving to work, you hit traffic. Do you make the decision to divert off the freeway because the freeway is at gridlock? Or do you stay the course on the freeway and hope that it clears up in the next couple of miles so that you can actually get to where you want to go?
- 1.77** Or you have a coworker who's being difficult. How do you navigate the situation? You have, IT has to come in because your computer breaks, for whatever the case may be. You need to be able to take a situation, break it down into its individual constituent components, solve those problems, and then recompile those solutions to solve the actual problem at hand. That's literally what mathematics is teaching you how to do. It's just doing it in a context which seems completely disjoint from the real world. But the skills that you develop in doing this translate immediately into the real world. I

would argue that the value is implicit, not necessarily explicit, when it comes to these things.

- 1.78** All right, so let's go after this one. How would we do this? The solution. Now, remember, the game is, look, I want to somehow eliminate this log base 3. I'm trying to get X. I'm trying to get X. If I could somehow just cover up log base 3 and log base 3 didn't exist, then I would say, oh, just subtract 4 on both sides. That's fine. Just subtract 4. If I have $X + 4 = 2$, then I just say subtract 4 and I win. I'd say X is negative 2.
- 1.79** The problem is that log 3 is in the way. I need to understand, well, how do I get log 3 out of the way? I come over here and I say, do I have any operations with exponentials and logarithms that would somehow eliminate a logarithm like this? Well, this rule is telling me, look, if I have a log base 3 of X and I want to access X, then I need to raise that logarithm to 3 to whatever that power is.
- 1.80** What I'm going to do here is, I'm going to say, oh, whatever I do to one side of the equation, I have to do to the other. I'm going to take 3 raised to the log base 3 of X plus 4, and whatever I did to one side, I have to do the other. This will be 3 to the 2 power. I'm just going to take both sides and exponentiate them with a base of 3. That's what I get.
- 1.81** Then I step back for a second and I'm like, OK, well, did I actually make my life worse or did I make my life better? Am I moving towards a solution or am I moving backwards? I step back and I'm like, OK, well, this is a 9, so that's great. That's 9. I'm OK there. But 3 to the log base 3, I say, OK, do I have a principle of that? Well, this says, look, B to the log base B is whatever's inside of that box.
- 1.82** OK, so I have 3 raised to the log base 3. These in some sense cancel, if you will. And what do I get? I get X plus 4. Well, that's a win. Because our intuition was, look, if I just had X plus 4 by itself, X plus 4 equals something, I could just subtract 4 and I win. OK, well, now I do. I have X plus 4 by himself, and that's equal to 9. So if I subtract 4 on both sides, I have minus 4, minus 4, what do I end up getting? Let's do it up here. If I, X plus 4 minus 4, I get X, and that's equal to 9 minus 4, that's 5. That should be my solution.
- 1.83** Now, as with anything, if you're claiming that this is your solution, you should probably check that your solution is actually plausible. Now, sometimes in mathematics, that's not possible. Sometimes you're like, well, I think this is going to be the solution. I think this is the best solution. Maybe I'm right, maybe I'm wrong. In these classes, you can actually check. You can plug your answer in and see what happens. Let's check.
- 1.84** If I take X is equal to 5 and I plug it in, what do I get? I get log base 3 of 5 plus 4 is 9. You're like, that's not helpful because I don't know what that is. But you step back over here and you say, oh, I have a rule here that says log base B of B to the Y is just the exponent Y. I'm going to say, oh, sweet. I have log base 3. Can I write 9 as 3 times something? 3 to the some power. The answer is, yeah, I can do that. This is log base 3 of 3 squared.

- 1.85** Then I say, oh, this principle says I just need the exponent. Log base 3 of 3 to the whatever, this is 2. Oh, that's what I wanted it to be. I wanted it to be 2. It looks like X equals 5 is actually a solution. It looks like X equals 5 is actually a solution. What about over here? This was, I have a logarithm equal to a number. How do I get rid of it? I raise both sides to the corresponding exponential with the corresponding base of the logarithm, 3 to this, 3 to that.
- 1.86** What if I go in reverse? What if I have an exponential equation and I want to somehow extract the X from the numerator here? What am I going to do? Well, if I exponentiate it here, I'm going to take the logarithm down here. How do I approach this one? What's my solution here? My base of my exponent is 5, so I'm going to take advantage of this rule over here. My base is 5. If I take logarithm base 5, I'll be able to extract the exponent and bring it down. That's the goal. I'm going to be able to extract the exponent and bring it down.
- 1.87** I'm going to take log base 5 of 5 to the $2x$ minus 6. Whatever I do to one side, I've got to do to the other. I've got to do to the other. I look at my properties over here. Log base B of B to the Y is Y , so log base 5 of 5 to this. Oh, sweet. The B s match, so this is just equal to $2x$ minus 6. All right, I'm happy with that. So what do we have? We have $2x$ minus 6 is equal to log base 5 of 4. Now, you might say, "Well, yeah, but, Robbie, you have this really annoying thing hanging off on the right side. I don't know what that is." I'd say, "I agree. I don't know what that is." It's some horrible number.
- 1.88** I can go to an online calculator and plug in log base 5 of 4, and it's going to give me some ridiculous decimal. That's fine. But at the end of the day, what is it? It's a number. I don't care what number it is. It's just a number. It doesn't matter. I could think of this number as being 7. If this number is 7 and I say solve for X , you're like, oh, just add 6 and divide by 2. OK, fine. Why does this thing have to be something that you like? Why can't it just exist as a number?
- 1.89** The principle applies. If this number were 9, you'd say, oh, add 6, divide by 2. The answer is 15 over 2. OK. Why does that not just translate directly here and say, look, if this is just a number, why can't I just add 6 on both sides and then divide by 2? Divide by 2? I'm going to get, what? X is equal to the log base 5 of 4 plus 6 all over 2. That is a horrible-looking number. It's absolutely atrocious. But at the end of the day, it is just a number. It is just a number.
- 1.90** If you take that X value and you go back and plug it into this particular equation, you'll get that it satisfies this equation. How do you see that? One thing that I always encourage anybody who's studying mathematics to do is, yes, there is value in writing things down in all of its glorious detail, just every single detail whatsoever. There's value in that.
- 1.91** There's also value in stepping back and seeing if you can manipulate the symbols in your head and see what's going to happen. I want to walk you through that, how you

could do this, just mentally. If I wanted to check that this horrible number was actually what it's supposed to be, it's supposed to solve this particular equation, what am I going to do? I'm going to plug it in for this X , and I'm going to see if the equality holds. What happens when I do that?

- 1.92** Well, look at this. This number is logarithm plus 6 over 2. If I multiply it by 2 first, what's going to happen? This 2 is going to disappear. I multiply it by 2, the 2 disappears. Now I have in my head the number which is just log plus 6. What am I then going to do? I'm going to subtract 6. What's going to happen? I first multiply by 2, the 2 goes away. I then subtract 6, the 6 goes away, and I'm sitting here. I now have log base 5 of 4. I have 5 to the log base 5 of 4. 5 to the log base 5 of 4 is equal to 4. Bingo. That's exactly what it is.
- 1.93** That exercise in trying to just do that mentally to check, "Oh, I multiply by 2, that'll do this, subtract 6, that'll do this, then I apply this property," is incredibly valuable. It's incredibly valuable for a lot of different reasons. OK, so let's do one more thing in terms of the relationship of inverse functions and the original function from which they came in terms of their graphs. I'm going to illustrate this with logarithmic and exponential functions. I'm going to illustrate this with those.
- 1.94** If I give you the graph of a function, how do I graph its inverse? That's a motivating question. It's like, well, why in the heck would you want to know what its graph is? Who cares? Well, go back to the original question that we had earlier when I said X squared was equal to 16. We couch that in a function theoretic way to say that the inverse function was the square root of X . The inverse function was the square root of X .
- 1.95** Why would knowing the square root of X graph be useful? Well, if I knew what the square root of X graph was, then I could say, well, OK, then I could just plug in any input value and just say, oh, here it is on the graph, and it'll give me my corresponding output. There's value to knowing what these graphs are. Let's look at graphs of exponential and logarithmic functions and see if we can extract just a general principle from this.
- 1.96** I'm going to do a concrete example because I think it's valuable to illustrate things with concrete examples, and then you can abstract this away from that if you'd like. I'm going to say B is equal to 2. I'm going to take my base to be 2. The function I'm interested in is F of X is equal to 2 to the X , and I know that its inverse is equal to F inverse of X is equal to log base 2 of X . I know those are the corresponding inverses of each other.
- 1.97** Now, what do their graphs look like? I'm going to do my best here. This is a little tricky. This is why Desmos is awesome. Desmos will help you do this for free, and it'll draw it way better than I ever could by hand. It'll do a far better job. But I'm going to do my

best. If I try graphing this particular function, I know that when X is zero, I'm up here at 1. I know I have this point here. This is 0, 1. I know I have that.

- 1.98** I know the function looks like this. It goes down like this, and then it spikes it up very quickly like that. I know that. The property of inverse functions is that their domains and their ranges switch. This is very beautiful. I claim that it's very intuitive, and we actually have a very intuitive understanding of that now. Let's think about it as this side. We're going to say this side of the room is the domain for F . This side of the room is the range for F . Domain and range, inputs, outputs. Inputs, outputs for F .
- 1.99** Well, think about that. What does that mean? That means I take an input from this side of the room, I run it through F , it gives me an output on this side of the room. What is the inverse function going to do? The inverse function starts where? Here in the outputs, and it's supposed to take an output and give me an input. That's what it's supposed to do.
- 1.100** So think about it. If this is the domain and this is the range of F , then this is the domain and this is the range of the inverse of F . They switch. So all I'm saying is, mathematically, if I say, what is the domain of F ? That's the range of F inverse. And what is the range of F ? That is the domain of F inverse. It's kind of this beautiful symmetry between the two. It's wonderful.
- 1.101** What do I notice here? Well, what is the domain of this function 2 to the X ? Well, the domain is all of the valid inputs, everything that you're allowed to plug into the exponential function. You start walking around the X -axis, those are your inputs, and you say, OK, well, what valid ones am I allowed to plug in? You're like, well, I could plug anything in. In particular, what do I know? I have that the domain of F is all real numbers.
- 1.102** So, I'm going to use minus infinity to infinity. But if you want to use this little symbol $\mathbb{R}$, that's fine, too, whatever you want to use. It's minus infinity to infinity. What is its range? The range are all of the elements that you can capture from outputs. It's everything that you can get in the outputs. That's along the vertical axis. That's saying, what values can I get going up the vertical side of things?
- 1.103** You quickly realize, I can get arbitrarily close to zero. I don't ever seem to get zero. Then I can go where? I can go arbitrarily high. My range is going to go from zero to infinity, where I put a parentheses on zero because it doesn't ever attain zero. If it did, then I'd put a bracket around it. Taking this into account, that means that the domain of F inverse needs to be the range of F . The domain of F inverse is zero to infinity, and the range of F inverse is equal to negative infinity to infinity.
- 1.104** The beautiful thing about all of this is that, if I simply just take the graph of F of X equals 2 to the X , and I take the line, which is Y equals X , and I imagine taking this graph and just mirroring it on this side, that's the graph of the inverse. I just take the mirror image of this on here. So you imagine this line is like Y equals X . It's like your mirror, like

literally a physical mirror. And you say, if this is like me looking in the mirror, what do I see on the other side of the mirror? This is going to ruin mirrors for the rest of your life.

1.105 When you look into a mirror, what are you seeing? Well, you see your face. But realistically, what are you seeing? You're seeing your face into the wall a certain period of time or a certain distance. What is that distance? It's the distance between your face and the mirror. The light bounces off the mirror, bounces off your face. That's what you actually observe. And so whatever that distance is, you're seeing that distance into the wall. You're seeing your face that far in.

1.106 So if I imagine this as being a mirror and I'm looking into the mirror, what is this going to look like? This point is going to see what? It's going to see a point over here. This point is going to see what? It's going to see a point over here. The point 0, 1 is going to see what point? It's going to see the point 1, 0. It's going to see this point. The point here is going to see this point down here and so on.

1.107 So these things are reflected, reflected like that. That's how it looks. This is the graph of my Y equals the log base 2 of X . This is the graph of my Y equals 2 to the X . And that's the relationship between exponential functions and logarithmic functions in terms of their graphs, and more generally, any function F and its corresponding inverse as being a reflection across this line.

Lecture 2: Complex Arithmetic

At a glance — Complex arithmetic.

- 2.1** All right, so now we're going to pivot a little bit, and we're going to basically do a deep dive into complex numbers. What are they? Why are they useful? Where do they come from? Things like that. Now, complex numbers are treated in various different ways. Typically, the historical context of them is, they're around the 1500s. So Gerolamo Cardano was the one who, I would say, people knew about them, but he was the one who put them forward. He was the one who put them in the mainstream.
- 2.2** And the idea behind them stems from quadratic equations. So if we start with a quadratic equation of the form $AX^2 + BX + C = 0$, where, again, A is non-zero because if A were zero, then this is not a quadratic. If A were zero, this is just a degree one thing, and I could attempt to solve that. If A is non-zero, the quadratic formula tells me that all of the roots are going to look like $X = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$.
- 2.3** OK, well, that's all fine and great, except for the fact that you step back for a minute and you say, OK, well, but what happens when this is a negative number? The natural question is, what happens? What is the square root of $B^2 - 4AC$, when $B^2 - 4AC$ is negative? You quickly realize, well, I don't actually know what that is.
- 2.4** For example, let's say, how would we make this thing small? Let's see. Let's suppose B is zero. If B is zero, and let's say, I don't know, A and C are both 1. Let's just choose values of that. $A = C = 1$. So what's the quadratic that I'm looking at? B is zero, so that term's gone. A is 1, C is 1. I'm looking at $X^2 + 1$. That's my polynomial, and I'm asking, what are its solutions? I set that equal to zero.
- 2.5** If I look at that, then I say, OK, B is zero, A is 1, C is 1. I'm looking at the square root of negative 4. The question is, well, what in the heck is that? If it were just the square root of 4, then I would be happy. I'd say, oh, it's 2. That's fine. Or the square root of 9, I'd say it's just 3, or the square root of 16, I'd say it's 4. That's fine. But the square root of negative, it's like, I don't know. I have no idea what that is. OK? OK.
- 2.6** Then it begs the question, can I make sense of it? That's what Cardano did. The way to make sense of this is with complex numbers. It's with complex numbers. The idea is you simply just define a symbol. You define the symbol " i ", and the way you define it is the square root of minus 1. That's how you define it. In particular, what you're doing is, you're saying, well, look, if I square both sides and add 1, I'm saying that $i^2 + 1 = 0$. So, i solves the equation that we had earlier, $X^2 + 1 = 0$. That's how it's defined.

- 2.7 Now, this i is called a complex unit. It's a complex unit. What it allows us to do is, it allows us to say, OK, well, what do complex numbers look like then? Just like we asked the question, if I give you integers and rational numbers, and then I say, what is a real number? A good definition of that is like, well, any number that you can write as a decimal. I could ask the same question now, and I could say, well, what does a complex number look like? The definition of a complex number is the following.
- 2.8 A complex number, that's what I'm defining, a complex number is something of the form, let's say, $A + Bi$, where A and B are real numbers. That's what a complex number is, something that looks like $A + Bi$, where A and B are real numbers. Again, any mathematical object that we get, we want to then define what are the arithmetic operations associated to it. Can I add, subtract, multiply, divide, do all of that? In this class, we're actually going to work through how you do each of these things from a very systematic approach.
- 2.9 Then we're going to go deeper into this and say, well, can I rewrite these complex numbers differently, or can I only write them ever this way? Can I only ever write them as $A + Bi$, or can I write them differently? Then if I can write them differently, can I somehow use that to my advantage to solve equations in a very precise way? That's the trajectory of where we're going.
- 2.10 What is the arithmetic of the complex numbers? What's my arithmetic? Well, addition and subtraction are what you would expect them to be. Multiplication is a little different, and division is just very different than what you would expect it to be. So this is the arithmetic of the complex numbers. This bold-faced C , remember we had the notation for R as this bold-faced R , and that was what we designated the symbol for the real numbers? The symbol for the complex numbers is this bold-faced C . That's the complex numbers.
- 2.11 The arithmetic of that. What I'm going to do is the following. I'm going to say, let's let Z equal $A + Bi$, and W equal $C + Di$. The two complex numbers, whatever they are, $A + Bi$ and $C + Di$. Then I want to understand, what is $Z + W$? How do I add the two? How do I add the two? The way you think about this is, you think about this like polynomials. You think about like terms. What terms look alike? What things do they have in common?
- 2.12 If I look at these two things, I say, well, look, they have a part here that doesn't have anything to do with i , and then they have a part here that has everything to do with i . Those are their distinct parts. So, I have the same thing here, something that has nothing to do with i , something that has everything to do with i . If I add them, you just combine those terms that look alike. I'm going to take what? $Z + W$, I'm going to get $A + C$. That's my $A + C$, and then plus $B + D$ times i . I combine the terms corresponding to what they look like.

- 2.13** The second one is, if I subtract them, how does this work? Same thing, subtract away corresponding parts. So Z comes first, and I subtract away W , so that's going to be A minus C plus B minus D times i . I'm going to have that. Now, multiplication is where it gets weird. Multiplication looks a little strange. If I took Z times W , so we're going to work this out. Z looks like A plus Bi , and W looks like C plus Di . When I distribute, corresponding to the distribution law, A times C , I'm going to get an AC , A times Di , I'm going to get an AD times i . Bi times C , I'm going to get a plus BCi , and then Bi times Di , I'm going to get a BDi squared. BDi squared. That's the idea.
- 2.14** But then I step back and I say, well, look, if I'm trying to make this arithmetic make sense, then this is concerning to me because the idea, what do I mean by make sense? I mean, if I take a complex number and a complex number and I multiply them, I should get a complex number back. I shouldn't get complex number times complex number equals something completely unrelated to complex numbers. It's like, that wouldn't be a good operation to have.
- 2.15** I step back and I say, well, OK, you told me that every single complex number can look like something plus something i . Now you're saying that this is what the multiplication looks like. This doesn't look like something plus something i . This looks like something plus—OK, this looks like something i . I could factor out an i and get AD plus BC . That's fine. But then this has an i squared. You said complex numbers. What's going on?
- 2.16** Well, you have to go back to the definition of what i is. Remember that i is equal to the square root of minus 1. So i squared is equal to minus 1. So I can substitute that in there. I can replace i squared with negative 1. Now, this just becomes what? This just becomes minus BD . Then I could take that number and combine it with the AC , and now it looks the way it should. This is going to look like AC minus BD plus AD plus BC times i . Now, it looks like a complex number, something plus something i . That's fine.
- 2.17** Unfortunately, division, I can't tell you right now because I need a concept before I can actually get to division. But so far, we have these three from here. What would that look like in terms of an example? Let's suppose Z is equal to 1 plus i , and W is, I don't know, 2 minus $3i$, something like that. Whenever you have abstract principles, it's incredibly useful to simply just take very small examples. Take very nice numbers, 1 and 1 , 2 and negative 3 , and work it out and see if you can get these things to start making a little bit more sense.
- 2.18** Because understanding it in the abstract might be useful, but to be brutally honest, I never remember this formula. If you gave me this and you said, "Hey, what's the multiplication?", I'm not going to be like, "Oh, it's AC minus BD , and then it's AD plus BC ." I'm like, no, I'm not going to do that. But I'm going to remember how to do this. I'm going to remember how to do the distribution and say, "Oh, I'm going to have to do this, and then I'm going to have to have an i squared, which is a negative 1 ."

- 2.19** Why do I know that? Well, because I've worked out tons of examples that look like this. I've developed that skill in terms of working on small examples. Having small examples is really beneficial when trying to understand pretty much anything. If I look at the first one, if I take Z plus W , what's that going to look like? Well, if I take Z and W , I add their like parts. So 1 plus 2 , that's a 3 . Then I have $1i$, and I'm subtracting $3i$, so I'm going to get minus $2i$. That's a 2 . There we go, 3 minus $2i$.
- 2.20** For number two, if I take Z minus W , so Z minus W . The real part, 1 minus this part, so 1 minus 2 is a negative 1 . Then I have $1i$, and I subtract off negative $3i$. OK, so what is that? Negative negative $3i$ is going to become plus. I'm going to get $1i$ plus $3i$, so I'm going to get a plus $4i$, like that.
- 2.21** Now, the multiplication is the tricky one because I have all the distribution that I have to do. OK, so let's do that. Number three, if I take Z times W , again, I never remember that formula, AC minus BD and so forth and so on. ZW is going to be what? Z is equal to 1 plus i . W is 2 minus $3i$. Now I just work it out. Now I just work it out.
- 2.22** What am I going to get? 1 times 2 , so that's a 2 , 1 times negative $3i$, that's a minus $3i$, i times 2 , that's a plus $2i$, and i times that is going to be what? Minus $3i$ squared. That's the first pass of just doing the distribution law for each of those. But again, now I have to use this principle of saying, the definition of i was the square root of negative 1 , so i squared is negative 1 . If I replace this i squared with negative 1 , what does this whole term become? Well, it becomes minus 3 times minus 1 , which is just equal to 3 . So I get a plus 3 on that.
- 2.23** OK, well, if I have a plus 3 here and I have a 2 here, 2 plus 3 is 5 . That's great. I have negative $3i$'s plus $2i$'s, that's going to give me a negative i , so I get minus i . OK? So that would be my corresponding multiplication. All right. So that's great. But I'm missing the operation of division. I haven't talked about division yet. How do we actually figure that out?
- 2.24** Well, in order to talk about division, you have to have the notion of what's called the conjugate. Let me give you that. Let me give you that. The conjugate is special with the complex numbers. It's very simple to compute in a concrete example, and it's very easy to visualize geometrically. We want to understand what the conjugate is and how to compute with it. Here we go.
- 2.25** Given, here's kind of a, I guess, vocabulary, if you will. Given a complex number Z is equal to A plus Bi , given a complex number that looks like that, we call A the real part of Z , and we denote it the real part, $\text{RE of } Z$ is equal to A . That's its name. It's just the notation associated to it. We call B the unfortunate name, imaginary part of Z , and denote it $\text{IM of } Z$ is equal to B .
- 2.26** Again, the reason why it's called imaginary is because when Cardano—it's clear actually in the paper that Cardano put forth when he originally proposed this idea of this letter i , is that he even struggled with it himself. He even struggled with this concept himself,

that there's somehow some number that's represented by the square root of negative 1. He even struggled with this idea. It wasn't very well accepted.

- 2.27** Again, given that numbers were typically geometric in nature and things like, oh, the square root of 2 is equal to this because I can draw a triangle that has this side length, this side length, and the hypotenuse is the square root of 2, it was actually a physical entity that you could represent. Because i was not physically representable, geometrically, people were like, it's just imaginary. It doesn't exist. It's not real. It doesn't exist in our real world.
- 2.28** And so, it kind of adopted this name of imaginary, which is unfortunate, because complex numbers are literally like how airplanes get off the ground. That's great. Those are very real things. Those are very real objects. Imaginary numbers, it's just the name that's kind of persisted through time. It's a little unfortunate. But I give them names. I say that A is the real part, B is the imaginary part, so the part that's attached to the complex unit i .
- 2.29** For example, if I gave you Z is equal to $3 - 5i$, then I would say that the real part of Z is 3, and the imaginary part of Z —well, be careful. When you have a negative, you have to bring that negative in. What is the imaginary part? The imaginary part is whatever the number is attached to i . It's negative 5, in this case. It's minus 5. That's what they are.
- 2.30** Now, why is it useful to designate them as the real and the imaginary part? It's because it allows me to graph them. It allows me to graph them. Graphically, it allows me to say that, well, if I call this my real axis, and if I call this my imaginary axis, then I can graph any complex point by simply going over X number in the real direction, Y up and down in the imaginary direction, and plot the point as if it's in the XY plane. For example, $3 - 5i$, I would go over 1, 2, 3, down 1, 2, 3, 4, 5, and I would say this point here is $3 - 5i$. That's my complex number there. I can actually graph these things in the plane, which is interesting.
- 2.31** Now, what's beautiful about them is that I have this notion of conjugate. I have this notion of conjugate. Here's my definition. If I give you any arbitrary complex number whatsoever, so it looks like $A + Bi$, the conjugate, sometimes called the complex conjugate because it's a complex number, but we're just going to call it the conjugate, the conjugate of Z , which is $A + Bi$, is $\bar{Z}$, and then you put a bar over top of it to denote that it's the conjugate. You just say Z , and then the symbol is, you just put a bar over top of it.
- 2.32** So $\bar{Z}$, which is equal to, you keep the A the same. You don't change A at all. You just change the sign on B . You change the sign on B , so you get $A - Bi$. That's the conjugate. Now, it begs the question, why in the world would you think about such a thing? It turns out, because this is exactly what you need in order to make sense of

division, which is not obvious that this needs to be a thing, but this is precisely what you need in order to make division possible.

- 2.33** What would this be here? Let's do this example here. If I have Z is equal to $3 - 5i$, then what would $\bar{Z}$ be? What's my rule? My rule says, identify B and then change its sign. In this case, what is B ? The imaginary part is negative 5, so I change its sign, negative 5 goes to positive 5, so the conjugate would be $3 + 5i$. You might be looking at this and being like, well, you're lying to me because Z is equal to $A + Bi$. You're telling me the definition, it needs to look like $A - Bi$?
- 2.34** And it's like, well, OK, it does look that way. Why does it look that way? Look, this guy really looks like what? This looks like $3 + \text{negative } 5 \text{ times } i$. Do we agree? That's what it looks like. Again, addition and subtraction are one and the same. $3 - 5i$ is the same as $3 + \text{negative } 5i$. What does this really look like? This is $3 - (-5i)$. That's exactly what that formula is telling me over there.
- 2.35** The moral of the story is that if I give you a complex number, look at the imaginary part, whatever it is, change its sign, that's the conjugate. Plus or minus, it doesn't matter. Just change the sign, that's the conjugate. What would that look like? Well, $\bar{Z}$ is $3 + 5i$. So what do I do? I go over 3, but now I go up 1, 2, 3, 4, 5. Now it's up here. If this were Z , this is $\bar{Z}$, which is $3 + 5i$.
- 2.36** What I notice is that, geometrically speaking, what does the conjugate do? It just takes this number and it reflects it across the real axis. That's all it does. It just takes this point here, and it just takes the point here, and just flips it up. That's it. All right? So there's a geometric intuition. So this is the conjugate, this is the reflection about the real axis. It's a reflection about the real axis. That's all it is.
- 2.37** OK, so why is the conjugate useful? Conjugate is useful for the following reason. What happens? What are the arithmetic operations that we have? We have addition, subtraction, and multiplication so far, and we're trending towards division. That's where we're going. That's what we want. What happens if I apply those operations to the conjugate, addition, subtraction, and multiplication? What happens if I give you Z is equal to $A + Bi$, so that $\bar{Z}$ is $A - Bi$? What happens to $Z + \bar{Z}$? Let's apply addition to it. What happens?
- 2.38** Well, look, $A + A$, that's $2A$, and then $Bi - Bi$, they cancel. What happens to $Z - \bar{Z}$? $A - A$ cancels. $Bi - (-Bi)$, so I get $2Bi$. What happens when I multiply them? $Z \cdot \bar{Z}$. I get $A + Bi \text{ times } A - Bi$. If I work that out, I get $A^2 - ABi + ABi - B^2i^2$, let's see, minus B^2i^2 , i^2 squared, I get that.
- 2.39** Well, that's nice. The i 's cancel. Those go away. And i^2 squared is negative 1, so negative 1 times negative B^2 squared is going to give me positive B^2 squared. What do I get? I get $A^2 + B^2$ squared. I get a real number. You stare at that for a second, you're like, whoa, OK, what's happening there? That looks familiar. $A^2 + B^2$ squared. Where is that coming from? Well, geometrically, let's look at it this way.

- 2.40 If I look at the point here, which is $A + Bi$, that means I went over A here and I went up B here. Do we agree? Which means, Pythagorean theorem tells me that's a right triangle, $A^2 + B^2$ is equal to C^2 , which means that that needs to be the square root of $A^2 + B^2$. That's the length of the hypotenuse. So this quantity on the right-hand side is really what? The square root of $A^2 + B^2$, the whole thing squared. That's what it is.
- 2.41 So is this quantity relevant? Right here, and the answer is yes. It's the hypotenuse of this triangle. So it's very relevant, geometrically speaking. So I should probably give it a name if it's relevant. The name of that thing, the name of this, here's my definition, is that the modulus of $Z = A + Bi$ is, it looks like absolute value bars. You just put vertical bars on either side of it.
- 2.42 The modulus is equal to the square root of $Z \bar{Z}$, which is simply the square root of $A^2 + B^2$. That's what it is. The modulus is a measure of how far away is the point $A + Bi$ from the origin. It's its distance. This thing here is the modulus of Z . The hypotenuse is the modulus of Z .
- 2.43 Again, why is this useful? Well, let's go to division. Let's go to division. If I say Z is equal to $A + Bi$, and W is equal to $C + Di$, and that's non-zero, I'm going to assume it's non-zero, because you can't divide by zero. Then what is Z divided by W ? What is this? This is $A + Bi$ over $C + Di$. Now, here's the trick. Here's the trick. How in the world do I divide by an i ? After all, you told me every complex number looks like number plus number i . You did not tell me that I'm allowed to divide by i .
- 2.44 For example, if I gave you 1 divided by i , I would say that's not a complex number. I would argue it is. We just need to know how to make it look like one. How do I do that? The method for doing that is the following. Take this quantity and multiply by 1 in a clever way. What do I mean by that? Multiply by the conjugate of the denominator. Multiply by the conjugate of the denominator. So multiply by $C - Di$ over $C - Di$. You're multiplying by 1 .
- 2.45 Think about it like fractions. If I had a fraction like three-fourths, and I said, multiply that fraction by 5 over 5 , I get 15 over 20 . I haven't changed the fraction. I've changed what it looks like, but I haven't changed its value. The same thing holds here. Multiply by the conjugate above and below, and what are we going to get? What does the denominator become? Well, we figured that out. What is the conjugate, a number times its conjugate? It's equal to the sum of the squares of its components.
- 2.46 What's the denominator? The denominator is $C^2 + D^2$. Notice what's the win there. There's no complex unit i down there anymore. We've gotten rid of it. It's just a real number. What does the numerator become? It becomes a disaster. I actually have to work it out. This is a complete disaster. What is it going to be? A times C , AC . What is this going to be? Plus BD . I get Bi times minus Di , so that's minus BDi^2 . i^2 is negative 1 , so that'll be a plus BD .

- 2.47** Then I'm going to get the mixed terms. Then I'm going to get what? Plus BC minus AD times i . You say, "Well, that, Robbie, that still doesn't look like a complex number. It doesn't look like number plus number i ." It's like, yeah, sure it does. This is a real number. You can now write this as AC plus BD , divided by C squared plus D squared. That's a number, plus BC , BC minus AD , all over C squared plus D squared times i . Now it looks like a complex number.
- 2.48** So division is incredibly complicated. Division is incredibly complicated, but what did we really say? What did we do symbolically? Z over W , we said, multiply by W bar over W bar. And then simplify. That's all we did. It looks like we've done something incredibly complicated, and in some sense we have. But the reality is, is we just took the fraction and multiplied it by 1. That was it. But by multiplying by 1, we changed everything in a way that now makes this something that we can write like this.
- 2.49** Let's just do a silly example. Again, let's just do a silly example. OK, so let's go back. Let's go back to an example. Let's say Z is equal to, we had that $3 - 5i$ earlier. We had that $3 - 5i$ earlier. Then let's just take another one. Let's take W is equal to, let's do something relatively small. We had $1 + i$ earlier as well. I want you to find, what is Z divided by W ? Find Z over W . That's the game.
- 2.50** I say, all right, Z over W , Z is $3 - 5i$. Fine. W is $1 + i$, fine. But now I have to play this game. What do I do? I look at the denominator. I multiply numerator and denominator by the conjugate. I multiply numerator and denominator by the conjugate, $1 - i$ over $1 - i$. All right. What does that do? The denominator, I know what it's going to be. It's going to be the sum of the squares of the real and imaginary parts. It'll be 1 squared plus negative 1 squared. It'll be $1 + 1$. I know the denominator is going to be 1 squared plus minus 1 squared. I know that's the case.
- 2.51** How do you do that? Look, 1 times 1 , 1 squared. 1 times minus i is negative i plus positive i , so the i 's cancel, minus i squared, but i squared is negative 1 , so that's plus 1 . What do I get upstairs? 3 times 1 is 3 . 3 times minus i is minus $3i$. Negative $5i$ times 1 is minus $5i$, and negative $5i$ times negative i , the negatives cancel, so that's plus. And then $5i$ times i is $5i$ squared. Oh, goodness. Keeping track of all the negative signs is the worst thing when doing complex arithmetic.
- 2.52** Because I have minus sign times minus sign becomes positive, but i times i is i squared, which is then negative. You have to keep everything straight. The bookkeeping is a little tough. Let's see what we get. Oh, boy. What do I combine? i squared is negative 1 . OK, so this is negative 5 . 3 minus 5 is negative 2 . Negative $3i$ minus $5i$ is minus $8i$. Divided by 1 squared plus 1 squared is 2 . Divided by 2 . OK, that's a 2 . But then I can write this as minus 2 over 2 minus $8i$ over 2 , and that's equal to minus 1 minus $4i$.
- 2.53** That looks like a perfectly good complex number. That looks like a perfectly good complex number. Arithmetic in the complex numbers makes sense. Just a quick rehash, what have we done? Addition, subtraction, multiplication, division, they all make sense.

We have these two other operations that we've talked about, the conjugate, so negate to the imaginary part, and then the modulus, which was the length of the hypotenuse of the corresponding triangle.

2.54 We saw that the conjugate and the modulus were intimately connected by saying the modulus is the square root of Z times $\bar{Z}$, its conjugate. That's what we have so far. But as with anything in mathematics, as with anything in mathematics, you want to try and understand it from all sorts of different perspectives. We had the geometry of saying, look, if I have a point out here, which is Z is equal to A plus Bi , I had a right triangle running around, did we not? I can drop that line straight down and I get a triangle.

2.55 Well, whenever you have a right triangle running around, it's screaming trigonometry at you. Sines, cosines, tangents, everything like that. They're screaming trigonometric functions. If I give that an angle of θ , well, this is the real part of the complex number. This distance is A . This is the imaginary part, which is B . We already said that this was equal to the modulus, the modulus of Z . We already said that's the modulus.

2.56 In particular, what do I know? If I take the cosine of θ , what do I get? Cosine is the adjacent over the hypotenuse, so this is A over the modulus of Z . And if I take the sine of θ , sine of θ is the opposite. Opposite is B over the modulus of Z . I get those two things. In particular, I can say, what does that mean? That means that A is equal to the modulus of Z times cosine of θ , and B is equal to the modulus of Z times the sine of θ . Do we agree? OK.

2.57 Then I can step back and be like, oh, well, that means that I can write this complex number differently. I could write this complex number, which is A plus Bi . I can really write it as A , which is this scary-looking thing, plus B , this scary-looking thing, times i . Do we agree? I could rewrite Z as A , which is the modulus of Z cosine of θ , plus i times the modulus of Z sine of θ . This is my A , and this is my B . Yes? We can do that.

2.58 Now, again, a natural question is, why in the heck would you do this? If addition, subtraction, multiplication, and division all make sense, normally, why in the world are you doing this and writing it differently? I want to motivate that. I want to motivate that. Look, the modulus of Z can come out, and I'm left with cosine of θ plus i sine of θ .

2.59 Now, there's a Swiss mathematician called Euler who came up with the following formula. This is Euler. One of the biggest pet peeves in mathematics is mispronouncing names of mathematicians. It's very natural to think that this name is Euler. It's not the case. His name is Euler. That's the proper pronunciation of it. What he came up with is, he said, "Look, I know what this is. I know what cosine of θ plus i sine θ is. It's actually equal to an exponential function." In particular, it's precisely that guy. It's precisely e to the $i\theta$. That's exactly what it is.

- 2.60 This is called Euler's formula. Euler's formula says $e^{i\theta}$ is equal to $\cos\theta + i\sin\theta$. That's what his formula says. All right? You step back for a second, you're like, OK, we're in the weeds. What are we doing at this point? Wait a second. You told me complex numbers look like number plus Bi , and now you're contradicting yourself and saying, no, complex numbers can look like modulus times $e^{i\theta}$. I'm saying, yeah, that's true. That's true.
- 2.61 That shouldn't be that farfetched. Because if you go back to real numbers, for example, you go back to real numbers, if I gave you $1/2$, and I said that's a perfectly good real number, it's actually a rational number, then I could say, well, I could write this in a lot of different ways. I could write this as $4/8$. I could write this as $5/10$. I could keep playing this game over and over and over again.
- 2.62 Alternatively, I could say, well, this is equal to 1×2^{-1} , or I could write it as 4×8^{-1} , and I could play this game over and over and over again. I don't think the argument is valid to say, oh, every complex number has to look like this. It's not allowed to look like this. It's like, sure, it is. As long as the mathematical principles are allowed and you're doing things precisely, you can write it however you want. That's fine.
- 2.63 This is taking advantage of trigonometric identities, whereas this is taking advantage of definitional properties of saying this is what complex numbers, this is how they were defined. Just like up here, one-half, I could represent it in a lot of different ways. It's still the number one-half. But I have lots of different ways of writing it. Now, I want to claim, so this form is called the polar form of the complex number Z . It's the polar form of the complex number Z . I want to illustrate why we care. I want to illustrate why we care.
- 2.64 Just because we can write something mathematically or we can do something mathematically doesn't necessarily mean that it's relevant. It doesn't necessarily mean that it's important. It doesn't necessarily mean that it has any utility associated to it. Does that make sense? I want to motivate why this has utility. I want to motivate why this is relevant. OK, that's the game.
- 2.65 The polar form is incredibly important when trying to solve equations. I'm going to motivate this in two different ways. Here's an example. I want you to find all solutions to $X^4 = 1$. I want you to find all solutions to $X^4 = 1$. OK? So bear with me. Bear with me for the next 60 seconds, let's say. I can look at this and immediately be like, oh, plus and minus 1 work. If I plug in 1 and I take 1 to the fourth, I get 1. If I take negative 1, negative 1 to the fourth, I get 1. Fine. OK, great.
- 2.66 Do you agree with me that this is equivalent, so double arrows back and forth just saying this is equivalent, to saying, find all solutions to this? If I just move one to the other side, then that's an equivalent statement. Now what I'm saying is the following. This thing is a polynomial. This is a polynomial of the degree equal to 4. Why is it

degree 4? Because that's the highest power that's occurring on the variable X . It's a polynomial of degree 4. OK.

- 2.67** Well, the fundamental theorem of algebra—big result, big theorem, big deal— says that if you have a polynomial of degree N , whatever N is, in this case, N is 4, then it has four solutions. Four roots. It's got four of them. When we first looked at this, we said, "Oh, there's two, plus and minus 1." I'm like, that's true. What are the other two? You gave me two. The fundamental theorem of algebra says this is degree four, therefore, there are four solutions. You've given me two of them. What are the other two?
- 2.68** Right now, it's not obvious. We can't just look at this and be like, "Oh, X equals, I don't know." This is where the polar form reigns supreme. This is where the polar form reigns supreme. How do we do this? Polar form. Polar form. I'm going to think of X as a complex number. Part of the fundamental theorem of algebra says that this polynomial has all of its solutions in the complex numbers. I may as well assume that X is a complex number.
- 2.69** If I assume that X is a complex number, then I can write it in polar form like that. Do we agree? That was my polar form equation for it. X is equal to the modulus of X times E to the i theta. I'm implicitly assuming here that I have this X value here. Here's my value of theta, and X is equal to the modulus of X times E to the i theta.
- 2.70** Now, similarly, let's look at this equation here, X to the fourth equals 1. What about 1? Can you represent the number 1 as a complex number? The answer is, well, sure. The number 1 is equal to $1 + 0i$. Do we agree? That's $1 + 0i$. The number 1 is, like, here. This is 1, which is $1 + 0$ times i . That's what that one looks like. What does that look like in complex form? Well, its modulus is 1. Its distance from the origin is 1. This is equal to 1 times E to the i times what?
- 2.71** Well, what is its angle? One doesn't make an angle. Its angle is zero. So this is like E to the i zero. Do we agree? The angle is zero. The key is the following, though. The downside of using polar form is that this angle is not unique. This angle is not unique. Why is it not unique? Well, look, the number 1, $1 + 0i$ has angle zero. It also has angle 2π . It also has angle 4π . It also has angle 6π . I can just go around and around and around and around and around, using the 2π -periodicity of cosine and sine.
- 2.72** Cosine of zero is 1. Cosine of 2π is 1. Cosine of 4π is 1. Cosine of negative 2π is 1. Cosine of negative 4π is 1. I go around and around in the circle, depending on the direction. I can represent this differently. What you do here is you say, well, this is technically angle zero plus any multiple, so $2\pi K$ of 2π . I could go around $2\pi K$, 2π , two times, one time, 20 times, or I could go around minus 2π , minus 4π , minus 6π , and so on, where K is an integer. OK? Where K is an integer. OK.
- 2.73** Then I have 1 is equal to the modulus of 1, E to the i times $0 + 2\pi K$. Then let's look at this equation and plug in these corresponding representations of these numbers into X and Y , respectively. Let's see what happens. Well, what happens when I take X and I

raise it to the fourth power? What do I get? I get X to the fourth is equal to the modulus of X to the fourth, E to the i fourth θ . That's how exponents work.

- 2.74** If I have the product of two things and I raise them to a power, what do I do? I take each individual piece and raise it to the corresponding power. So the modulus of X to the fourth, E to the i θ to the fourth, exponential rules tells me that that just brings the 4 upstairs. Well, if that's equal to 1, 1 is equal to $1 E$ to the $i0$ plus $2\pi K$, and that's equal to 1.
- 2.75** Bear with me. I know we're off into the weeds and it's like, what the heck is going on? The punchline is coming soon, I promise. The punchline is coming. Let's look at that equation. Let's look at that. What does that mean and what does that do? It means that the modulus of X to the fourth power has to be 1. That has to be true. The modulus of X to the fourth is equal to 1. In particular, the modulus of X is 1. The modulus is a length. It's a length. It's the distance from the origin out to X . It has to be 1. This is telling me the modulus of X is equal to 1. I know that.
- 2.76** What does it also tell me? It says that E to the $i 4 \theta$ is equal to E to the i zero plus $2\pi K$. It means that those two quantities have to equal each other. Those two quantities have to equal each other. I also know that 4θ is equal to 0 plus $2\pi K$. In particular, θ is, what is this? Divide by 4, so $2\pi K$ over 4, which is really πK over 2. That's what θ is.
- 2.77** What that means is that when I take that θ and I plug it back in up here, I get that X is equal to the modulus of X , which is 1, E to the i times πK over 2, where K is an integer. K is an integer. K is an integer. I'm claiming that these are all of my solutions. Well, think about this for 2 seconds. Think about this for 2 seconds. I know this is a lot. So think about this for 2 seconds.
- 2.78** Look, how many integers are there? Well, integers are what? Negative 3, negative 2, negative 1, 0, 1, 2, 3, 4, 5, so forth. They go on forever. They go on forever. So I have infinitely many integers, and I'm claiming that if you give me a specific integer, this is going to be a solution to this equation, X to the fourth equals 1. You sit back on that for a second and you're like, that's insane. That breaks math because you just told me the fundamental theorem of algebra says this is a polynomial of degree 4. There should be four solutions in the complex numbers.
- 2.79** But you're claiming that this infinite family of numbers, every single one of them, is a solution. Then what is it? You've got infinitely many things, and only four of them should work. That's what I'm claiming, which seems completely insane. The answer is, yes, this is an infinite family of things, but they repeat. They repeat over and over and over again. Every four integers, they repeat. If you give me K is equal to zero, K is equal to 1, K is equal to 2, K is equal to 3, that's a string of four consecutive integers. Those will give me solutions, four solutions in particular.

- 2.80 The second I move to K equals 5, and I plug that in, that's already a solution I have. The second I go to K equals 6, that's already a solution I have. K equals 7, already a solution I have. K equals 8, already one I have. They just repeat. They cycle through the same thing infinitely many times. Let me show that to you. Let me show that to you. Let's write these down.
- 2.81 Let me put a little subscript on this guy. Let me put this $X_{\text{sub } K}$, so that I know that, once I choose an integer K , so say K is equal to 2, if I replace that K with a 2, $X_{\text{sub } 2}$ is the solution corresponding to K equals 2. It's just a label, it's just a tag, so that I can keep track of things. Let's do that. Let's figure this out. Here's our table. Let's say here is K is my integers, and here's $X_{\text{sub } K}$, which is E to the $i \pi K$ over 2. That's what it is.
- 2.82 Now, remember what this thing is. Euler's formula tells me that this is cosine of πK over 2 plus i sine of πK over 2. That's what it tells me. That's what Euler's formula tells me. It's equal to sine and cosine. These are values on the unit circle. If K is equal to zero, what do I get? $X_{\text{sub } 0}$ equals—this should be what? K is zero, so this is cosine of zero, plus i sine of zero. I go to the unit circle, I go to the unit circle, which is like the bane of everybody's existence, because the unit circle is miserable to deal with, but it is a thing, and we have to be comfortable with it.
- 2.83 I go to the unit circle and I say, if I have an angle of zero, where am I located on the unit circle? That's 1, 0, which means cosine of zero is 1, sine of zero is zero. So this is just 1. Then I go back and I say, hey, that was one of the solutions I knew. I already guessed plus and minus 1 were solutions. That's good. I recovered that one. What about if K is equal to 1? What is $X_{\text{sub } 1}$? If K is 1, I get cosine of π -halves plus i sine of π -halves. I go to the unit circle at π -halves. That's at 90 degrees, so that's true north, 0, 1. So cosine of π -halves is zero. So that's zero.
- 2.84 Sine of π -halves is 1, so this number is i . That's new. That was one of the ones I didn't find. I had guessed automatically that plus and minus 1 were solutions. This one I had no clue was a solution. That's new. That's good. So we recovered a new one. What about K is equal to 2? $X_{\text{sub } 2}$ is cosine of π plus i sine of π . Again, the unit circle on π , I'm hitting this point, which is minus 1, 0. Cosine is negative 1, sine is 0. So this is minus 1. Those are the two that I recovered. Those are the two that I knew, plus or minus 1. Before doing this entire thing, those are the ones that I knew.
- 2.85 What about K is equal to 3? $X_{\text{sub } 3}$ is cosine of 3π over 2 plus i sine of 3π over 2. Again, the unit circle says cosine of 3π over 2, that's down here. So this is 0, minus 1. Cosine is zero, sine is negative 1. So what am I going to get? Minus i . Those are the two new ones that I didn't know. Those are the two new ones. But that was only for K is 0, 1, 2, 3. What about K is 4? X_4 . What do I get? I get cosine of 2π , plus i sine of 2π . OK, so what happens? Sine of 2π , cosine of 2π , I'm 1, 0. This is equal to 1. I already have that one. I already have that value.

- 2.86** What do you think X^5 is going to be? X^1 . What do you think X^6 is going to be? X^2 . X^7 , X^3 . X^8 , X^0 . They just cycle back on each other. In particular, X^0 will be X^4 , will be X^8 , will be so on. X^1 is X^5 is equal to so on. X^2 is X^6 is so on, and X^3 is X^7 , and so on. That's why I say, yes, they occur infinitely many times, but they cycle back on themselves.
- 2.87** What are my proper roots? The roots/solutions to X to the fourth equals 1, our X equals minus 1, 1, i , and minus i . Those are my solutions. Those are my solutions. Where were they located? Notice that they actually occur on the unit circle in those locations if I view those as complex numbers. Complex numbers over the complex plane, this point here is 1, which is 1 plus $0i$. This point here is i . This point here is minus 1. This point here is minus i . That's exactly where all four of my solutions happened. That's where they happened.
- 2.88** This particular equation is more general than this. It's more general than this. These things are called the fourth roots of unity. That's their name. In general, a root of unity a solution to X to the N is equal to 1. That would be considered, a solution to that is considered an n th root of unity. What we just found were the fourth roots of unity, 1 minus 1, i minus i .
- 2.89** The only way that we were able to solve for those was to use the polar form of the complex number, and therefore the polar form is relevant because it actually gave me a systematic method for solving algebraic equations. What did I notice about the geometry of these things is that these things lived on the unit circle. They lived on the unit circle. They're all equidistant from each other. Angle-wise, they're all 90 degrees spaced away from each other.
- 2.90** I've got this one here, 90 degrees later, this one, 90 degrees later, this one, 90 degrees later, 90 degrees later. They're all 90 degrees away from each other, which is nice. Maybe that's a fluke. It turns out it's not. They're all equidistributed on the circle. Same angle, distance away, 2π over N . Why are these relevant? Roots of unity, it's not just this parlor trick of trying to solve equations like this. Roots of unity show up in things like harmonic analysis, where you're trying to understand audio signals, visual signals, things like that.
- 2.91** If you're trying to break these down into, how do I look at sampled points on this particular audio signal, and then reconstruct it in particular ways? Reconstructing these things goes under the guise of what's called the fast Fourier transform and the discrete Fourier transform. Specifically, the discrete Fourier transform uses roots of unity to actually reconstruct signals.
- 2.92** It's related to the fast Fourier transform, which is considered one of the best mathematical advances in applied mathematics in the 20th century. This parlor trick of solving for roots of polynomials, which complex numbers facilitates, is actually an integral part of one of the deepest branches of applied mathematics and one of the most influential and impactful branches of applied mathematics of the 20th century.

Lecture 3: Matrix Arithmetic

At a glance — Matrix arithmetic.

- 3.1** Now that we have complex numbers, we're going to take a little bit of a 90-degree turn and start talking about vectors and matrices and things like that, which would typically be couched in terms of linear algebra. But the way I'm going to spin it is, matrices and vectors are a way of taking things that we already know, solving linear equations, so something that looks like $AX + BY = 0$, something along those lines, and trying to figure out what solutions to that are.
- 3.2** It's a way of taking that and saying, let's consider multiple of them at a time and see if I can find a solution to all of them simultaneously. So a matrix is really just a mathematical notation for, a mathematical object for incorporating all those things into one whole. Now, what does that have to do with precalculus? Well, precalculus is all about trying to understand the mathematical foundations necessary for calculus. Calculus is all about rates of change, figuring out how things change on a very small scale.
- 3.3** Well, one of the topics in a calculus course is optimization. Optimization is crucial in everything that we do. I would argue that everything that we do in the world is an optimization problem, even from how do I get from point A to point B the most efficiently, the quickest? How do I get from, how do I purchase this object in the cheapest way? There's always some sort of optimization that we're trying to figure out.
- 3.4** Well, the optimization is tied very nicely to a problem called linear regression, basically, which has its basis in matrices. There's this beautiful interplay between optimization, on the one hand, and linear regression, and it all has matrices involved. That's the topic for today, is that we want to talk about what are matrices, where do they occur, how do we do things with them, and what can we say? And that will help lead us into calculus eventually.
- 3.5** We want to look at matrices. What is a matrix? Now, the way I'm going to motivate this is the following. If I give you the following equation, let's say $2X + Y = 4$. I give you $2X + Y = 4$. I ask you, what does that look like geometrically? Well, then, geometrically, what I can say is, well, I know what things look like when they look like $Y = MX + B$. This is a line. If I can make this equation look like a $Y = MX + B$, then I can say, oh, it's a line. I can graph it. I can do certain things. I can represent it visually.
- 3.6** How does that work? That's going to give me $Y =$ equal to, if I subtract $2X$ on both sides, I'm left with $Y = -2X + 4$. Now it's in the form of $Y = MX + B$, where M is equal to negative 2, B is equal to 4. Now I have a shot of actually graphing this. Now I have a shot of graphing this.

- 3.7** If I put it on my XY plane, I'm going to say I go up to my Y intercept. My Y intercept is up here at 0, 4. My slope, which is this number, that's my slope, that tells me I'm going to go down to and over 1. If I want to try and look at that, I'm going to go down 2, over 1, down 2, over 1. I'm going to go down to here. So this point would be 2, 0, and I'm going to connect those two with my line. OK?
- 3.8** OK, well, that's all fine and good. But what if instead of just giving you a single linear equation like this, what if I give you two? What if I give you two? And instead of saying, I want to know what the solution to each is individually, I want to say, does there exist a solution to both of them simultaneously? That's the question I want to ask.
- 3.9** If I gave you a second equation, what if my second equation looks something like, let's say, minus X plus Y is equal to, I don't know, let's just say 1. Minus X plus Y is equal to 1. I can run the same program and I can say, OK, let's solve for this. If I solve for this in terms of the Y equals MX plus B, well, then I'm going to get what? Y is equal to, if I add X on both sides, I get X plus 1. Now, here my slope is 1. My Y intercept is 0, 1. It's going to tell me, go to this point, which is 0, 1, go up one, over one, up one, over one, and so on, and draw the corresponding line.
- 3.10** Let's label these. This one is Y equals minus 2X plus 4. This one is Y is equal to X plus 1. Now I want to ask the following question. If I pair these two equations together, if I think about these as being two equations, to which I want to figure out, does there exist a unique solution to this? The question I want to ask is, is there some X and Y which satisfies, satisfying both one and two. Is that true? Is there an assignment for X and an assignment for Y for which this thing has a solution?
- 3.11** I could stand back and I could say, look, I could stare at the first equation all day long and say that I could just start picking off solutions to that. For example, if X is 0, Y is 4, that solves it. If X is 1, Y is 2. If X is 2, Y is 0. If X is and so forth, I could just start choosing values of X and solving for Y. But for example, if I choose the point X is 0, then Y is 4, well, if I plug X is 0 into that and Y is 4 that, that wouldn't be true. Do we agree? If I plug in 0, I get 0. If I plug in 4, I get 4, and 4 does not equal 1.
- 3.12** Now, I can see that from the graph because if I look at the point 0, 4, 0, 4 is up here and it only lives on this line. It only lives on this line. It doesn't live on this line, Y equals X plus 1. I can't expect it to solve the equation given by Y equals X plus 1. What I'm really looking for is, I'm looking for this point here. That intersection point would be my unique solution to the system. When you consider multiple linear equations at the same time, this is called a system of linear equations. It's just a system of linear equations. It's a collection of linear equations.
- 3.13** How would I solve this? Well, look, if I take Y here, which is my minus 2X plus 4, I take my Y here, which is X plus 1, at this point, whatever this point is, those two Y values have to be the same. Do we agree? They intersect. If they intersect, they have to be the exact same Y coordinate. In particular, I know that, at this point, so at this point, I have

what? This Y value, minus $2X$ plus 4, has to equal this Y value, X plus 1. That has to be true. That has to be true.

- 3.14** You can think about it as, like, imagine you're crossing an intersection. Imagine you're crossing an intersection, and so you walk across the intersection, and maybe you walk across the intersection at, say, 3 PM. Then somebody else is crossing that same intersection, maybe they're going the other direction, but they're crossing the intersection at 4 PM. Well, if you imagine saying, look, this is the path that I was traveling at 3 PM, and then at 4 PM, somebody else came this way across the intersection. At some point, your guys' paths had to intersect. You guys crossed the same physical location in space.
- 3.15** That's exactly what we're saying here. I've got this line here, this line here. If they intersect, then their coordinates, their physical locations, have to be the same. Their X 's and Y 's have to be the same. But that gives me a method because if I set these equal to each other, I can do what? I can add $2X$ on this side. I'm going to add $2X$ on this side. What does that give me? 4 is equal to $3x$ plus 1 . I'll subtract 1 , subtract 1 . What do I get? I get $3x$ is equal to 3 , divide by 3 , divide by 3 . Tells me X is equal to 1 . Tells me X is equal to 1 .
- 3.16** But this is a location in space. It has an X coordinate, it has a Y coordinate. What's my Y coordinate? Well, it shouldn't matter. If I take X is equal to 1 and I plug it in here, what do I get? Minus 2 times 1 plus 4 , that's 2 . But if I took 1 and plugged it in here, I'd better get the same value because it's in the same physical location. 1 plus 1 is 2 . I get that those two values were the same. And so my Y coordinate is going to be Y is equal to 2 . This unique point right here is $1, 2$.
- 3.17** Now, what does that mean? That means that if I take X is equal to 1 , Y is equal to 2 , it's going to solve this and it's going to solve this. So let's check that. Let's check to see if that actually works. If I take X is equal to 1 , Y is equal to 2 , what do I get here? 2 times 1 is 2 , plus 2 is 4 . That checks. Negative 1 plus 2 is 1 . That checks. It solves both. This is my unique solution to my system of equations.
- 3.18** What I want to do is, I want to say, well, this is all fine and good, but it's not very systematic. It's just like, OK, here's what my equation looks like. Here's what my other equation looks like. Here's what they actually look like geometrically. They're lines. Now I'm going to set them equal to each other. It's like, that's not going to generalize very well. If I gave you now an equation that had 20 variables in it, I can't even write it down. I can't even graph it. It doesn't make any sense.
- 3.19** How do I make this more systematic? That's where matrices come into play. That's where matrices come into play. Let's see how I can write this as a matrix. The definition: a matrix is an array of numbers or objects or whatever you're going to be doing arithmetic with. It's an array of numbers. Let's denote it by something. They're typically denoted by a capital letter. We're going to say it's denoted by capital A , which

has, say, some number of rows, M rows, and N columns. I'm going to have a certain number of rows associated to it and a certain number of columns associated to it.

3.20 Just as an example, a silly example, but just as an example, I could take the matrix A , which looks like, let's say it has two rows and three columns. I've got two rows. Let's say I've got 1, 2, minus 1, 0, 3, 4. I have two horizontal rows and three vertical columns. What we would do is, we would say this is a 2 by 3 matrix. Two is the number of rows. Three is the number of columns. In this case, when I'm doing this abstractly and I'm saying a matrix is M rows and N columns, I'm saying it's an M by N matrix. Number of rows, M , by number of columns, N . That's what we're talking about. This matrix here would be an M by N matrix. Again, number of rows by number of columns.

3.21 Now, they don't have to be shorter on this side and wider this way. They could go vice versa. They could do something like, let's give it a new letter, let's call it capital B . I could have something that's like, 1, 1, minus 1, 3, and 2, 0, 1, 4. That's completely fine. It's taller than it is wide. How many rows do I have? I have 1, 2, 3, 4 rows. This is a 4 by, how many columns? I've got 1, 2. It's a 4 by 2 matrix.

3.22 It's simply a way of collecting data in a very precise way. We're just laying the groundwork for this. Again, we want to understand, here's a new mathematical object. It's a matrix. We want to understand, are there arithmetic operations I can do with this thing? Can I take matrices and add them? Can I take matrices and multiply them? Can I take matrices and divide them? Does that make any sense?

3.23 The answer is, adding them and subtracting them makes a lot of sense, and it's actually very intuitive as to how you do it. The answer to multiplication is, sometimes you can multiply them. A lot of times you can't. The answer to can you divide them is, most certainly no. You can never divide them. Let's think about the arithmetic of matrices, and then I'll tie it back to the system up here. So, arithmetic of matrices.

3.24 Just to motivate this, this might seem somewhat random, but are matrices useful? Are matrices useful? What are they used for? I'll give you some examples as to why matrices are useful from some relatively simple applications up until things that you guys use every single day. Things that you use every single day.

3.25 As a starting example, say you wake up in the morning and you're trying to plan out your day and you want to understand, what is the weather going to be like? What do I need to wear? What do I need to wear throughout the day? Is it going to be cold? Is it going to be warm? Is it going to be rainy, sunny? Things like that. You probably open your phone and you say, all right, well, what's it going to be? The forecast says it's going to be this. OK, great.

3.26 How are they figuring that out? Matrices are used in weather forecasting, where you take past data and you say, look, what's the probability it's going to be sunny today? Well, what was it last year at this time? What is the current weather pattern looking like? If it was sunny yesterday, then it's probably likely it's going to be sunny today. If it

snowed yesterday, it's probably likely it might snow today. If it rained yesterday, it's more likely to rain today. It's going to be very rare to be, like, well, it was 90 degrees and sunny yesterday, and so tomorrow it's going to snow. That's not going to happen, most likely.

3.27 What you do is, you store those probabilities of those events happening. What's the probability that it's going to be sunny tomorrow, given that it is sunny today? You store those probabilities inside of a matrix, and then you take higher and higher multiplications of that matrix to determine probabilities going forward. This is an elementary model of what's called a Markov chain that you'd see in a probabilities or dynamics course, but that's where matrices are used.

3.28 The example that you see every single day, the example that you see every day, when you go onto a search engine like Google, and you just type something in. I want to know what is, fill in the blank, the best food in this location, the best activity to do in this location, fun things to do, whatever, it gives you this list of options. It gives you this list of options. It gives you the top hit, the second hit, the third hit, all the way down. It just rank-orders them.

3.29 There are certain numbers associated to matrices. They're special. They're called eigenvalues. If the matrix is square, so what do I mean by square? I mean the same number of rows as columns. You can imagine if I had three rows and three columns, it looks like an actual square. It's got nine elements in it. Corresponding to square matrices, there's a certain name. It's called an eigenvalue.

3.30 What Google did was they found out that they could model the World Wide Web and all of its webpages and things like that with matrices, with matrices. When you do a search, it's taking your particular query and it's finding a certain eigenvalue associated to that. Then it's rank-ordering the pages based on that. Matrices are used everywhere in the world. They're one of the most important mathematical objects in existence.

3.31 What is the arithmetic that we can do with these things? Well, how do we add them? Adding. You can only add matrices of the same size. It would not make sense to say, take this matrix, which is 2 by 3, and add it to the matrix, which is 4 by 2. They're not the same size. This one has two rows and three columns. This one has four rows and two columns. You can only add matrices of the same size. This is term by term, if the same size.

3.32 Let me give you an example of that. If I give you the matrix, let's call it A, let's make them relatively small. Let's say it's 1, 2, 3, 4, 5, 6. It's a 2 by 3, 2 rows, 3 columns. And I give you a matrix B, which has to be the same size. It's got to be 2 by 3. I don't care what the entries are, but they've got to be 2 by 3. Let's say 0, minus 1, 2, minus 1, 0, 1. Again, this is a two by three. As long as they're the same size, I can add them.

3.33 What do I mean term by term? I mean, well, here's this entry in the upper left corner. Here's this entry in the upper left corner. Do the operation on that entry. What do I

mean? Take 1 plus 0. OK, 1 plus 0 is 1. Great. Move to the next entry. 2 plus negative 1. 2 plus negative 1 is 1. 3 plus 2, 5. 4 plus negative 1, 3. 5 plus 0, 5. 6 plus 1, 7. There you go. That's what you would call the matrix $A + B$. This would be $A + B$. That's how you do it.

- 3.34** Now, what is subtraction? Remember, subtraction and addition are one and the same. How would I do $A - B$? You do the exact same thing, just term by term. It would be 1 minus 0, 2 minus minus 1, 3 minus 2, 4 minus minus 1, 5 minus 0, 6 minus 1, and you get the same size matrix, 2 by 3. The same exact thing. So addition and subtraction, I'm going to say, same as addition. It's term by term.
- 3.35** Again, they have to be the same size. It would not make sense for me to say, look, take the matrix A and the matrix B and say, what is $A - B$? Just think about it. If I say it's term by term, and here's A and here's B , I could go upper left corner, 1, minus upper left corner, 1. Fine. I then move over to the next entry, 2 minus 2. Fine. I then move to negative 1, and I say negative 1 minus what? There's nothing here. It wouldn't make sense to say, do this term by term, if they're not the same size. If they're the same size, it's term by term. That's all you can do.
- 3.36** Matrix multiplication, though, is messy. Matrix multiplication is incredibly messy, and it's very tricky to figure out. So matrix multiplication is done as follows. If I give you two matrices, A and B , let's say A is M by N , and B is something else. In order for A times B to be defined, AB is defined, if what? The number of rows of B is equal to the number of columns of A . If the number of rows of B is equal to the number of columns of A .
- 3.37** What does that mean? That means that, look, if N is the number of columns of A , then the number of rows of B has to be the same thing. Those two things have to equal each other. Otherwise, my multiplication is not going to make sense. It's not going to make any sense. Let's do a small example to figure this out. Let's do a small example. For example, look, I've got B here, which is 4 by 2, and I've got A here, which is 2 by 3. Let's just do this really quick.
- 3.38** Would it make sense to say A times B in this example? In this example, well, let's think about it. What is A ? A is a 2 by 3. I've got three columns. B is a 4 by 2. This is a 4 by 2. That number is not equal to that number, so that is not going to be defined. It's not going to be defined. The multiplication doesn't make sense, which is why I said the arithmetic operations when I said multiplication, sometimes.
- 3.39** But look, if B had been, like, 3 by 2, A is 2 by 3, B is 3 by 2, now they match. Now I would be able to multiply the two. You have to be very careful when it comes to matrix multiplication. Let's do an example. I think writing down a formula for matrix multiplication is not super enlightening. I'm just going to give you an example for matrix multiplication.
- 3.40** Let's take a matrix A , and let's say it's 2 by 3. Two rows, three columns. I'm going to take the matrix, which is two rows, 1 minus 1, 2, 0, 1, 1. Let's keep small numbers. This

is 2 by 3, two rows, three columns. Now, in order for B to multiply A, on the right-hand side, I need B to be a three by something. Let's say it's just three by two. Three rows, two columns. Let's say 1, 1, minus 1, 1, 0, 2. This is 3 by 2. 3 by 2. I've got a 2 by 3 and a 3 by 2.

- 3.41** Well, then what is A times B? I'm deliberately keeping this example up from the beginning because I want to show you how you can understand this using matrices. A times B. What do we have? A times B. A is 1, negative 1, 2, 0, 1, 1, and B is 1, 1, negative 1, 1, 0, 2. That's B.
- 3.42** Now, here's the trick. You'll see immediately why the number of columns in A has to equal the number of rows in B. The way you do the multiplication is as follows. You imagine taking this row of numbers. Take this row of numbers, rotate it vertically, and multiply it component-wise by these three numbers. What do I mean by that? Take this row, rotate it vertically, and multiply by this. Really what I'm saying is, take 1 times 1, so that's a 1, plus negative 1 times negative 1, that's a 1, plus 2 times 0, that's a 0.
- 3.43** What is that number? It's a 2. Where does it go? It goes right there. Then imagine taking that row, and as you rotate it vertically, just propagate it for all of the columns that you have in B and do the same thing. That's what's going to fill out all of the rows in the first entry, in the first row of A times B. What am I going to do next? Take this row of A, rotate it, and now multiply it by 1, 1, 2. What do I get? 1 times 1, 1. Negative 1 times 1, minus 1. 2 times 2, 4. I end up getting 4.
- 3.44** Where does that 4 go? It goes right here. It goes right there. Now, what am I going to do? I'm going to jump down. Now that I can't take this row, I've run out of columns to go, what do I do? I get rid of this row, I jump down to the next row, and I repeat the same process. I take this row, rotate it, multiply, that'll be this entry, multiply, that'll be this entry. Let's figure out what that is.
- 3.45** Take the bottom row of A, 0 times 1, 0, plus 1 times negative 1, minus 1, plus 1 times 0, 0. This is a minus 1. That entry goes right here, minus 1. Last one, take this bottom row, turn it sideways, multiply by this. 0 times 1, 0, plus 1 times 1, 1, plus 1 times 2, 2, 0 plus 1 plus 2 is 3, and I get 3. That is the product of A and B.
- 3.46** It's wonky and a little weird that that's how it's defined. But do you agree that, when you actually see the matrices, it's actually not that horrible to compute? It's a little time-intensive, but it's not the worst thing in the world. Now, one thing I want to notice is, what is the size of A? A was two rows by three columns. The size of B was three rows, two columns. The fact that those two numbers were equal to each other meant that the product of A and B made sense.
- 3.47** And what was the result? What's the size of AB? This is 2 by 2, and that's how it is always. If you have A and B, A is 2 by 3, B is 3 by 2, the result, the product, is going to be this guy, the 2, by this guy, the 2. So, if I'd had, for example, A was 2 by 6, and B was 6

by 7, then the result would be 2 by 6, 6 by 7, 2 by 7. The result would be 2 by 7. Two rows, seven columns.

- 3.48** What in the world does this have to do with anything associated to the original system that we had? Well, now that we understand matrix multiplication, we can say, all right, can I write this using matrix notation? Can I write this using matrix notation? The answer is yes, you can. This is how we make things a little more systematic. Remember, the motivation was, look, given this system of equations, I can look at their lines geometrically, but that's not very systematic. If I give you more variables, you can't even graph it anymore. That's not very robust.
- 3.49** You'd like to be able to say, look, given this system, are there mathematical objects by which I can define these things in a more systematic way? And matrices give you that. How do we do this? This is the matrix form for systems. The way that you do it is you line up your variables. Let me get the solution out of the way. X equals 1, Y equals 2. Let me get that out of the way. You line up your corresponding variables.
- 3.50** What do I mean by that? I write $2X$ plus Y equals this. The next one, I'm going to write something X plus something Y equals something, because the order of the variables. I'm not going to write this as Y minus X equals 1. Even though it's the same equation, the variables wouldn't be lined up anymore. When the variables are lined up, I can write the matrix—this is called the coefficient matrix—as the coefficient of my variables.
- 3.51** What do I get here? I get a 2, I get a negative 1, I get a 1, and I get a 1. That's my coefficient matrix. The corresponding coefficients here, the 2, the negative 1, the 1, and the 1. That's where those come from. Now, if the first column of this corresponded to the coefficients of my X variable, then I can put my X variable at the top of my next matrix. If the next variable was Y , then Y then follows, and so forth and so on.
- 3.52** You can imagine, if I had more variables, I would just keep tracking them all the way down, in the order in which they came. This will now be equal to the right-hand side, which is my 4, 1, which is my 4, 1. Now, I want to check. Does this actually make sense? Well, let's think, what is the size of this matrix? The size of this matrix is, I've got two rows and two columns. The size of this "matrix"—when you only have one column, you don't call them matrices anymore, you call them vectors. That's why I keep saying "matrices".
- 3.53** It's still a matrix. It's just, there's a new name that's given to it. It's called a vector. That's fine. The size of this vector is what? Two rows, so 2, by 1 column. Is this multiplication defined in terms of matrix multiplication? Well, sure. Those two things are equal to each other. Therefore, it's defined. The result will be a 2 by 1. Well, that's good because this is a 2 by 1. If I'm claiming that this is equal, they had better have the same size.

- 3.54** Now, let's just do the multiplication. The multiplication says what? Take this row, rotate it, and multiply it by this column. What do I get? 2 times X, plus 1 times Y, is equal to what? 4. Well, that's equation one. $2X$ plus Y is equal to 4. If I take this row, negative 1 times X, I get minus X, plus 1 times Y, plus Y, is equal to 1. Well, that's this equation. That's nice. That's nice.
- 3.55** OK, so let's give these things names. Let's give these things names. All right. Now I have some notation with matrices and vectors for being able to write down systems of equations. Writing down systems of equations. Again, maybe this is helpful, maybe it's not. I don't know. But at the very least, you can imagine, look, if I gave you five equations, and each of those equations involved six variables, well, then I'm going to have a 5 by 6 matrix here, a 6 by 1 vector here, and a 5 by 1 vector over there, independent of having to graph them because I can't even visualize what they look like.
- 3.56** If they have six variables, what in the world does that look like? We can't visualize that. At least I have a mathematical notation for being able to handle it. How would I recover solutions to this thing? That's the natural question. Look, this matrix equation is representing this system of equations. At the end of the day, I don't care how you write things. I just want, what is the solution? I want X is 1, Y is 2. That's what I want. I don't care if you write it with matrices; I don't care if you write it this way, if you graph them as lines. I just want the answer.
- 3.57** The question now becomes, how do you use this to find the answer, if the answer exists? OK. Let's see how we would do that. Let's give this a notation. Let's call A equal to our 2 by 2 matrix. This is 2, 1, minus 1, 1. Now, how do I denote a vector? How do I denote a vector? The classical way of denoting a vector is just by giving it a variable name. Variables are typically given as an X or a Y or a Z or something to that effect. It wouldn't make sense for me to say something like this. It wouldn't make sense for me to say something like this.
- 3.58** Why? Well, because X is here and X is here. That doesn't make any sense. What I'm going to do is, I'm going to give this one a new label by putting a line underneath it to indicate that this is actually a vector and it consists of these two entries, these two variables, X and Y. I'm giving it a new name. I'm giving it a new name. I'm just putting a line underneath it. Sometimes you'll see people use things like X with a little vector on top of it. You'll see people use that instead. That's fine. My notation for it is, I just put a line underneath it.
- 3.59** Now, on the right-hand side, I need to name this vector, too. I'm going to call that, the classical way to do this is to just call it B, but as a vector, so I put a line under it, and that's 4, 1. This system now looks like what? Here's A times X equals B. This now looks like A times X is equal to B. Do we agree? I'm just calling this guy A, this guy X, this guy B, and so it's AX equals B. OK?

- 3.60** OK, so let's stop for two seconds and say, how do, OK, so I have a matrix now. How would I try and solve something like this? Well, the only intuition I have is going back to, well, when I was given equations like $3X$ equals 5, I could say, oh, I know how to do that. I know how to solve that. What do I do? If I want to solve for X , I just say, oh, OK, divide by 3 on both sides, divide by 3 on both sides, and that tells me X is equal to five-thirds, yes? And I'm happy. It's like, that's my solution.
- 3.61** It's very tempting to say, look, if I'm trying to solve for this vector of variables, it's very tempting and very natural, actually, to say, oh, how would I solve that here? I would just say, OK, X is equal to B divided by A . That seems a very natural thing to do. Just like over here, I said $3X$ equals 5 divided by 3. OK, so I just divide that coefficient out. I get X is five-thirds.
- 3.62** The problem is that this makes zero sense. This doesn't make any sense at all because dividing by a matrix doesn't make sense. You're not allowed that arithmetic operation. We're not going to abandon all hope here. We're going to say, all right, well, can I somehow maybe recast this? Can I recast this in a different light? If I start with $3X$ divided by 3, what am I really doing? I'm dividing by 3, yes, but I'm really multiplying by one-third. Do we agree? I'm really saying, take both sides and multiply by 1 over 3 . Let me put this on the left, actually. 1 over 3 times 5 .
- 3.63** It's kind of what I'm doing. That's my divide out by 3. Why am I writing it as multiplication? Because look, I'm only allowed to multiply matrices. I'm not allowed to divide them. If I can somehow understand divide by three as a multiplication problem as opposed to a division problem, then maybe I have a shot of actually solving this. Maybe I have a shot of solving this.
- 3.64** The question then becomes, does there exist a matrix? Let's call this, what do we call this number, one-third, from a multiplication standpoint? We call one-third equal to what? 3 upper minus 1 is one-third. We say that, look, this is the inverse of 3 . Do we agree? We say that's the inverse of 3 . What do I mean by that? I mean that it is a number such that, when I multiply it by 3 , I get 1 . That's really what I mean.
- 3.65** We want to say, does there exist a matrix similar to this thing as being the one-third? Does there exist a matrix A inverse? Let's give it the same notation, A inverse, such that, when I take A inverse times A , I get 1 . Well, what is 1 in the matrix world? 1 in the matrix world is just a square matrix with 1 's down the diagonal. OK? Does there exist a matrix that does that? Well, I don't know. But let's see. Let's see if we can do that.
- 3.66** If there does exist a matrix, A inverse, that we give that name, then we call A inverse, the inverse of A . We give it a name. We call it the inverse of A . Now, there's a very beautiful result. There's a very beautiful result which says the following. If I give you a 2 by 2 matrix, like the one I have, 2 by 2 , two rows, two columns, I can determine if its inverse exists by computing a specific number. The result is the following.

- 3.67** If a matrix A looks like A, B, C, D , if A equals A, B, C, D , if A is invertible, if A equals this, then, let's say this, A inverse exists if and only if AD minus BC is non-zero. A inverse exists if and only if the upper left corner times the lower right corner minus the offset diagonal, B times C , AD minus BC is non-zero.
- 3.68** Moreover, I can say that A inverse is equal to, I can write it down. It looks like 1 over AD minus BC times, change A and D , there's D , there's A , and take the negative of B and the negative of C . I can even tell you what it is. I can even tell you what it is. Let's go back to our situation over here. Let's go back to our situation over here.
- 3.69** The matrix that I have here, A , is $2, 1, \text{negative } 1, 1$. Let's go back to this example. I have A is equal to $2, 1, \text{negative } 1, 1$. That's my 2 by 2 . I know what A, B, C , and D are. Let's compute this number, A, D , minus B, C . It has a name. It's called the determinant of the matrix. It actually has a name, this thing does. It's called the determinant, the determinant of A . Let's compute it. Let's see what it is. Well, what do we get?
- 3.70** A times D , so that's 2 times 1 . So 2 times 1 minus the product of those two, minus 1 times 1 . What do I get? I get 2 plus 1 . That's 3 . That's non-zero. I like that. That's non-zero. In particular, that means from this result that the inverse exists. The inverse exists, and moreover, it tells me what the inverse is. It says it's 1 over that number, so 1 over 3 , times what? I switch A and D ; I switch those. OK, so D is 1 , so that 1 goes here, the 2 goes there. And I take negatives of B and C . So I get a negative 1 here and a negative negative 1 . I get that. That's what I get.
- 3.71** Now, why in the world do I care about this? Well, again, remember, we're trying to solve this system, AX is equal to B . We're trying to solve this. If A inverse exists, and I multiply by A inverse on the left, that's what I get. The A inverse is meant to cancel the A , so I get X is equal to A inverse B , which would allow me to solve my system. Let's compute that now.
- 3.72** We now know what A inverse is. It's this thing. It's this matrix. We know what B was. That was what? The matrix $4, 1$. I can now multiply A inverse by B , and I hope that I can recover the solution that I found using the lines at the beginning. That's my hope. We'll see if this works. All right, so let's work it out. Let's see what happens.
- 3.73** I get X is equal to A inverse times B . A inverse is that thing, so it's a one-third, 1 minus $1, 1, 2$, times what was it? $4, 1$. That's what B was. That's what we get. Now, we have to make sure that the multiplication makes sense. This matrix is 2 by 2 . This vector is 2 by 1 , so the twos match up. So multiplication is legitimate. All right, so this is one-third; I'll leave the one-third out. And what do I get? I take this row, I turn it, and multiply by this.
- 3.74** What do I get? 1 times 4 , it's a 4 , plus negative 1 times 1 is minus 1 . This row now times this. I get 1 times 4 , it's 4 , plus 2 times 1 is 2 . I do the arithmetic, and this is one-third times 4 minus 1 is 3 , 4 plus 2 is 6 . I divide everything through by 3 . I get $1, 2$, but that was equal to my XY , which means what? Which means X is equal to 1 , Y is equal to 2 .

And that's exactly what we just figured out. That's exactly what we figured out up here, was that X was 1 and Y was 2.

3.75 Now we have a systematic method. We have a systematic method for being able to solve square systems. OK, square systems. AX is equal to B , provided that this A is invertible. We have a method for being able to do that. Now, why would this be useful? Why would this be useful? Let me motivate this in the following way. Again, maybe this is just one of those parlor tricks that we had when we talked about complex numbers, or maybe this is actually relevant in the real world. This is incredibly relevant in the real world. But I need a notion for matrices before we can get there.

3.76 The notion is what's called the transpose. Given a matrix, let's say this. A , I'm just going to illustrate this with an example. Let's say $\begin{pmatrix} 1 & 2 \\ -1 & 3 \\ 2 & 1 \end{pmatrix}$. Given a matrix A , so I have three rows, two columns, so it's a 3 by 2, the transpose of this matrix, the transpose of A , the transpose of A is A^T . And what is that? Here's how you compute it.

3.77 You look at the original matrix A , you look at the first row of A , and you make that the first column of A^T . So you take $1, 2$, you rotate it down, and make it $1, 2$ vertically. You then go to the next row of A , negative $1, 3$, rotate it, and I'm going to get negative $1, 3$. Next row, $2, 1$. OK? That's what the transpose is. You take the rows of A and you make them the columns of A^T . That's how that works. That's what the transpose is.

3.78 In particular, A is 3 by 2, which means that A^T has to be what? If it's 3 by 2, then A^T has to be 2 by 3. It's going to change. Those interchange. Those interchange. Why am I interested in the transpose? I'm interested in the transpose for the following reason. I'm going to give you the following question. Given the points, let's say $(0, 0)$, $(1, 1)$, and $(2, 0)$, given those three points, I can graph them. What do they look like? $(0, 0)$, $(1, 1)$, $(2, 0)$. Those are my points.

3.79 Now, I could ask the question, do these three points live on a single line? You quickly realize, of course not. These two points live on this line, these two points live on that line, and these two points live on the X axis. There's no one line on which all of these points live. That's a hopeless problem. But what I can do is, I can say, find M and B such that the line Y equals MX plus B is closest to the points.

3.80 If the points don't live on a single line, then I can say, OK, find a line which is closest to all of them. If we eyeball it, we would say it's probably this one. Do we agree? It's probably this line in the middle that's going to be closest to all of them. How do I approach this? Well, the way I can approach it is the following. I want to find M and B such that I'm closest, Y equals MX plus B is closest to the point. What I can do is, I can say, well, let's just plug these points in. Let's plug these points in and see if something actually makes sense. See if they make sense.

- 3.81** They're not going to satisfy this equation necessarily. They're going to be off of it. But let's just pretend that they do satisfy this equation. What can I get? Let's just pretend that. We're going to pretend there's a very precise way to deal with this. You would see this in an optimization course. But we're just going to pretend that the points satisfy Y equals MX plus B .
- 3.82** What does that mean? That means when X is 0, Y is 0. When X is 1, Y is 1. When X is 2, Y is 2. What is that going to give me? That's going to tell me 0 is equal to M times 0 plus B . It's going to tell me, for Y , let's see, for this point 1, 1 is equal to M times 1 plus B . For this point, 0 is equal to M times 2 plus B . It's going to give me that system, which I think of as a system of equations in the variables M and B . That's what I think of.
- 3.83** Let's model this as a system. What is this in matrix form? Well, it looks like what? I've got three rows, and I've got two equations. It looks like this will be a 0, a 1, a 2, a 1, 1, 1, multiplied by an M , a B is equal to 0, 1, 0. Do we agree? If I take this row, this times this, I get $0M$ plus $1B$ equals 0. $1M$ plus B equals 1. $2M$ plus B equals 0. I'm literally capturing these equations back.
- 3.84** Now, I'm a little sad because this matrix is not square, so it doesn't make sense to talk about its inverse. The only matrices that have inverses are square ones. Even then, that's not sufficient. You have to make sure that a certain number associated to it is non-zero, the determinant, which is difficult to compute. But what I can do is, I can say, look, if this is my matrix A , if that's my matrix A , what if I multiply by A transpose on both sides? What if I multiply by A transpose on both sides? What's going to happen?
- 3.85** If I multiply by A transpose on both sides, I see, OK, A transpose times A , let's call this my vector X and this my vector B , to keep with the same notation that we had before. So A transpose AX is equal to A transpose B . What do I get on the left? A transpose is 0, 1, 1, 1, 2, 1, times A , 0, 1, 1, 1, 2, 1, times X , which is MB , is equal to 0, 1, 2, 1, 1, 1 times B . What was B ? 0, 1, 0. We work it out.
- 3.86** How do I do this? I take this row, I turn it, multiply by this one. I'm going to get 0 plus 1 plus 4. 4 plus 1 is 5. This times this, so 0 plus 1 plus 2, 4. 0 plus 1 plus 2, 3. 1 plus 1 plus 1, 3. Times M and B , which is equal to, OK, what am I going to get on this side? This times that, I get 0 plus 1 plus 0, so I get a 1. 0 plus 1 plus 0, I get a 1.
- 3.87** Now, I'm exactly in the same situation that I was to begin with. I have a 2 by 2 matrix now, so I have a hope of it being invertible. Do I know if this is invertible? Let's call this new matrix, let's call it capital B . How do I know if its inverse exists? I need to take A times D , minus B times C . I need to compute that. What is B inverse in this case? B inverse is 1 over 15, minus, let's see, what is this? This is 3 and 3, so that's a 9, so 15 minus 9, whatever that is.
- 3.88** Then what do I do? I switched A and D , so this is 3, 5, and I negate these, minus 3, minus 3. That's what I get. If I take B inverse and multiply on both sides on the left, and

on the left-hand side of the equation and on the right-hand side, what do I get? I get what? I get my vector, MB , is equal to, OK, what is this inverse here?

- 3.89** So 1 over 15 minus 9 , what is that? 15 minus 9 is 6 . So this is 1 over 6 . What is the matrix? 3 , negative 3 , negative 3 , 5 . I multiply that by 1 , 1 . We get a 1 over 6 times what? 3 minus 3 is 0 . OK. So that's OK. 3 minus 3 is 0 . What do I get here? Negative 3 plus 5 is a 2 . If I bring the one-sixth in, I get 0 , one-third. But that's equal to M and B . What does that mean? That means that M is 0 , B is one-third, and the line that I was looking at was $0X$ plus one-third, which is just one-third.
- 3.90** The line that I wanted, which went through those three points, was literally the constant Y is equal to one-third, which is what we thought it would be. That is the line that's going to give me the line that is closest to all of the points in my set. This forms the basis of what is called linear regression.
- 3.91** Linear regression is basically saying, it's one of the most important questions of the day, which is saying, if I give you more and more and more data, based on that data, can you predict what's going to happen next? Can you predict what's going to happen next? This forms the basis of what's called supervised learning in machine learning, in data science, and AI, and it's completely revolutionizing the world. This is exactly where it starts.

Lecture 4: Sequences and Series

At a glance — Sequences and series.

- 4.1 All right, so today we're going to get the topic of sequences and series. Now, my personal opinion is that these are the most important things. One of the most important things in all of mathematics is a sequence and a series. Unfortunately, it's also considered to be one of the most difficult. And so what we're going to try and do is take away that difficulty, strip it down to what are the base pieces that you need for sequences and series, and then see what we can do with these things.
- 4.2 What's going to happen is that these things are going to lead us naturally into the concept of a limit. A limit is the, one of the most fundamental principles of all of calculus in terms of understanding what's happening in the very small distance, small scales, but also being able to say what's happening long term. We want to think about sequences and series as giving us an insight into what's going to happen long term.
- 4.3 The idea is basically the following. Sequences and series. A sequence you want to think of—what's kind of nice about sequences is that it gels with our intuition, at least from what a sequence is in real life. If I say I have a sequence of steps that I'm going to be doing to accomplish a particular task, then that says that I'm going to do task one, I'm going to do task two, I'm going to do task three, and I'm just going to follow that sequence of steps.
- 4.4 You can think of it like a recipe, for example. If I'm cooking a certain dish, it says, OK, do this first, then do this, then do this, then do this. It's a sequence of steps for me to be able to cook that dish. It gels with our intuition. Now, what is a sequence, mathematically? A sequence is a function. It's a function such that, if you give me a natural number, 0, 1, 2, 3, 4, so on, my function is going to give me the corresponding term of my sequence.
- 4.5 A sequence is a function F on the natural numbers. What do I mean by natural numbers? I mean 0, 1, 2, 3, and so forth. Sometimes people don't include 0 in the natural numbers. They just say it starts from 1. I'm going to say that they start from 0. That's completely fine. But a sequence is a function on the natural numbers, such that the output F of N , whatever that is, if N is one of my natural numbers, and I plug that in as my valid input into my function F , then it's going to give me the output F of N , such that the output F of N is a real number.
- 4.6 For us, we're going to care about being real numbers. You don't need to restrict it to be real numbers. You can say it's whatever output you want. You could say that they're complex numbers. You could say that they're integers. You could say that they're rational numbers. It doesn't matter. I'm simply just going to say that F of N is my corresponding output. A common notation for this, a common notation, is that we denote $A_{\text{sub } N}$, we call that F of N . That's the notation for it.

- 4.7 We don't necessarily want to be pulling around this F of N in every single computation we're doing. We just want a little bit of a nicer notation for it. It's just $A_{\text{sub } N}$, and we call this the n th term of the sequence. OK? Let's just create some of these things. Let's see what they are. Here are some examples. Here are some examples.
- 4.8 Number one, basically, if you just start writing any list of numbers whatsoever, any list, that's a sequence. You just give me a list of numbers. Let's say 2 minus 1, 1 over 3, 5, 6, so on—whatever list of numbers you want. Then what I'm going to do is, I'm going to declare that this one is $A_{\text{sub } 0}$, this one is $A_{\text{sub } 1}$, this one is $A_{\text{sub } 2}$, and so on, A_3 , A_4 , and so on. Similarly, I could give you a sequence that looks something like, let's start with negative 3, let's go to 5. Let's go to 13, 21, 29, so forth and so on.
- 4.9 The negative 3 will be my A_0 , A_1 , A_2 , A_3 , A_4 , so forth and so on. You can keep playing this game. If I take, let's say, one-half, one-fourth, one-eighth, one-sixteenth, and so forth, that's my given sequence. Now, one question you can ask with sequences is, you can say, look, this first one was random. We were just pulling numbers out of the sky. We were saying 2, negative 1, and then suddenly a fraction shows up, and then it goes back to 5 and 6.
- 4.10 It's like, I have no clue what the next term is going to be. We were just choosing numbers at random, just to illustrate that you can define a sequence by just choosing random numbers, more or less. Whereas this one, if I look at this, I can say, well, is there some pattern? Is there a pattern that I can make out from this? I stand back for a little bit and I say, OK, I started at negative 3, and then I went to 5. So what did I do to go from negative 3 to 5? I added 8 to negative 3. Negative 3 plus 8 is 5.
- 4.11 And then I went from 5 to 13, and I'm like, well, I added 8 again. Then I went from 13 to 21, and it's like, oh, I'm just adding 8 every single time. So if I pick up on that pattern, now I could generate however many terms I want. Do we agree? If I know that the pattern is, the next term is simply going to be 8 plus the previous term, then I can generate as many terms as I want. I'm going to get 37, and then 45, and then 53, 61, so forth, and so on.
- 4.12 Similarly to this one, I say, look, if I start off with this one, I started off with one-half, and then I went to one-fourth, and then I went to one-eighth, and then one-sixteenth. It stands to reason it's probably going to be 1 over 32, 1 over 64, 1 over, so forth and so on, because what am I doing? To go from here to here, I multiply by a half. To go from here to here, I multiply by a half. Multiply by a half. Multiply by a half.
- 4.13 So one question we want to ask is, look, this one is somewhat random. That one's random. It's not clear what the next number should be. To go from 2 to negative 1, I subtracted 3, and then magically, I somehow went positive and then divided by 3. It's like, OK, and then I went from one-third to 5, and then I added 1, and then it's like, well, what are you actually doing? It's like, these are these silly puzzles that people say,

what is the next term in the sequence? Given these 5 previous terms, what's the next one going to be?

- 4.14** This one is random. We don't know. We don't know. This one, it seems like, oh, I'm just adding 8 each and every single time. So maybe, sequences that are generated by simply adding a fixed number every single time are in some way special. Maybe those are special. Similarly, these sequences, I start with a number and I just multiply by the same number over and over and over again. Maybe those are special in some way, and it turns out that they are. It turns out that they are.
- 4.15** We're going to give them names. We're going to give them names. So sequences 2 and 3, so again, the first example that I gave you was basically random, fairly random. These ones are special. So you add a fixed number each time. In this example, we're just adding 8 each time, over and over and over again. In this one, what am I doing? I'm multiplying by, multiply by a fixed number each time.
- 4.16** We want to try and understand, if I think about those as being like, that's my defining characteristic of these sequences— you add the same number over and over again, or you multiply the same number over and over—well, can I give a name to them? That's the name that we're going to give them. Here's the definition.
- 4.17** If I have a sequence such that, the next term, I simply just add the same constant over and over again, like the plus 8 plus 8 plus 8, we call that an arithmetic sequence. A sequence is arithmetic if the next term is obtained by adding a fixed constant to the previous term. We call them arithmetic sequences. Arithmetic sequences.
- 4.18** We call a sequence geometric, it's geometric, if the next term is obtained by multiplying a fixed constant to the previous term. That's going to be a geometric sequence. Down here, we would say, this guy here, number 3, is geometric. Number 2, we would say is arithmetic. That's arithmetic. Now, arithmetic and geometric sequences, you want to give their corresponding constants.
- 4.19** For example, in number 2, how am I going from the previous term to the subsequent term? I'm adding 8. How do I go from negative 3 to 5? I add 8. How do I go from 5 to 13? I add 8. 13 to 21, add 8. We're going to call that D, and the common difference is going to be 8. Call it D for difference.
- 4.20** How can you decide what the corresponding number is here? You look at this term and you subtract this term. What is 5 minus negative 3? It's 8. What is 13 minus 5? It's 8. What is 21 minus 13? It's 8. You notice that there's a common difference of 8 between subsequent terms. That's how you figure that out.
- 4.21** How do you decide what the geometric constant is going to be? We're going to call that one R. Why are we going to call it R? Well, it's because it's a ratio. It's a ratio where you take the subsequent term, one-fourth, and you divide by the previous term. If I were to take one-fourth divided by one-half, what do I get? I get two-fourths, which is a half. If I

take one-eighth and I divide by a fourth, I take one-eighth and I divide by one-fourth, what do I get? I get four-eighths, which is one-half. I notice that common ratios between these, I'm multiplying by a half now.

- 4.22 You define a sequence to be arithmetic if terms have a common difference of 8, common difference of D , and you say that the sequence is geometric, if the common difference, in this case, is a half, common difference is, a common ratio is one-half. The common ratio is R .
- 4.23 Let's look at arithmetic sequences. Let's look at this one. Well, if sequences are functions, if sequences are functions, then a natural question to ask is question one, does there exist a formula for $A_{\text{sub } N}$? Is there a formula for it? At the end of the day, $A_{\text{sub } N}$ is a function. It's a sequence, after all. So it's a function, and functions have definitions. They have formulas associated to them. So it makes sense to say, well, what is F of N equal to? What is F of N equal to?
- 4.24 And the answer is surprisingly simple. The answer is surprisingly simple. It says this. It says, well, the formula is given by $A_{\text{sub } N}$. $A_{\text{sub } N}$ is equal to the following thing. It's equal to A_0 plus N times D . That's all it is. It's equal to A_0 , the first term of my sequence, the very first term of my sequence, plus N times D . Let's see if that pans out. Let's see if that pans out for down here. Let's look at that example.
- 4.25 The example down here, we said that, what, A_0 was negative 3. That's the first term of my sequence. So A_0 is equal to minus 3. And we said the common difference was equal to 8. That was equal to 8. If this formula is actually true, then it should say that any term of my sequence is equal to A_0 plus, let's see, D is 8, so this is $8N$. Any term of my sequence can be obtained like that. And A_0 is minus 3, so this is really a minus 3 plus $8N$. That's what we're trying to understand.
- 4.26 So what happens? What is A_0 ? What does that mean? A_0 means, take N and substitute it in for 0. If I take N is equal to 0, what do I get here? I get negative 3 plus 8 times 0, which is negative 3. That pans out. A_0 is right. A_0 is negative 3. What is A_1 ? A_1 , what does this mean? It says take N , plug it in for N equals 1. What do I get? I get negative 3 plus 8 times 1. Negative 3 plus 8, 5. That checks out.
- 4.27 Next term was 5. Let's keep going. What is A_2 ? That says take N , substitute in 2. So this is minus 3 plus 8 times 2. So that's negative 3 plus 16. What is that? That's 13? 13? That checks out. That's my next term. You can see you're going to keep getting the terms of the sequence. Why is this valuable? Why is this valuable? Well, what if, for whatever reason, I want to know, what is A_{100} ? I want to know, what is A_{100} ?
- 4.28 What I don't want to do is say, OK, well, my sequence starts at negative 3, and I know that I just add 8 every single time. I know how to answer the question. I'm going to start at A_0 , which is negative 3, I'm going to add 8. That's 5. I'm going to add 8 again. That's 13. I'm going to add 8 again. That's 21. That's 29, 37, blah, blah, blah, so forth. I don't want to do that. I don't want to do this 100 times. It's a complete waste of time. If

I have a formula for this, then I can say, oh, what is A_{100} ? OK, the formula tells me, take N and substitute in 100. So this is minus 3 plus 8 times 100.

- 4.29 OK, so that's what? Negative 3 plus 800. So that's what? 797. Done. I didn't need to start from the beginning and add 8, add 8, add 8, blah, blah, blah, blah, all the way down. I didn't need to do that. I have a concrete formula which tells me, look, if you want the 1,500,000th term, OK, replace N with 1.5 million, that'll tell me what the term is. I don't have to compute the previous million-plus terms in order to do that. This is incredibly useful. When you have arithmetic sequences, they simply look like the first term plus N times your common difference. That's kind of nice.
- 4.30 Now, what about geometric sequences? What do geometric sequences look like? Do they have a concrete formula? Just like this one that I have down here. I started with one-half. I then went to one-fourth, one-eighth, one-sixteenth, so forth and so on. Do I have a formula for that? Same question, just different context. Can I find a formula for geometric sequences? The answer is yes.
- 4.31 What is A_N going to be in this case? Well, A_N is going to be the first term of your sequence, A_0 . Since subsequent terms are generated by multiplying by a ratio, not by adding a difference, this will now be equal to times R to the N power. That's going to be my formula for geometric. Let's see that in an example. Does that actually pan out?
- 4.32 If I say A_0 is one-half, that's my first term of my sequence, and I'm simply going to multiply by a half every single time, the ratio is one-half. They just happen to be the same. That's a fluke, but that's OK. The formula tells me it should be A_N is equal to A_0 times R to the N . So A_0 is one-half times one-half to the n th power. So, let's see what happens.
- 4.33 What is A_0 ? Replace N with 0, so I get one-half times half to the 0, half to the 0. That's 1, so this is one-half. Good. I get my first term. Next one, A_1 . I get one-half times one-half to the 1. That's one-half times a half, that's one-fourth. Good. I get my second term. What do I get for A_2 ? I get A_0 times one-half squared. That's one-half, one-half, one-half. That's 1 over 8. Good. I get my third term, one-eighth. It looks like this is the formula that I need.
- 4.34 Now, again, why is this useful? Well, again, if I had said, what is A_{100} ? I don't want to have to start from one-half and just keep multiplying a half over and over and over again a hundred times. To get to what I want, I really just want to be able to say, this is one-half times one-half to the hundredth power, whatever that number is, basically 0. Arbitrarily, I mean, that's really close to 0.
- 4.35 Arithmetic sequences and geometric sequences have very nice formula associated to what their general n th term is going to be. But there's all sorts of sequences out there. The first sequence that I wrote down was essentially random. It was essentially random. Where do sequences occur in nature or in the real world? Well, think about it this way.

- 4.36 What if instead of looking at the stock market, let's just say you look at the Dow Jones Industrial Index. It's up, it's down, it's up, it's down, and so forth. Rather than looking at it as all of the possible jagged edges of going up and down and up and down, what if you say, I only just want to know what is its value every hour, something like that. I don't want to know what it is right now. I just care what is it when they open at that hour, and then 60 minutes later, 60 minutes later, 60 minutes later, and so on.
- 4.37 What am I doing? I'm creating a sequence of values of the Dow Jones industrial average, which is, I'm saying at this hour, it was this. At the next hour, it was this. The next hour, it was this. It's just a sequence of points. It's giving me an approximation as to what the index is doing throughout the day.
- 4.38 Now, I think that's a terrible example, to be honest, because if you're interested in the stock market, you want to know what is it right at this moment so you know what to do actionably. But it illustrates the point that sequences are all around us. All right, so that's what sequences are. The question we want to ask now is, what are series? Are they different? Are they the same? Are they related? What can we say about those?
- 4.39 So a series, a series is a sum of terms of a sequence. By a sum, I mean an infinite sum. This is going to be an infinite sum, which seems a little crazy because you're like, well, I've never really encountered an infinite sum. I've never been asked to add numbers together infinitely many times. It's a new concept.
- 4.40 The notation for this, the notation is going to be the following. It's a little scary when you see it the first time, but it's going to be this. You're going to say K goes from 0 up to infinity, of $A_{\text{sub } K}$, whatever that is. What this represents is it's a very compact way of saying, look, start with K is equal to 0. Plug that value in for that K right there. What am I going to get? I'm going to get A_0 . Write that down, A_0 .
- 4.41 Now, this Greek letter sigma, capital sigma, just means you're adding. That's all it means. It says once you've plugged in K is equal to 0 in for K , move K up to 1. Plug K equals 1 in now. What do I get? I get A_1 . Now move K up to 2. What do I get? A_2 . K is 3. Move it up. Keep going. How far do you go? You go until you get to the top. Well, you can never reach the top, so you just put a dot, dot, dot, and say you're just going to go on forever. You're just going to go on forever.
- 4.42 That's the notion of a series. Notice that it's similar to a sequence in the sense that this thing here is a sequence. The $A_{\text{sub } K}$ is a sequence, but it's different in the sense that a sequence just said, what, you give me a K value, K is equal to 5. I'll give you a number. You give me K is 5, I give you A_5 , that's a number.
- 4.43 The series is different. It's saying, no, I don't want to just give you the number A_1 . I want to say, what's the sum of everybody? They're different objects. They're very different objects. A question that would be a natural thing to ask, it seems a little strange to ask this question, actually, because you're adding infinitely many things

together. It's like, what the heck does that even mean, to add that up? In our day-to-day lives, we don't do this.

4.44 We say, I made \$5,000 this month, and I have \$3,000 in expenses, so I take 5,000 minus 3,000. Oh, I've got 2,000 at the end of the month. Whatever. We do basic arithmetic with finitely many things. We don't say, I made a dollar here, a dollar here, a dollar here. I want to know, if I do that infinitely many times, how many dollars do I have? We don't really comprehend that very well. That's partially what makes sequences and series so difficult. It's trying to get into that headspace of what does it actually mean to add infinitely many things?

4.45 Well, I'm going to tell you what it means to add infinitely many things. A question we want to ask, which seems crazy on the surface of it is, is the sum $\sum_{k=0}^{\infty} A_k$ a finite number? Is that a finite number? If I add these numbers up and I do it infinitely far away, do I get a finite number or not? It seems a little strange. Let me illustrate this. Let me illustrate this. Here's an example.

4.46 Let's look at the sum $\sum_{k=0}^{\infty} \frac{1}{k+1}$. We'll start from 0. We'll start from 0, and I'll keep it with K . K goes from 0 to infinity, of $\frac{1}{K+1}$ plus $\frac{1}{K}$ plus 2. That's what I'm going to define. That's what I'm going to define. I want to understand what is this series. On the face of it, what I could do is, I could say, oh, OK. I could start plugging in values of K . Do we agree? K is equal to 0. I plug it in. What do I get?

4.47 I end up getting 1 over, OK, if K is 0, this is a 1, that's a 2. So I get 1 over 2. Plus when now K is 1, what do I get? This is 1 plus 1, that's a 2. 1 plus 2, that's a 3. So this is 1 over 2 times 3. Plus 1 over, OK, if K is two, 2 plus 1 is 3, 2 plus 2 is 4. So this is 3 times 4. Next one will be 4 times 5, and so on. The question I want to ask is, is that a, is that a finite number? It stands to reason that it's at the very least plausible that this could be a number. This could actually be a number.

4.48 Why? Look at the denominators. The denominators are getting bigger and bigger and bigger, which means that these numbers are getting smaller and smaller and smaller. It's like the further out you go when you add these terms on, are they really contributing anything to the number? Or are they so small that they're basically contributing nothing? That's what we want to try and understand.

4.49 In order to understand this, though, we need to understand, what is the partial sum of this? This is what our brains can actually understand. Our brains can actually understand this part, because the partial sum says the following thing. The partial sum says, look, I don't want to go all the way to infinity. I only want to go to 100 or 1,000 or 5,000. I only want to go to a finite number of terms. That's the partial sum. That's why it's called a partial sum. You're not taking all of the sums, you're just taking some of them. That's what we need to understand.

4.50 Partial sums, what are they, and how do we compute them? The n th partial sum of the series, $\sum_{k=0}^{\infty} \frac{1}{k+1}$ is, you sum up K goes

from 0 to N of $\frac{1}{K+1} + \frac{1}{K+2}$. You say, look, I can't do something infinitely many times, so I'm just going to cut it off at a certain point that's finite. I'm going to call that N. N is allowed to whatever, say 100, 200, 300, 400, 1,000, 1,000,000, it doesn't matter. And that's going to give me a finite thing. That's going to give me a finite thing.

- 4.51** The hope is that I can somehow say something about this. I can say something about that. Can I come up with a formula for it? That would be a miracle. That would be a miracle. If I could actually write down this number is equal to $\frac{5}{N} + \frac{1}{N}$ or something like that, that would be incredible. That would be incredible. In this example, I can. In this example, I can, and I want to show you how we can do that. Let's look at the following thing.
- 4.52** If I look at the partial sum, K goes from 0 to N, $\frac{1}{K+1} + \frac{1}{K+2}$, I claim that I can write that like this. Something in here, something in here, where this something is going to help me figure out what the formula is. Now, in order to do that, I need to figure out how do I take this object and break it up into pieces that I can handle one at a time. How can I break it up?
- 4.53** The question I want to really ask is, can I take $\frac{1}{K+1} + \frac{1}{K+2}$ and write this as something over $K+1$, plus something over $K+2$? Can I do that? Why do I want to do that? It's unclear at this point, but we'll see that it's exactly the right thing to do. Let's ask that question. I really want to say, is there some constant A here and some constant B here, such that this is going to be true? Well, that's our question mark.
- 4.54** If it's true, then when I multiply both sides by $K+1$ times $K+2$, the equality should still be true. Do we agree? If I have the number like one-half is equal to $\frac{X}{Y}$ over something plus $\frac{Y}{Y}$ over something, then when I multiply both sides by 2, the equality is still true. OK? If I multiply the left-hand side by the denominator, what do I get? I get a 1. I'm canceling out the denominator. If I multiply the right-hand side by $K+1$ times $K+2$, what happens? The $K+1$'s cancel, and I get $K+2$.
- 4.55** What happens with B? If I multiply $\frac{B}{K+2}$ by this thing, the $K+2$'s cancel, and I get B times $K+1$. What am I doing? I'm saying, multiply by $K+1$ times $K+2$ on both sides. That's what I'm doing. Do we agree? I multiply by $K+1$, $K+2$. This side, I get a 1. This side, the $K+1$'s cancel. I get A , $K+2$. This side, the $K+2$'s cancel. I get B , $K+1$. And I'm good.
- 4.56** Well, here's the point. Look, if K is equal to minus 1, if K is equal to minus 1, what happens to this term? It goes away. If K is equal to minus 1 here, $-1 + 2$ is 1, I get A is equal to 1. So I get A is equal to 1. If K is equal to minus 2, this term goes away. $-2 + 1$ is negative 1, B is equal to negative 1.
- 4.57** So think about that for a second. This is now equal to what? $\frac{1}{K+1} + \frac{1}{K+2}$, is equal to A , that's a 1, over $K+1$. Plus B , that's a minus 1, all over $K+2$. And so now I can write this partial sum as what? $\frac{1}{K+1} - \frac{1}{K+2}$.

- 4.58** You step back and you say, "OK, wait a second. Why are you doing this?" The reason why I'm doing it is the following. Now write out some of the terms. The terms are no longer going to look like this, one-half plus 1 over 2 times 3 plus 1 over 3 times 4 plus 1 over 4 times 5. What are they going to look like? These are going to look like the first one when K is zero, 1 over 1 minus 1 over 2 plus now K is 1, one-half minus one-third. Plus one-third minus one-fourth.
- 4.59** What's my last one? When K gets all the way up to the top at N, what do I get? 1 over N plus 1 minus 1 over N plus 2. Why is that useful? Well, look at it from here. You're adding finitely many terms. I see a negative one-half and a positive one-half. That cancels that. I see a negative one-third and a positive one-third. That cancels that. My next term here would have had a one-fourth minus one-fifth, negative one-fourth, one-fourth. What's the pattern? The previous term, the negative, cancels the positive of the next term.
- 4.60** In particular, what's going to happen? The previous term here is going to cancel the 1 over N plus 1. And so what's my total partial sum going to be? What's my formula? My formula becomes the only terms that survive are the 1 and the minus 1 over N plus 2. So it's 1 minus 1 over N plus 2. That's my concrete formula. That's a miracle that you can actually write down what is the corresponding partial sum associated to this? What is the partial sum associated to that? It's 1 minus 1 over N plus 2. That's what it is.
- 4.61** Here's a question, is I can say, look, I was originally interested in understanding when the upper part goes off to infinity, when it's arbitrarily large. I just showed, if I only go to a finite number of terms, what do I get? I get 1 minus 1 over whatever N was. The question then becomes, what happens when N gets huge? What happens when N gets huge? i.e, what happens when n goes off to infinity? What's going to happen?
- 4.62** If N were, say, a million, I get 1 over a million plus 2, I punch that into a calculator and I get 0.000. I get a lot of zeros. And then I say, OK, let N be a billion. I get 1 over a billion plus 2. And what am I going to get? 0.000, even more zeros. Do we agree? And then I plug in 100 billion, then I plug in a trillion and so forth, and I get bigger and bigger and bigger and bigger ends.
- 4.63** And what's going to happen? I'm going to get 1 over larger and larger and larger values here. If I plug that into a calculator, I'm going to get more and more zeros in my decimal expansion. Where is this term going? This goes to zero as N goes to infinity. Do we agree? That term is getting arbitrarily small as N gets arbitrarily large.
- 4.64** What does that mean? That means the larger and larger this N becomes, the closer and closer this quantity gets to, what? Closer and closer it gets to 1. So what does that mean? This series is equal to 1. That's what it equals. So we've added now an infinite number of numbers, and it's equal to 1. This thing is equal to 1. That's what it is.
- 4.65** Now, we have to make this very precise. In order to make this precise, we need to make sense of what in the world does this mean? It goes to zero as N goes to infinity.

What the heck is that? What does that mean mathematically? Intuitively, we can make sense of it. We can say, as N gets arbitrarily big, as N goes to 100 million, billion, trillion, so forth, where does this quantity go? Oh, it's going to zero. It's getting close to zero.

4.66 But is there a mathematical notion that actually talks about these things? The answer is yes. This is the notion of what's called a limit. This is what underpins all of calculus, understanding limits. We're going to get into limits. We're going to get into limits. But I wanted to give you this to motivate how adding infinitely many numbers, even though intuitively, it doesn't seem to make sense with our brains.

4.67 We look back at it and we're like, "Wait, I don't understand. What do you mean? You're adding infinitely many numbers and you're telling me you get the number 1? This is the goofiest way I've ever seen somebody represent the number 1." You want the number one? OK, add a half to one-sixth plus one-twelfth and so forth. That's absurd.

4.68 But this notion of series and corresponding sequence of partial sums is one of the most important aspects of mathematics in terms of how you can represent certain objects with infinite quantities. You're adding infinitely many terms and you can represent the number 1 using things like that. Let's look at certain types of series that are special. In a calculus class, this would be made incredibly rigorous.

4.69 We would say things like, when does a series—we'd talk about the notion of converge. What does it mean for a series or a sequence to converge? Well, it means that the limit of a certain quantity exists and it's equal to something very precise. But right now, we're just trying to get the intuition for what are sequences, what are series, what can we do with them, and so forth.

4.70 Let's look at another class of series. These are called the geometric ones, and they come from geometric sequences. So geometric series. The geometric series. A geometric series is the analog of a geometric sequence. Remember, a geometric sequence looked like what? Its terms looked like A_0 . The next one was obtained by taking the previous term and multiplying by R . The next one was obtained by taking the previous and multiplying by R and so on. That's what I get. That was a geometric sequence.

4.71 What is a geometric series? It's going to say, add all of that up. Because remember, series are adding, sequences are points. They're just functions. A geometric series is going to be something of the form, the sum, and goes from zero to infinity, A times R to the N . That's what it's going to be. Actually, if we want to use our K 's, we can say these are K 's. We can stick with the same notation that we've had. K goes from 0 to infinity, AR to the K .

4.72 It doesn't matter what you call this. You just have to be consistent. You have to be consistent. Whatever you want to call this little counter, you have to make sure that that's consistent throughout. It doesn't make sense to say K equals 0 and then have this be an N , or say N equals 0 and then have this be a K . As long as they match up, they're

fine. But you can call them whatever you want. I, J, K, whatever. I just would recommend not using "i" because of the complex numbers that we I talked about earlier. i had a different designation; it was the square root of negative 1. But you could use J, K, M, and whatever you want to use.

- 4.73 But what is this saying? This is going to be, OK, when K is 0, I get AR to the 0. R to the 0 is 1. So this is just A plus K is 1. I get AR plus AR squared plus AR cubed plus so forth and so on. It's literally just saying, take this sequence and add them up. Add them up. A question we want to ask, a question is going to be, do geometric series add up to a finite number? Just like the previous example we had, we had this weird cancelation that was happening. We had a negative half plus half, negative third plus third, negative fourth plus fourth. There was some weird cancelation going on, and that magically allowed us to say that it gave me a finite number.
- 4.74 We want to ask the same question with geometric series and say, well, do geometric series give me a finite number? Mathematically, we would say, do they converge? But do they give me a finite number? So, do geometric series yield finite numbers? And the answer is yes and no. It depends. Yes and no. It depends.
- 4.75 For example, if I gave you this series, the sum K goes from 0 to infinity of, say, 10 to the K, what is this? This is 1 plus 10 plus 10 squared plus so on. You'd be like, well, that's not going to be a finite number. I'm going to take 1 plus 10 plus 100 plus 1,000 plus 10,000 plus 100,000 plus million plus so and so on. You're like, that's never going to be finite. That's going to go off to some goodness knows huge number that we're going to call infinity.
- 4.76 Versus if I gave you something like this, K goes from 0 to infinity of $1/2^K$, so this is what? 1 plus half plus fourth plus so on, eighth, 16th, 32nd, so forth. It's not clear that that number is equal to 2, that it's finite. There's something weird going on. There's something weird going on. This one goes to something massive that we're going to call infinity. This one's going to the number equal to 2. What the heck is going on?
- 4.77 We need to understand, how do we add these things up? How do we add these things up? What am I going to do? Let's look at the partial sum. Nth partial sum. What is that? Remember, what does that mean? We take the upper limit, which is infinity, and just say, cut it off at a finite value. We're going to call that value N. That's all it means. Take the sum from K goes from 0 up to N of AR to the K. What does that look like? Well, I can write it down. It's finite after all. It looks like A plus AR plus AR squared plus all the way out to AR to the N.
- 4.78 The last one that I get is when K is equal to N, so that's AR to the N. OK, that's fine. I still don't know what—does this have a nice formula associated to it? The one that we had previously, it was $1 - 1/2^{N+1}$. That was a nice formula associated to it. This one's a mess. It's like, I've got all of the N terms. I don't see anything that cancels nicely.

It's not like my minus half plus half minus third plus third and so on, that those are somehow canceling. So what can I do? Well, there's a trick. There's a trick to it.

- 4.79** If I take this quantity and I give it a name, let's call it S_N . Let's give it a name. What happens when I multiply S_N by the common ratio R ? What happens? So if I take what is R times S_N , think about this in terms of this expansion here. If I multiply this by R , what happens? What happens? Well, I'm just multiplying every term by R . I'm going to get AR plus AR^2 plus AR^3 plus dot, dot, dot, plus this one times R is what? AR^N plus 1.
- 4.80** That's how exponents work. R times R , I add the exponents of 1. So R times R is R^2 . R times R^2 , 1 plus 2 is 3, so R^3 , and so on. So how do I take R times R^N ? R is R^1 , R^N is R^N , so R^{N+1} , that's what I'm going to get for that. Now, what's beautiful is, let's subtract the two. Let's subtract the two. What happens when I take S_N and I subtract R times S_N ? What happens? I'm saying, take this quantity up here and subtract away this quantity here. What's going to happen?
- 4.81** A ? Do I see anything that cancels A down here? No. So A is going to survive. I'm going to get A . Plus what? So I've gotten rid of A . AR minus, oh, AR . Those cancel. That's nice. Those cancel. AR^2 minus AR^2 . Those cancel. That's nice. AR^3 minus AR^3 . AR^N minus—what's the term that's prior to this one? It's AR^N . This guy is going to cancel with that one. Then I'm going to get minus this guy here.
- 4.82** What do I get? I get A minus AR^N plus 1. Do we agree? Well, that gives me a nice formula now because, look, if I factor out the S_N on this side, I get a $1 - R$. I can simply just divide both sides by $1 - R$, and that gives me a formula for what S_N is, in a nice compact little form. A nice compact form.
- 4.83** The formula we get is the following. S_N , which again, remember, is this. K goes from 0 up to N of AR^k is now equal to what? A times $1 - R^{N+1}$ all over $1 - R$. That's my cute little formula. Now, what I have to be very careful about is what happens to this term when N gets big? When N gets big. That's not great. When N gets big, that thing can cause a lot of problems.
- 4.84** Because, for example, look, if R is equal to 10, like I had over here, if R was equal to 10, when I take 10 to the N , if N is infinity, or if N is a million, for example, I'm taking 10 to the million plus 1, that's a gigantic number. Even if N is small, like N is 1, 2, 3, 4, these are 10, 100, 1,000, 10,000, so forth. These numbers are massive.
- 4.85** But what happens if R is small? Like R is equal to a half. If R is equal to a half, what's going on? If R is a half, what's half to the first? That's half. What's half squared? That's a fourth. Half cubed, that's an eighth. Half to the fourth is a 16th. This is getting small. This is getting very small. So much so that I can pretend it doesn't exist. It's going to go to zero, kind of small. That's the idea.

- 4.86 And again, there's a very rigorous way of saying that. We're saying that the limit as N goes to infinity of this quantity will be zero, if R is a certain value. It's a calculus concept. And again, we're going to introduce that in this course with limits. So what we can say is that the sum K goes from zero to infinity. This is a fact. We have all of the ingredients necessary to actually prove this mathematically, which is amazing.
- 4.87 But the fact is that this sum, AR to the K , is equal to—if this term weren't here, if this term were gone, what would I have? This would be A divided by 1 minus R . It would be equal to that if this term were gone. In order to make this statement, I need to say, when is this equality true? This equality is true, the example that we had over here, we saw, look, if R is small, like one half, the result is true. If R is big, like 10 , the result is false. So where's the cutoff? For what values of R is it true, and for what values of R is it false?
- 4.88 The values for which it's true, provided minus 1 is less than R is less than 1 . Provided R is in between that range, negative 1 to 1 . This formula is true. This formula is true. Let's use it. Let's see what happens. Let's look at this one. What do I have in here? R is equal to one half. Is that in this range? Is that in this range? Yeah, R equals one half, that's less than 1 , and it's bigger than negative 1 . OK. So that means what? This particular geometric series should equal what? The sum, K goes from zero to infinity. So here are some examples.
- 4.89 Of 1 over 2 to the K is equal to what? This is equal to 1 plus one half plus one-fourth plus so on. This is equal to, what is the first term? Well, the first term is right here. So that's A . A is equal to 1 , and R was equal to a half. So if this formula is true, which we've provided all the evidence to suggest it is, we haven't proved it technically, but we've given it enough diligence, let's say, to actually believe that it's plausible, then A is 1 and R is a half. So this is 1 over a half, which is equal to 2 . This infinite series is equal to 2 .
- 4.90 Now, if I hadn't I clarified this, if I hadn't said, look, this equality is only true if R is in between negative 1 and 1 , and I just start throwing this formula around, I just start throwing it around. I start saying, look, A over 1 minus R for any R whatsoever. It doesn't matter. It doesn't matter. Let's go back to this series. The sum of 10 to the K , where K was equal to zero to infinity. This is my 1 plus 10 plus 10 squared plus so forth and so on.
- 4.91 What's my first term? A is equal to 1 . What's R ? That's 10 . What would happen if I plug this in? If I plug that in, what am I going to get? 1 divided by 1 minus 10 . That's negative 1 over 9 . Yeah, that doesn't make any sense. You're taking 1 plus 10 , that's 11 , plus 10 squared, that's 100 , so that's 111 , plus $1,000$, so that's all ones, $1,111$ plus $10,111$ plus $100,000$, $11,000$, all of this, all the way down, all a sequence of ones, and you're telling me that that's negative 1 over 9 . That's nonsense. That's nonsense. This is not true.
- 4.92 Why is that not true? Because R equals 10 does not satisfy being in that range. It's too big. So this formula fails if R is not in that range. But if R is in that range, R is equal to

one half, that's in between negative 1 and 1. When I plug this in, I'm going to get 1 over 1 minus a half, which is 2.

4.93 So that's the beginning of sequences and series, which then provide us an avenue for talking about these things as what happens when N goes to infinity? What does it mean to converge? What does it mean to diverge? What does it mean to have a limit? Which is the beginning of calculus.

Lecture 5: Average Rates of Change

At a glance — Average rates of change.

- 5.1** We're now in the position where we can start building the tools necessary to get into calculus. A natural place to start is with what are called average rates of change. Average rates of change. What I want to do to motivate this is, the first question we have to ask ourselves is, what is a rate of change? What is a rate of change?
- 5.2** Well, it's something that we encounter every single day. You get in your car, you drive, you look at your speedometer, your speedometer tells you you're traveling 50 miles an hour. That is a rate of change, 50 miles per hour. But one way that I like to think about it is viewing a rate. Let's look at units. Let's look at units. Choose a quantity and then give yourself a time aspect to that particular quantity.
- 5.3** For example, we already have miles per hour, which I'm going to think of as fractions, is miles per hour. That's telling me that if I'm going 50 miles per hour and I maintain that speed, then in one hour, I should have traveled the distance of 50 miles. That's what that's saying. But we have other quantities like this as well. For example, we have any sort of distance per time. We can generalize this to say any distance per time. That's fine. Feet per second, meters per second, kilometers per hour, whatever the case may be.
- 5.4** But we have other things as well. For example, if you are an athlete and you train with a heart rate monitor, what is that actually giving you? What data is that giving you? That's giving you beats per minute. Beats per minute, typically, so BPM. It's telling you how many heartbeats are you getting per minute based on the level of exertion that you're experiencing. Every single one of these things is a corresponding rate, and it's a rate of change.
- 5.5** What do I mean by that? It's a change in, a change in something, let's say Y , divided by the change in T , time, or X , or whatever the case may be, whatever your variables are. It's a change in one variable per change in another variable. Up here we would say, Y is being measured in miles, that's the change in miles, whereas T is measured in time, so change in miles per change in time. This one is saying the change in beats, so heart beats per time unit of minutes. Change in Y , divided by change in time.
- 5.6** That's what a rate of change is. If we think about it as kind of this quotient, if we think about it as this being this quotient, then we can actually compute these things if we were given, say, a table of data. Let's suppose I give you the following table. I love cycling. I think it's absolutely amazing. Let's suppose you go out for a bike ride and you decide to do, say, 50 miles. That's what you want to do.
- 5.7** What you do is, you just make a table and you say, OK, here's going to be time. Let's say that variable is T . Then we'll say D is distance. D is distance. This is going to be

represented by miles. That's going to be represented by miles. Let's say at T equals zero, so zero hours, you start off. So your distance is zero. You haven't gone anywhere. You just literally just get on your bike, you start your stopwatch. OK, great. And now you start taking off.

- 5.8** And then every single hour, you check what is your distance. So let's say, after the first hour, it's a slow warmup. So maybe it's like, you've gone 16 miles in the first hour. And then in the second hour, well, you've already gone 16, so let's say you go an additional, you're starting to feel a little better on the bike. You go an additional 20. Let's say you go 36 now. Then in the third hour, you start trying to calm down a little bit. Maybe you've turned back around, you're heading back home, whatever the case may be, and you go your additional 14. Let's say you go your 14, so you end up going 50.
- 5.9** The question we want to ask ourselves is, well, what is our average speed? What is our average speed? What this is telling me is that, within a three-hour period of time, I rode 50 miles. OK, great. That's fine. I could say that my average speed is equal to what? Well, how far did I go? Divided by how long did it take me to go that distance? Really, what that's saying is, it's saying, how far did you go? You went 50, but you started at zero. But these are changes in that quantity.
- 5.10** My average speed will be the change in, let's say, position divided by a change in time. What's my change in position? Well, I started at zero and I ended at 50. My change in position is going to be, I ended at 50, and I'm going to subtract off where I started, so 50 minus zero, divided by what is my change in time? Well, I started at zero and I ended at 3 hours, so this is going to be 3 minus 0. This is going to be 50 divided by 3 miles per hour. All right?
- 5.11** So what is that? Well, 3 goes into 50, what? 3 goes into 5 once, remainder of 2, and then it goes into 26. So it's like 16 miles and change, 16 miles per hour in change. OK, a little bigger than that. All right? That's what it's saying my average speed was over the course of the entire trip. But is that actually an accurate reflection as to how the whole bike ride went? The answer is, well, no. Why not?
- 5.12** Well, I wasn't going 16 miles an hour the entire trip. When I actually just get on my bike, I climb on my bike and I start pedaling, I'm not suddenly going from a dead standstill to immediately 16-plus miles an hour. I have to get to 5 miles an hour, 7 miles an hour, 10, 12, 14, 15, 16, and so on. There's a gradual build into it. Then realistically, I'm not maintaining 16 miles an hour the entire time. Maybe I come up to a stop light. I have to come down from, maybe I'm going 18 miles an hour and I drop down back down to zero to stop at the stoplight.
- 5.13** So, the averages are missing certain important details. Now, what I could even do further is I could say, well, this is the average speed, what? Over the whole trip, over the whole ride. That's the average speed that I have over the whole ride. What if instead I'm interested in saying, look, the first hour of my ride, that's a warmup. First

hour of my ride is just a warmup. I'm just trying to get my muscles going, trying to get my heart rate going, trying to get things going. Really, where I'm putting in all the work is in between hours one and two. This would be a warmup.

- 5.14** This would be working. I'm actually trying to get a workout in within that period of time from hour one to hour two. A natural question to ask would be, oh, well, then what was my average speed when I was actually doing work? What was my average speed there? And so I could say, OK, what was the average speed during my working phase? What's that going to be? Well, it's going to be the same quantity. It's going to be the change in the position over the change in time. But I have to be careful.
- 5.15** What is my change in position now? My change in position is not the, I went 50 miles when I started from zero. Now it's what? I end up at 36. I started at 16. And so what's my change in position going to be? Well, I ended at 36. I started at 16. And I ended at hour 2, and I began at hour 1. So the change in my time is going to be 2 minus 1. So what do I get? 36 minus 16 is what, 20? And 2 minus 1 is 1. I was going 20 miles per hour in my working phase.
- 5.16** Clearly different from the average speed that we had over the entire duration. But even that is somewhat misleading. Because I don't know about you, but at least, think about this when you drive your car. When I drive my car and I, say, travel 15 miles, wherever it is, at the end of it, there's a little display that pops up on my car and it says this was your average speed, this was average fuel consumption, and they give me all these stats.
- 5.17** But if I look at that and I say, OK, my average speed was, say, 45 miles per hour, well, does that mean that I was going 45 miles per hour the entire time? And it's like, no, I didn't just get to 45 and hit cruise control and just sit there at 45. I've got to speed up, I've got to slow down, I've got to do things like that.
- 5.18** So even just like this, in my working phase, it tells me that my average speed was 20 miles per hour, but that doesn't mean that I was literally on my bike holding 20 miles per hour the entire time. There are times where it dips below and I'm maybe going 18, 19. There are times where it dips above, where I'm 21, 22. But on average, it averages out to 20 miles per hour. That's the notion of average rate of change.
- 5.19** How do we make this more systematic? What we've done now is, we've said, look, here's a table. We're able to compute some values based on that table and interpret those values physically in terms of what's happening. Can I make this more systematic, mathematically? The answer is, of course you can. How do we do that? How do we make this more systematic? Well, it's this quotient that I'm talking about, this change in Y , divided by change in time. That's how we make this rigorous.
- 5.20** Here's our definition. The average rate of change of F on the interval, let's say, A, B , is, let me call it F sub average. I'm just giving it a name. It's a number, but I'm just going to give it a name, F sub avg, so F average. What is it? Let's go back to the examples that

we had. How did I get the average speed? I said, well, where am I at the end of my journey, minus where was I at the beginning of my journey? Divided by what was the time I finished the event minus the time that I started the event?

- 5.21** It's the change in distance divided by the change in time. What is this going to be? It's going to be, where is my function at B? Well, I don't know what it is, but I can symbolically represent it as, where is F at B? That's F of B. Minus where was F at A—that's the change in F on that period of time, divided by the period of time, which is B minus A. I want you to think about as being, this is the change in F over the change in the input, whatever we want to call the input, whether that's the variable X, whether that's the variable T, whether that's the variable whatever, whatever the input is, that's what we want to understand.
- 5.22** That is my average rate of change. That's all fine and good. Now we have a notion of finding the average rate of change. We've said that that's maybe a good measure of something. Maybe that's a good measure of something. For example, a big thing in cycling or in just endurance training in general is heart rate training. Heart rate training. You want to be able to train your heart at a certain rate. It's very beneficial for increasing endurance and how your body burns fuel and all that stuff.
- 5.23** For example, if your heart rate training level is at, say, 150 beats per minute, and that's what you want to maintain, and you've got your heart rate monitor on and you get on your bike, and you get yourself up to 150 beats per minute, well, then what are you trying to do? You're trying to maintain being in that 150 beats per minute for as long as you possibly can. That's the idea.
- 5.24** Well, are you always going to be at 150 beats per minute? Well, not really, because you're going to pedal faster, you're going to increase resistance, and so maybe your heart rate ticks up a little bit and you get up to 152, 53, 54. And then maybe you start going downhill and you start spinning a little easier, and it's like suddenly it starts dropping down to 151, 150, 149, 148, 147. And then it fluctuates above and below that. Just like our cars when we're driving, if I say my average speed was 50 miles an hour, it's like, yeah, sometimes I'm above 50, sometimes I'm below 50, but on average, it averages out.
- 5.25** It gives us a coarse measurement of the physical things that we're interested in. How could I interpret average rates of change graphically? How would I do that? Graphical interpretation of a rate of change. I'm just going to call it a rate of change right now because we don't have any other rate of change, other than the average version that we have for rates of change.
- 5.26** But what we're building towards, and what calculus is going to talk about is, the way I want you to think about this is, look, going back to the cycling example, if you look at, say, I don't know, the Tour de France, and you look at a particular stage of the Tour de France. Let's say it's a sprint stage. It's a relatively flat terrain, and at the end of the

stage, it's this big, massive sprint where everybody's just going as hard as they possibly can to try and get to the finish line.

- 5.27** During the first three or four hours of the event, they're probably cruising at a relative 27, 28 miles an hour. And that's relatively fine for an elite cyclist like that. They can maintain that for hours. But then once they get to the end, suddenly they're sprinting as hard as they can and they're going 45, 47, 48, 49, 50 miles an hour, trying to sprint for the finish line, right? Well, I can look at their average and say, OK, they averaged about 28, 29 miles per hour over the entirety of that event.
- 5.28** But what I really want to understand is what happens right in those last few seconds. What happens in that last 15 seconds of that event? Everybody's pushing super hard. They're going 45 miles an hour. I want to understand that. That's an instantaneous rate of change. That's saying, in that instant, you are going 45 miles per hour on your bike, not over the course of four hours, you are doing 28 miles an hour. I want to understand instantaneous rates of change, and that is what calculus is all about.
- 5.29** But the way that you get there is, you understand average rates of change, and then you start shortening your time interval in between that very, very, very small. For example, in the bike event, I could say, just like we had in our table of data, we said when time was equal to zero, I wasn't moving, and when time was equal to 3, I was at 50 miles. My average speed was 50 divided by 3, 16 and change.
- 5.30** What if instead of saying, even just cutting it up by hour, 0, 1, 2, 3, what if I start subdividing it into 30 minutes? Every 30 minutes, I take a measurement, or every 15 minutes, I take a measurement, or every 10 minutes, I take a measurement, or every 5 minutes, 2 minutes, 1 minute, 30 seconds, 15 seconds, 10 seconds. If I look at a small interval of time, which is 10 seconds, think about this just from a biking standpoint. Let's even shorten it a little bit further.
- 5.31** If I take a 5-second window and I say, what was my average speed in that 5-second window? That's probably really close to what my actual instantaneous speed is. Because in a 5-second period of time, can I really ramp up my speed that quickly unless I start sprinting as hard as I can? Probably not. If I'm just riding normally, 5 seconds down the road, my speed at point time 0 and time 5 seconds later, probably about the same, more or less. Somehow, shortening this time interval, this window, allows me to get some control on what's happening instantaneously.
- 5.32** That's the notion of a limit, and that's the notion of calculus and derivatives. That's where we're headed. That's where we're headed. Let's look at the graphical interpretation of rate of change. Well, I want to give you an example of this. I want to give you an example. I'm going to give you the function F of X . I'm going to make it a nice quadratic. I'm going to make it a nice quadratic. I can show you how I come up with these examples because I think this is actually a very valuable skill, number one.

- 5.33** One of the best things that you can do to actually learn a particular discipline is not just read a book on how to learn it or do homework exercises on how to learn it. That's valuable. That's completely valuable. But what's better, I would argue, is if you can step back and say, OK, I want to create an example that illustrates fill-in-the-blank, whatever that principle is. I want to create an example that illustrates blank. What do I need to do? How could I do this?
- 5.34** I could say, all right, I want to create a function. I want to make it relatively simple, so polynomial, quadratic, something like that. I want it to go through points that I get to control. I'm going to choose points. I'm going to say this point here, 0, 0. I'm going to say this point over here, which is 2, 0. Maybe we'll just go right in the middle for the top of it. We'll say 1, 1. Those nice points, 0, 0, 1, 1, 2, 0, and I want to create a function that goes through those points and points upside down. It's going to look something like, something like that. That's what I want to do.
- 5.35** I can say, all right, well, I know quadratics are completely—they look like this. They look like $AX^2 + BX + C$. That's what quadratics look like. I know that if A is negative, then it points down. I know if I go through 0 here and I go through X is equal to 2, those are the roots of the polynomial. In particular, this thing could look like F of X is equal to minus X times X minus 2. When X is 0, the function is 0. When X is 2, the function is 0. That's capturing these two points, 0, 0, and 2, 0.
- 5.36** The negative out front is simply saying, turn this upside down. I can look at this function as minus X squared plus $2X$. That's what its graph is going to look like. All right. So F of X is equal to that. What is F average, so I want to say, find F average on the following intervals. Number one, we're going to take the interval from X is 0 to X is 2, so 0 to 2. Number two, we're going to take the interval from, let's say, 0 to 1. We're going to take, number three, the interval from 1 to 2.
- 5.37** Then number four, we're going to take the interval and say, so what is this point here? This is where X is equal to 1. What if I just choose this point is one half and this point is what, 1.5? So this is what, 5 over 2? Let's just choose that interval, one half to five-halves. One half to five-halves. Let's look at those. I want you to compute F average on each of those intervals. OK, so how do we do that?
- 5.38** Well, I need to understand what the question is asking. It's saying, find F average. OK, so I need to ask myself the question, do I have a formula for the average of a function on a given interval? I step back and I say, OK, here it is. F average is F of B minus F of A over B minus A . It's the change in F , divided by the change in input. In this case, my input is X . So change in F over change in X . I can start computing these things.
- 5.39** For number one, I'm going to get F average is equal to F of, OK, so what do I have? F of 2 minus F of 0, all over 2 minus 0. So this is F of 2, minus F of 0, all over 2 minus 0. All right. So I have to go up and say, all right, what is F of 2 and what is F of 0? Well, what

happens when I plug in 2 for my function? That's going to give me a 0. What happens when I plug in 0 to my function? That's going to give me a 0.

- 5.40 What am I going to get in the numerator? I'm going to get 0 minus 0 over 2. Well, that's just 0. That's just 0. That makes sense. I started at the origin, I ended back on the X axis. It's like, my rate of change is like, I didn't really go anywhere. I didn't really go anywhere. What does that mean physically? What does that mean physically? Look, if this Y is distance, if this Y is distance, you can think about it like this.
- 5.41 Look, if I am walking in this period of time, from 0 out to T, 0 out to 2 on my X axis, I think about this as time, and my Y is representing distance, I can think about this as, OK, in between 0 and 1, I walk maybe a mile, but I walk the mile in which direction? I walked a mile from, say, this point to that point. But then what did I do? I started from that point and I walked back to this point. I walked one mile going away from the point, and then I walked one mile coming back to the point.
- 5.42 If you think about this in terms of that scenario, then the average rate of change makes perfect sense because it's saying the following. If I start at this physical location and I go and walk a mile, and then I walk back, and I stand in this physical location, have I moved from this location from a net effect? The answer is, well, not really. My net change in my position is zero. I started here, I ended here.
- 5.43 Now, did my distance change? Of course, my distance changed. I walked a mile that way, I walked a mile this way. So my distance changed, but there's some directionality to it. It's like, if I walk to my right, I count that positive. If I walk to my left, I count that negative, and they somehow cancel each other out because my net change is zero. My average rate of change is zero.
- 5.44 Let's compute the next one. We've done this one. The F average was zero. In between 0 and 1, in between 0 and 1. So, F average is F of 1 minus F of 0 over 1 minus 0. What do I get? What happens when I plug in X is equal to 1? We need to do some computation for this, so I'm going to erase the definition that we have. What happens? What do I get for F of 1? That's minus 1 times 1 minus 2 minus 1 times minus 1. That's positive 1.
- 5.45 We already said F of 0 is 0, so this is 1 over 1, so that's 1. My average rate of change is equal to 1. What does that look like graphically? That looks like, if I started at this point and I went to this point, it looks like all I did was, I went up 1 and over 1. That was my rate of change. Do we agree? My F average was like, if I take the line that goes between these two points, the slope of that line is equal to 1. I go up one, over one. This slope here, the slope is equal to F average. The average rate of change is given by my slope of that. OK?
- 5.46 All right, so let's see what happens from 1 to 2. If I look at 1 to 2, I'm looking at, so F average, this will be what? F of 2 minus F of 1 all over 2 minus 1. We've already computed some of these things. We already know F of 2 was 0. That was computed up

here. So this is 0 minus F of 1. F of 1 was computed here. That's 1. This is going to be negative 1 over 1, which is minus 1.

- 5.47 Well, again, that tracks in terms of our interpretation of this as saying, look, if I start at this point and I go down to this point and I draw the line between those two things, that line has what slope? Slope is equal to F average. I go down one, over one. That's negative 1 slope. It's negative slope. Now, you interpret this, you're like, well, wait a second, Robbie. You're telling me that if the Y axis is representing distance and the X axis is representing time, then you're telling me my F average is negative 1? I have a negative rate of change?
- 5.48 I have a negative feet per second, negative miles per hour, a negative—what the heck does that mean? What do you mean by negative? That doesn't make any sense. Of course it does. It means, look, I'm walking away from this point in a direction that I call positive, and then what am I doing? I'm returning to the point in a direction that I call negative. What does the negative mean? It means I'm going at a rate of one unit per time just in the negative direction. I'm returning back to where I started.
- 5.49 That's what the negative is indicating. The negative is indicating that you are somehow returning back to wherever you were. All right? That's one interpretation of it. There's lots of ways of interpreting these things. OK, so that's this one. Now, this one is a little different. So I'm going to do this one up here. How do I compute, for number four, F average for the interval one half to five-halves. So this is, where is F at five-halves minus F at one half, all over five-halves minus one half, OK? I want to understand that.
- 5.50 All right, so let's see what happens. I need a formula. Oh, boy. OK, so that's my formula for what F is. So I got to plug in five-halves to that and one half into that. OK, so this is messy, but that's OK. The real world is messy, right? So mathematics that's supposed to be describing the real world being messy doesn't seem all that too out of place.
- 5.51 When I plug in X as a half, I get minus a half times half minus 2. This is why you're thankful that you understand algebra now, right? Because these algebraic principles are very, very important. When I started learning mathematics, my calculus teacher told me, "You don't learn algebra until you take calculus," and I love that. That was one of my favorite quotes of his because I think that's true.
- 5.52 You have all these concepts, and suddenly you're like, "Oh, in algebra, it's like, I remember how to distribute. I remember how to gather like terms and do order of operations and all that stuff." Then suddenly it's like, yeah, now you're doing that, but you're just doing it in the context of doing these computations. It's nice. Everything builds nicely, mathematically. Let's see what we get.
- 5.53 OK, so this is 4 divided by 2. That is 2. I'm going to hold on to that. The denominator is a 2. OK. Oh, boy. OK, what do we get up here? What is this? Five-halves minus 2. That's one half. I get minus five-halves times positive one half. That's minus five-fourths. This

one is a minus five-fourths. Minus, what do I get on this one? I get minus a half times, I have one half minus 2. So that's what? Negative three-halves. Negative three-halves.

- 5.54** And so that's what? That's three-fourths. I get negative five-fourths minus three-fourths. OK, so that's nice. That's minus eight-fourths. Right? So this is what? Minus 8 over 4 divided by 2, and this is negative 2 over 2, so this is minus 1. Let's just double-check to make sure everything worked. If I take one half minus 2, that's really one half minus 4 over 2, so that's negative three-halves. That's my negative three-halves. I then multiply by a negative one half, and that gives me a positive three-fourths. So this quantity is three-fourths. It's giving me this. I subtract that off.
- 5.55** What did I get here? Getting five-halves minus 2, that's really four-halves. So 5 minus 4 is 1, so that's one half. And one half times negative five-halves is negative five-fourths, which is what we have here. If I subtract those off, I get a negative 8 over 4. Negative 8 over 4 is negative 2, divided by 2. That gives me a negative 1. All right. So that's giving me the average rate of change.
- 5.56** Now, what we want to try and understand is, how do I make sense of smaller and smaller intervals, smaller and smaller intervals? In the example I gave you with cycling, I said, look, if you take an hour period of time, your velocity is fluctuating drastically, just like your heart rate is going to be fluctuating drastically. If you're doing an uphill climb, if you're climbing a mountain for an hour, your heart rate is probably going to be pretty elevated.
- 5.57** Versus if you're descending a mountain for an hour, I don't know what kind of mountain that would be. That would be a massive mountain, if you're descending literally for an hour. That's a very long time. But let's suppose you're descending the mountain for an hour. You're probably not doing a whole lot of work. Gravity is just pulling you down the hill. So your heart rate is probably pretty low. Over that hour period of time, your heart rate is fluctuating significantly. It's either high or it's either low.
- 5.58** But if I shorten that down and I say, OK, I only want to know what is my heart rate within this 10-second period of time, it's probably pretty constant. Your heart rate is not going from, say, 60 beats per minute, in a 10-second period of time, from 60 beats per minute to 130 beats per minute. It might go from 60 beats per minute to 67 beats per minute because it's elevating, or 60 to 70, maybe. But overall, it's probably relatively stable at that point.
- 5.59** We want to make sense as to what's happening when I shorten the time window that I'm considering. Let's go with this notion that we had of slopes of lines, these slopes of lines for these particular objects. Let's look at this and see what we can say. See if we can abstract things a bit, to try and get us into calculus a little more simply. Here's what I'm going to, here's what I'm going to start with. I'm just going to start with the graph of a function.

- 5.60 The function is going to look something like this. I'm going to make it look something along those lines. Do I need to know what the equation is for this function? No. What I'm trying to do is, I'm trying to give myself this playground to just explore and decide if the things that I'm going to do are, one, mathematically sound, and if they're mathematically sound, can I somehow abstract them to more general concepts? That's the idea.
- 5.61 Here's what we're going to do. We're going to stick with a point here. We're going to call it A, whatever that point is. If this is the graph of Y equals F of X , if that's the graph of Y equals F of X , then I can say, well, at this point A, if I go up to my graph, at this corresponding point, I know what that point is. It's got to have coordinates A, F of A. Do we agree? You give me the input of A, the graph has to give me an output of F of A, so that point has to have coordinate A, F of A. That's fine.
- 5.62 Now I want to say, OK, well, what happens if I move, say, over here, and I look at the corresponding point there, and I say, can I come up with the equation of the line which connects those two points? Can I come up with the equation of the line that connects those two points? The answer is yes. Let's call this point B for the time being, so that this point here is B, F of B.
- 5.63 I want to say, can I come up with the equation? If I were to extend this infinitely, I'd have this infinite line going in both directions. The point that I'm really only interested in is really just this little segment that's connecting the two. But the first question I want to ask is, what is the equation? What is the equation of the line connecting the point A, F of A, to B, F of B? That's the first question I want to ask.
- 5.64 At the very least, geometrically, this is a plausible problem. It's like, I've got a point here, I've got a point here, I can connect it by a line. It's no different than if I drew a point on the ground here and I drew a point on the ground there and I said, hey, connect the points by a line. There's no difference. What ingredients do you need to find the equation of a line? At the end of the day, I'm looking for the equation of a line. I need two bits of information.
- 5.65 Remember, lines look like Y is equal to MX plus B , where what? M is my slope, and B has something to do with the Y intercept. Has something to do with that. All right, so in order to figure out what the equation is then, then I need to say, all right, well, what is the slope? That's M . What is the slope? Well, what is this telling me to do? It's saying, how sloped is this? It's saying, go down a certain distance in the Y and then run over a certain distance on the X axis. That would be my slope. That's like how tilted it is.
- 5.66 You go down a certain amount and you go over a certain amount. Well, what is that distance from here down to here? Well, that's the difference of F of B and F of A. That's what that difference is. My slope is going to be F of B minus F of A, divided by what? That's my change in my Y variable. What's the change in my X variable? Well, that's just

B minus A. It's B minus A. What is that quantity? That's the average rate of change of F. That's the average rate of change of F.

- 5.67** This line, which is special, this line, which is special, it has a special name associated to it. We'll give it that special name. It's called a secant line. It's called the secant line. But that line has a certain slope associated to it, and that slope is equal to what? It's equal to this change in F, divided by change in B, or change in F divided by change in X, which is my average rate of change in my function. That's kind of interesting, that that pops up somehow geometrically. Maybe that's important. That's my slope.
- 5.68** Then what is the corresponding B here? What is the corresponding B here? I guess, maybe this B, being called B is maybe not the greatest because this notation using B here, let's call this guy D so that we're not conflating the two. Let's call it a different letter because we called B the X coordinate here. The Y-intercept is going to be someplace up here, which is going to be our B. Let's call this thing D. My slope is M, my Y intercept is 0, D. We found that the slope was equal to F of B minus F of A over B minus A.
- 5.69** If I wanted to, I could find the equation of the line. How do I do that? Well, I'm going to say, I now know that Y is equal to, I'm going to replace M with F average. I'm just going to say F average times X plus D. Then how would I solve for the variable D, whatever that is? Well, look, this line has to go through the points A, F of A, and B, F of B. In particular, if I plug in X is equal to A, I could say that F of A, that's the Y coordinate, is equal to the number F average times A plus D. Then just move this over so that I get D is equal to F of A minus F average times A. That's what D is.
- 5.70** And then I could substitute that back into this to actually get the equation. If I actually wanted the equation, I could write it down. But the important thing for us is saying, look, the slope of this line is the average value, the average change of my function F. That's the key. That's the key. The secant line, this has slope given by F average on the interval A, B. Here's what we're going to do. We're going to say, look, go back to the cycling example.
- 5.71** If A is equal to one hour and B is equal to two hours, then this F average is representing my average speed in between those two hours, hours one and two. How would I get it closer to saying, what is my speed now? What is my, if you have a little GPS headset on your bike and it's telling you your speed, your heart rate, all these different things, how far you've gone, if you're looking down at that particular thing and it says, oh, you're going 18 miles an hour, how do you actually approximate that? How do you get closer to that?
- 5.72** Well, the way you get closer to it is by doing the following. This is introducing the notion of a derivative, which we're not going to give it that name yet. We're going to get there. But we're going to do the following thing. This seems weird because it's like,

well, I've never actually ever seen that before. This is actually a very common thing in mathematics in general.

- 5.73** But look, if I'm interested in taking the hour-long time frame that I had when I'm taking my data and shortening that to 30 minutes or shortening that to 15 minutes or 10 minutes or 5 minutes or 1 minute or 30 seconds or 15 seconds, then what I'm really doing is, I'm saying, look, start with A and move A away from itself by a certain amount that you get to control. That you get to control. What do I mean by that?
- 5.74** Think about starting your stopwatch right now at time equals zero, and then moving off of zero slightly by something I get to control. I could either shorten that time frame to be closer to 10 seconds. I could increase the time frame to be something like 10 minutes, but I get to control that aspect of it. What does that mean? Move from A to a point that looks like A plus some value, H, where you have complete control over H. You get complete control over H.
- 5.75** H is what's called a parameter, and it means you get to tune it. You get to choose values for it. If you want to move A away from itself by, say, 1, you get to choose H is equal to 1. If you want to move A away from itself by one half, you get to choose that H is one half. If you want A to be away from itself by 50, you get to choose H is equal to 50. Does that make sense? You guys get to choose what H is.
- 5.76** But now, if I follow the same procedure that I did over here, A plus H, if I go down to the graph, this point has coordinates A plus H, F of A plus H. Do the same procedure. Do the same procedure. What is the equation of the line connecting this? I'm going to change this to saying what is the slope of the line? I'm going to change this to slope. We could ask for the equation, but we're really interested in rates of change.
- 5.77** We're going to say, what is the slope of the line connecting A F of A to this new point we're going to call A plus H, F of A plus H. What is that thing? Well, what's the slope? The slope is the change in the vertical direction, divided by the change in the horizontal direction. The change in distance, divided by change in time. The change in beats of your heart, divided by the change in time, your beats per minute.
- 5.78** What is my corresponding slope? Let's call this M again. It's going to be the change in my Y's, my output, so F of A plus H, minus F of A, divided by the change in my horizontal variable, so that's A plus H minus A. Thankfully, the denominator simplifies a little bit. That's nice. The A's cancel each other and I just get H. This is equal to F of A plus H minus F of A all over H. That's what the slope is.
- 5.79** Now, it seems like a very complicated expression, and it's scary, I mean, to be brutally honest, it is scary. However, I want to allay any fears that are lingering about the A plus H. What does F of A plus H mean? And I see this all the time. This freaks people out, and understandably so, I'm not going to lie. If I give you a function F and I say, what is F of A plus H? Suddenly it's just like, I don't know. I don't even know how to handle that question anymore.

- 5.80** So let's do an example. Let's do a quick example to show you that this notation actually isn't something to be that scared over. It's very natural for this to be very intimidating, by the way. How do I think about that? Let's think of F of X . Let's just look at something like X squared plus $3X$ plus 2 . Let's look at something like that.
- 5.81** If I asked you, what is F of 2 ? Well, what would you do? I'm going to say, OK, the input is now 2 . So that's really saying X is equal to 2 . I'm just going to go through and say, everywhere I see an X , I'm going to replace it with a 2 . So this is going to be what? 2 squared plus 3 times 2 plus 2 . And then I work out the solution to this. What is that going to be? 4 plus 6 is 10 plus 2 is 12 , and I'm happy.
- 5.82** Then we can keep playing this game over and over again. I say, OK, what is F of zero? That's going to be what? Zero squared plus 3 times zero plus 2 . That's going to be 2 because those are zeros. Then I'm going to say, OK, what is F of minus 3 ? You say, OK, I need to plug in negative 3 for all of my inputs. So this will be what? Minus 3 squared plus 3 times minus 3 . That was a convenient choice because negative 3 squared is 9 plus negative 9 . Those cancel, so that's nice. Plus 2 , so I get 2 again. And so forth and so on.
- 5.83** You can keep playing this game. I feed you an X value, you know how to give me the output, F of that X value. You just plug it in and then go from there. Do we agree? But then suddenly, I say something like, I want you to find what is F of, let's say, A plus 2 . And suddenly, your brain just shuts off. You're like, nope, I don't know how to do that. I don't know how to do that. I claim that you do. You just need the right way of visualizing it.
- 5.84** F of 2 said what? Take the input of 2 for X and plug it in for X . That's all it said. F of zero, take X of zero, replace the X 's with zero. F of negative 3 , replace the X 's with negative 3 . That's all you're doing. This is going to seem completely ridiculous. But if I asked you, what is F of rectangular box? You're like, what the heck are you talking about? A rectangular box? That doesn't make any sense. That's not a number. But just run with it.
- 5.85** If I said, what is F of rectangular box? You're going back to this example and saying, well, how did I compute F of 2 ? I just took 2 and replaced all the X 's with 2 's. Then symbolically, this would at least be plausible to say what? OK, I'm going to take a rectangular box and I'm going to plug it in for X . I'm going to get a rectangular box squared plus 3 times rectangular box plus 2 . Do we agree? That's symbolically what's happening.
- 5.86** Then I ask you the question, what is F of A plus 2 ? For some reason, our brains shut down. We're like, I don't know. It's like, yes, you do. Yes, you do. Because if you know how to plug in a rectangular box in for something, then you know how to plug in A plus 2 . It's the exact same procedure. It's the exact same procedure. What is F of A plus 2 ? I

go and I find all of my X values and I replace the X with A plus 2, and I put the A plus 2 in parentheses.

- 5.87** What is it going to look like? I'm going to have an A plus 2 squared plus 3 times A plus 2 plus 2. That's what it's going to look like. I know this example up here of saying stick a rectangular box in is ridiculous, but I think it illustrates the point. All you're doing is taking something that goes in this spot and you're replacing all the things for which that's taking the place. That's the quantity defining it. That's X.
- 5.88** Let's go back up to this example here. The slope is given by this F of A plus H minus F of A all over H. What is that giving me? That's giving me the slope of that line. That slope is equal to this quantity that we have over here. It's got that quantity over here. So here's what I want you to think about now. I don't want you to think about things as being static. This is one of the profound ways of thinking about calculus. I don't want you to think about things as being static.
- 5.89** I take one measurement at T equals 1 hour in my bike ride, one measurement at T equals 2 hours in my bike ride, and I only use those two data points to compute the corresponding average speed that I was interested in. What I want you to think of is, what if I allow H to go to zero, to get small? What is that going to do geometrically? Well, think about it.
- 5.90** If H were 1, this says that I am one unit away from A. A and A plus 1, they are one unit away from each other. If H is now one half, then it's saying A plus one half is like here. I'm half a unit away. If H is one-fourth, I'm now one-fourth away from A. If H is one-eighth, or 1 over 10, or 1 over 100, or 1 over a million, what's happening is, as H gets small, this point starts migrating back to H or to A. It starts migrating back to A as H gets small. Do we agree?
- 5.91** You imagine this as being this point here, and as H goes to zero, what's happening, it's sliding its way back towards A. That's what's happening. What's happening to the slopes? What's happening to the slopes? Well, the slope in between A and A plus H is this thing. If I move this point to here, I'm now looking at this slope. Yes? If I move this point further to here, I'm now looking at this slope. Yes? Keep moving it back. What's going to happen? If I move it back up to here, I'm looking at this slope.
- 5.92** What's going to happen when this point gets all the way to A? As this point moves to here, what's the slope I'm going to get? I'm going to get the slope of that line. After all, these are all lines, yes? I'm computing slopes of these lines. That's really what I'm doing. We're finding slopes of those lines. As H gets closer and closer and closer and closer to A, I'm still computing the same quantity. I'm still computing this, just for different values of H.
- 5.93** Those slopes are giving me the slope of this line, this line, this line, this line, until eventually it hits this limiting factor when H is as close to zero as you can imagine. Suddenly, I get this line that has this slope. Do we agree? What is the slope of this line?

This line is special. This line is called the tangent line to Y equals F of X at X is equal to A . It's called the tangent line there. Tangent, just meaning that it just kisses off the graph in this one point.

- 5.94** It's the tangent line there, and its slope is what you're reading on your speedometer. Its slope is your instantaneous rate of change. The slope of that line is what's being read when you're driving on the freeway and suddenly you're going 65 miles an hour, and you look down and it says you're going 65 miles an hour. That is the slope of this tangent line. It is your instantaneous rate of change. The slope of this line, slope of tangent line, so TGT for tangent line, this is equal to my instantaneous rate of change. The slope of my tangent line is equal to my instantaneous rate of change.
- 5.95** So if I'm on my bike and I look down at my headset and I see, oh, you're going 23 miles an hour, that's my instantaneous speed. At that moment in time, 23 miles per hour. What is that? Associated to the curve, which is giving me my distance per time, it's telling me the slope of the tangent line at that point is 23. That's what it's telling me. But what are we missing? If you think about it, what are we actually missing here?
- 5.96** We just waved our hands to some extent and said, well, if I just allow H to get small and get close to zero, then this is going to approach this and that's going to give me a slope and that's going to be what, we interpret that as being this type of quantity. That's not mathematically rigorous. I'm literally saying, look, let this thing move to this. Where in this motion is there math? There's not. It's just an intuition. I need to make sense of the following quantity.
- 5.97** The question is, what does it mean, mathematically, to let H go to zero? The answer, the answer is limits. Limits are how we make sense of certain quantities going to other quantities. If you go back to the sequences and series that we talked about earlier, we said things like, where is my series going? Why does it give me the number of 2 as I add up infinitely many things? I'm saying, as I go arbitrarily large, as N goes to infinity, I was implicitly saying I'm computing a limit.
- 5.98** What we're going to explore next is, how do I make sense of limits? What is their notation, what is the algebra associated to them, and what is their utility in trying to understand this picture further?

Lecture 6: Understanding Limits

At a glance — Understanding limits.

- 6.1** Now that we've thoroughly covered average rates of change of functions, we want to try and make sense of this instantaneous rate of change, which we motivated last time in terms of these slopes of tangent lines and things like that. In order to do that, mathematically, we need to understand the notion of a limit. What is a limit?
- 6.2** There's a very rigorous mathematical way of thinking about limits, but there's also a very nice intuitive way about thinking about limits. We're going to focus on the intuitive notion of what is a limit. Let's think about that. If I say, given a function, F , let's say it goes from the real numbers into the real numbers—I'm going to use that notation that we've been using before—given a function F and some value A in the domain, any domain, the limit of F is denoted the following.
- 6.3** The notation is this L - I - M and then subscript X goes to A of F of X . What I'm going to do is, I'm just going to give that thing a label. I'm going to call it L . I don't necessarily know if L exists. If it does, I'm going to call it L . If it doesn't exist, I'm just going to forget about this. I want to try and understand what does the limit actually mean. What does it represent?
- 6.4** Now, the idea is the intuitive one. Here's the idea, if you will. As we say that the limit of F as X approaches A , and that's how you read this, you say the limit of F as X approaches A , or the limit as X approaches A of F , those two things are the same. We say that that limit, the limit of F as X approaches A , means that the values of F get arbitrarily close to L as the values of X get arbitrarily close to A .
- 6.5** Morally speaking, we're really saying the following, when X is effectively indistinguishable from A , it's arbitrarily close to A , what is F of X getting arbitrarily close to? We say that it gets arbitrarily close to L . We can see silly examples of this. Silly examples of this. Let's say, here's one. If I give you the function F of X is equal to X squared, and I just draw the graph of Y equals X squared, so I have this parabola. Here's Y , here's X .
- 6.6** I approach the point, let's say, I don't know, 1. I'm saying, as X gets arbitrarily close to 1, what is F getting close to? Well, I go, OK, if I get close to 1, what is F getting close to? I give myself this little window, if you will, in my X direction. Say, if my X values are in that little window, where are my F values? My F values look like they live somewhere in here. It's a wide range.
- 6.7** But what I notice is that as I get closer and closer and closer and closer to X , what's happening is that these horizontal windows are going to get smaller and smaller and smaller and smaller, until what? I notice that the value of my function is getting very

close to this particular value, which is also equal to 1. That's the intuitive notion of a limit. You want to think about, as X gets close to A , what is F getting close to?

6.8 Now, I gave the motivation for this by saying that A is in the domain. A is in the domain, but it begs the question, does A really need to be in the domain? Does it need to be in the domain or not? That's a natural question. Does A need to be in the domain? Let's see if we can motivate that. Let's see if we can motivate an answer for or against being in the domain.

6.9 Well, again, all we really have is this graphical interpretation of this. But what if I gave you a graph that did something like this? What if I gave you a graph that does something like this? Here's X , here's Y . It's just going to do its thing for a little while. It's just going to do this. Then at this point, suddenly, I have an open circle, and here's going to be A . What do I mean by this open circle?

6.10 I mean that my function is not defined at X is equal to A . My function is defined, suddenly at X equals A , not defined, and then it goes from there. What would be an example? You might say, well, I've never seen a function that does that before. The function is not defined at a certain point. It's like, OK, well, you can think about this as saying, look, maybe this function is representing a particular quantity. Maybe it's representing, I don't know, let's say, power to a particular building.

6.11 The power is fluctuating at a certain point, and then suddenly there's a power outage, but it's only incredibly brief. I don't have a measurement for what the power is at that instant in time. You can think about it as like, here's the power fluctuating here, and then at X equals A , there's a power outage, but it was instantaneous. I don't have a reading for what the power was, so I have this little gap, and then it just keeps going on from there. You can think about things like that.

6.12 The question would be, well, look, if I go back to this intuitive notion of the idea behind limits, it's saying that the values of the function F get arbitrarily close to L as the values of X get arbitrarily close to A . I can imagine like, OK, if you imagine yourself as walking on the X axis, right? You're just walking this direction towards A , or you're walking this direction towards A . You're walking on those sides and you say, OK, well, if I start approaching A from this side, what is my function getting close to?

6.13 My function seems to be getting close to whatever this value is. Let's call this value L . As I walk to A from the left-hand side, it looks like my function is approaching L . But if I approach A from the right-hand side, it looks like my function is approaching, it looks like it's also approaching L . But then the question is, look, X equals A is not in the domain of my function. My function is not defined there, but the limit still makes sense. The limit still makes sense.

6.14 We can still say that L is equal to the limit of F of X , as X goes to A . This still makes sense. Why does that still make sense? Well, because the limit doesn't care about what happens at A . It just cares about what happens to the function close to A . Even if I'm

not defined at A , I can say, well, maybe if A were equal to 2, for example, I don't have a value at 2, but I have a value at 2.00001. That's close to 2. I have a value at 1.999999. That's close to 2.

- 6.15** Limits only care about what happens when you're close to A , not when you're equal to A . They don't care about that. We'll get into what it means to care about the value at A . That's a notion of what's called continuity, and we'll talk about that in a little bit. But right now, we only care about what's happening close to A . Let me give you an example. Let me give you a concrete example on how to compute limits. We'll start from the basics, and then we'll build from there. Let's just do the first example.
- 6.16** Let's say the limit as X goes to minus 2 of, let's say, X squared plus $3X$ minus 1. Let's say, evaluate this. I want to evaluate this limit. The first thing you always do in every single limit, your first step always is literally just to take this value of X and plug it in and see if you get something that makes sense. See if you get something that makes sense. Step number one, this is always the case, OK, so step one, check if plugging in X is equal to minus 2 makes sense.
- 6.17** Now, you might be thinking, well, you just told me that the limit doesn't care about what the value is when you are equal to A . I'm like, OK, fair. I did say that. But look, if my function were defined at A , if my function were defined here, so this were filled in, then the value of my function at A is equal to L , and that's also equal to my limit. There is some value in simply be saying, OK, plug in X is equal to A and see what you get. See what you get. OK, maybe I get something, maybe I don't.
- 6.18** All right, so step one, I plug in. What do I end up getting? This thing is equal to, I plug in negative 2, so minus 2 squared plus 3 times minus 2 minus 1. That's what? Negative 2 squared is 4, 3 times negative 2 is minus 6 minus 1. 4 minus 6 is negative 2, minus 1 is negative 3. Negative 3 makes sense. OK, negative 3 makes sense. What do I mean by something that doesn't make sense? I mean something that looks like zero divided by zero, or something that might look like an infinite quantity minus another infinite quantity, or things like that.
- 6.19** Things that make sense are things that when I plug in the value, I actually get a number back that I'm used to dealing with. In this case, I plug in negative 2, I get negative 3, and it's like, OK, I know what that is. That makes sense. That's concrete. In particular, this limit is equal to, I plugged in negative 2, I got negative 3. The limit is equal to minus 3.
- 6.20** Let me give you an example of something that doesn't make sense. Let me give you an example of something that doesn't make sense. Just direct substitution, that is. I would argue that these are probably the limits that you're going to encounter the most, are the limits that don't present to you as actually making any sense. You're going to get a zero divided by a zero and things like that. I would argue that the majority of limits you run into are going to be that way.

- 6.21** Here's another one. Evaluate, we're going to say, the limit as X approaches zero of the absolute value of X divided by X . Remember, this absolute value function is this function that you plug anything into it and it gives me the positive version of that. You give me 5, it gives me 5. You give me negative 5, it gives me 5. It always outputs the positive version of whatever I give it.
- 6.22** How would I approach this? How do I approach this? If I plug in directly, what happens? Direct substitution gives 0 over 0. The absolute value of 0 is 0 divided by 0, and that doesn't make any sense. 0 divided by 0. This thing has a name, and you'll encounter it in a calculus course, this thing is called an indeterminate form. The proof is in the name. It's indeterminate. It is not determinable until you do something else.
- 6.23** What I want to try and understand is like, OK, maybe the limit doesn't exist. Maybe the limit is like, I can just say it doesn't exist in this case. Maybe it does. How would I actually figure this out? You have to go back to your intuitive notion of a limit. X is equal to 0. Let's say A is equal to 0, to stick with our notation from before. A is equal to 0. That's the point I'm approaching. This is A .
- 6.24** I want to understand, when I'm close to A is equal to 0, what is my corresponding function doing? That's my function, F of X , to use the notation that we've had previously. That's my function, F of X . OK, well, think about it. Let's just make a table. Let's just make a table of some values. If I have a table here, if I have F of X is equal to the absolute value of X over X , and I just start choosing values of X over here that are close to 0, then maybe I can get some intuition as to what the limit is actually giving me.
- 6.25** Let's say close to 0 would be what? Like 0.1, maybe, and then like 0.01 and then 0.001. Maybe I have those three. Those are going to be on this side of zero, 0.1, 0.01, 0.001. But then I could be on this side of it, so I could also be negative. Let's plug in minus 0.1, minus 0.01, minus 0.001. Let's just see what's happening. Let's just see what's happening.
- 6.26** If I plug in 0.1, so one-tenth, if I plug in one-tenth into this, what is the absolute value of 0.1? Well, 0.1 is positive, so the absolute value is just 0.1. I'm going to divide by 0.1. I'm going to get 0.1 divided by 0.1, which is 1 in this case. All right? This is equal to a 1. Great. Same reasoning for 0.01. It's positive, so its absolute value is unphased. So 0.01 divided by 0.01, I get a 1. Same thing here, I get a 1. Same exact reasoning.
- 6.27** I'm plugging in positive numbers into an absolute value. The absolute value has to give me the positive analog of that number. Therefore, the absolute value of that thing gives me the same number. But now here's the tricky bit. This is where absolute values need to be treated very carefully. Remember, if I gave you what is the absolute value of minus 2? You told me the answer is 2. Whatever you feed into it, you have to give me the positive version of that.

- 6.28** How do you make that rigorous? My input here is negative 2. What am I really presenting to you? I'm giving you minus minus 2. Do we agree? That's giving me my positive 2. If you give me the input of minus 5, then I'm giving you minus minus 5, which is positive 5. That's really what I'm doing. The mantra is, look, if you give me a negative input for my absolute value, I'm going to give you negative 1 times what you gave me.
- 6.29** You hand me a negative 9, I'm going to multiply your negative 9 by negative 1 to get a positive 9. You hand me a 5, I'm not going to do anything to it because 5 is already positive. But if you hand me a negative number, I have to multiply it by a negative. When I come over here and I say, OK, what is the absolute value of minus 0.1 divided by minus 0.1, what do I end up getting? Well, the absolute value is what? The absolute value of a negative number is minus that negative number. Let's see, this is 0.1 all over minus 0.1.
- 6.30** The minus minus becomes positive. So this becomes 0.1 over minus 0.1. And what is minus 0.1 over negative 0.1? It's negative 1. So this is minus 1. But there's nothing special with this thing being minus 0.1. The same is going to hold if I plug in minus 0.01 or 0.001 and so forth. What am I going to get here? I'm going to get minus 1, minus 1, minus 1. It's like, what in the world is happening here?
- 6.31** If I give myself my axes now, and say this is X, this is Y, and now I'm approaching A is equal to 0, my function here is not defined at 0 because if I plug in 0, I get 0 over 0. That's an indeterminate form. It's not defined there. But every time I'm to the right of it, the value of my functions are 1. Every time I'm to the left of it, the value of my functions are negative 1. What am I doing? I'm not defined at 0, and I'm 1 from there on out. I'm not defined at 0, but I'm negative 1 from there on out.
- 6.32** That's what the graph of this function looks like. This is what's called a jump discontinuity. My function is doing something here, and then suddenly it jumps to something else and then takes off. It's a jump discontinuity. We'll talk about continuity soon. But it begs the question, OK, go back to your intuitive notion of a limit. A limit is what? It's some quantity that F is getting close to when X is getting close to whatever this is.
- 6.33** If I get arbitrarily close to X is 0, F should be getting arbitrarily close to what? If I imagine myself standing in a little neighborhood around 0, I've got 0's here on the ground. I am standing at 0. I am the function, and I am standing at 0. I'm not defined here. I step off of 0 slightly in the negative direction, what's my value? I'm negative 1. Suddenly, I'm negative 1. If I'm standing at 0 and I move in the positive direction ever so slightly, suddenly, I'm positive 1. Suddenly, if I move off to the right, I'm suddenly positive 1.
- 6.34** It begs the question, am I close to anything? Go back to your intuitive notion of a limit. We say that the limit exists if the function gets arbitrarily close to one fixed thing when

its inputs get close to one fixed thing. The fixed thing is A is equal to 0. I am standing on A equals 0. If I move even the slightest amount to my right negatively, suddenly I'm negative 1. I move positively, suddenly I'm positive 1. It's like, what in the heck am I close to?

- 6.35** The answer is, you're not close to anything. On the one hand, you're 1. On the other hand, you're negative 1. It's like you're not a fixed thing. So the limit doesn't exist. This limit does not exist. DNE. Does not exist. It doesn't exist. But what it does, what this example does is, it says, OK, there's something special going on here in terms of the direction at which you approach A . There's something special here.
- 6.36** Because if I just imagine myself as just putting blinders on, so to speak, if I put on blinders and I say, look, I'm not even allowed to know that this side exists over here. I'm only allowed to approach A from this side. If I'm only allowed to approach A from this side, then I would say, oh, well, the limit's negative 1. Because I don't even know that this world over here exists. I just pretend this world doesn't exist, and I say, if I approach A equals 0 from the left-hand side, I'd say, well, yeah, the limit's negative 1. And vice versa.
- 6.37** If I imagine this side of the world doesn't exist, and I only approach A from the right-hand side, I'd say the limit is 1 because that's all I know. That gets into the notion of one-sided limits from the left, from the right, and it ties in beautifully to allow you to say, when do limits exist in general? We're going to talk about one-sided limits here for a second. One-sided limits. One-sided limits. What do I mean by one-sided limits?
- 6.38** We say, the limit of F of X as X approaches A . Be very careful with this notation. Be very careful with this notation, because in order for me to designate what direction I'm approaching A , I have to do the following. I say the limit as X approaches A , and then I give it an upper superscript of a negative sign. I give it a negative sign.
- 6.39** This is not A to the negative 1, and therefore, 1 over A . This is not some inverse. This is nothing other than this negative is saying you're approaching from the left. That's it. You're approaching from the left. We say the limit of F of X as X approaches A from the left equals, let's call it, M , if when X gets close to A from the left.
- 6.40** What do I mean from the left? I mean X is less than A . When X gets close to A from the left, the values of F get close to M . That's what I mean. Graphically speaking, what am I saying? Come back over here. If I were to say, what is the limit of the absolute value of X over X as X approaches 0 from the left-hand side, what I'm saying is that, if I imagine myself walking along the X axis towards this way, I'm approaching 0, but I'm only approaching from the left-hand side, so X is less than 0, X is negative. What is my function approaching? My function is approaching negative 1. So this limit does exist, and it equals negative 1.
- 6.41** Then that begs the question, the limit from the right-hand side. We say this, if X gets close to A from the left, the values of F gets close to M . Number two, we say the limit of

f of x as x approaches A . So the notation here, you can imagine, if I put a negative symbol, a minus sign, to indicate that I'm approaching from the left, then I'm probably going to put a plus sign to indicate that I'm going to approach from the right. That's exactly what you do.

- 6.42** So, limit of f of x as x approaches A from the right equals, let's call it N . If when x gets close to A from the right, from the right, what do I mean from the right? I mean when x is bigger than A now, the values of f get close to N . The values of f get close to N . If I go back to this example and I say, all right, if I imagine myself as approaching 0 from the right-hand side now, that's the only direction I can approach 0. I say, if I approach 0 from the right-hand side, what are the values of my function f ? Well, they're getting close to 1.
- 6.43** I'm going to say, the limit as x approaches 0 from the right-hand side of the absolute value of x over x , that's going to equal 1. If I approach from one direction, I get 1. If I approach from the other direction, I get negative 1. What do those things have to tell me about the limit in general? What could we say? If we wanted to come up with an abstract mathematical statement, what could we say at this point?
- 6.44** We said, look, we originally considered the limit of this function as x approached 0. I didn't tell you a direction. I didn't tell you left, I didn't tell you right. I just said, as you approach 0. We said that limit doesn't exist. The argument that we made was, we made a table and we said, if I get close to 0 from the right-hand side, from the positive side, I'm 1, i.e., the limit as x approaches 0 from the right is 1, as I approach 0 from the left-hand side, I get negative 1. So, I get the limit as x goes to 0 from the left is negative 1.
- 6.45** I made this silly argument where I stood here and I said, look, if I move this way, I'm negative 1. If I move this way, I'm positive 1. So what can I possibly be close to? And the answer was nothing. But what is that saying mathematically? It's saying, this limit, as you approach A , in general, cannot possibly exist if the directional limits are different. If these limits are different. If the limit as you approach from the left is different from the limit from the right, this limit will never exist.
- 6.46** That makes sense if you think about it. If the limit as I approach from the left is 5, and the limit as I approach from the right is 7, then what is the limit as I approach that point? It's like, well, on the one hand, it's 5, on the other hand, it's 7. So it's like, what the heck is it? It can't make up its mind. It's like, am I 5 or am I 7? It's like, well, it doesn't exist. It doesn't make sense. You can't simultaneously be 5 if the wind blows you over here, and then 7 if the wind blows you over here, and then somehow be some coherent entity in the middle. That doesn't make any sense.
- 6.47** If I'm 5 on this side and I exist, then I should be 5 on this side so that when I approach in general, I'm 5. That's the intuition behind these things. Now we have a statement that we can make, that says that the limit doesn't exist if the directional limits are different.

By the way, how I'm approaching this is to try and give you the intuition for how mathematics is done.

- 6.48** I think there's a big misconception that mathematics is done in isolation in this little vacuum chamber of these brilliant mathematicians who just sit there and it just, I don't know, knowledge distills from the heavens or something like that. That's sometimes true, but fairly rare, if I'm being completely honest. Typically, mathematics is done by simply saying, I want to ask some questions. I'm going to goof around with the symbols legitimately.
- 6.49** You have to use legitimate mathematics to goof around with those symbols and then say, can I abstract out to an actual general fact? So what we just did with left- and right-hand limits is, we just proved the following fact or gave, we didn't prove it, but we gave strong evidence to suggest that this is true. The fact is that the limit of $f(x)$ as x goes to a is equal to L , exists if and only if the limit of $f(x)$ as x approaches a from the right is equal to the limit of $f(x)$ as x approaches a from the left is equal to L .
- 6.50** The limit will exist if and only if the left-hand limit and the right-hand limit are equal to each other and equal that corresponding value of L . That's what we just found out. Let's do another example. Limits are hard. Limits are incredibly difficult. Give yourself some grace when learning these things. It takes time. It takes some time to let all this stuff digest, if you will.
- 6.51** Here's another example. I want you to evaluate the limit as x goes to 2. The limit as x goes to 2 of $x^2 - 5x + 6$ over $x^2 - 4x + 4$. That's what I want you to evaluate. How do you approach a question like this? How do you approach a question like this? Well, you have to ask yourself the right questions at the end of the day. Part of making progress in anything, I think, is asking yourself the right questions.
- 6.52** When I see a question like this, the first question I ask myself is, what am I trying to compute? Oh, I'm trying to compute a limit. OK. Well, what's my first step in computing limits every single time? Well, maybe just plugging in x is equal to 2 actually gives me the answer. If I plug in x is equal to 2 and I get 3 over 7, OK, then the limit is 3 over 7. Great. Let's see what happens.
- 6.53** Direct substitution, what do I get? I get 2 squared minus 5 times 2 plus 6. OK, so what is that? 4 minus 10, that's negative 6 plus 6. OK, I get 0. I get 0. That's the numerator. The denominator, I get 2 squared minus 4 times 2 plus 4. What do I get? I get 4 minus 8. 4 minus 8 is negative 4 plus 4. OK, I get 0. Great. Fantastic. This presents to me in the form 0 divided by 0.
- 6.54** Remember, that's my indeterminate form. I gave you this warning that the majority of limits that you run into are going to be of the form 0 over 0. Then what do you do? You're like, OK, well, great. I don't want to do a table. I don't want to make a table and say, oh, x is 2.1 and then 2.01 and 2.001. If you have a calculator, that's all fine and

great, and you can guess what the limit is going to be. But that's not going to be super helpful.

- 6.55** You can then say, OK, well, maybe I can look at left- and right-hand limits. I can say, what happens as I approach from the left? What happens as I approach from the right? You run through that experiment a little bit and you're like, OK, but at the end of the day, if I approach from the right, I'm still plugging in things like 2.1, 2.01. I need a table. If I approach from the left, I'm plugging in things like 1.9, 1.99, 1.999. It's like, I still need a calculator and a table.
- 6.56** It's like, all right, that might be valuable. I don't know. But I don't necessarily want to go down that path right now. I'll just table it for now. If I can't find another way to solve this problem, then we'll go back to the table. One thing to notice is, these things are quadratics. These things are quadratics. And so maybe I can factor them. Maybe I could factor them and see what happens. Let's try factoring. Let's try factoring numerator and denominator.
- 6.57** After all, we spend time in algebra figuring out how to factor things. If I'm going to develop that skill set, I'd better be able to use it someplace. Let's see what I can get. I get the limit as X goes to 2 of $X^2 - 5X$, so I'm just rewriting right now, all over $X^2 - 4X + 4$. It's very important that you carry the limit through. I'm going to be very picky about this particular point because it's very easy to be lazy when doing the computation. I'm going to be very picky here.
- 6.58** First thing I'm going to do, so I rewrite the limit part, and now I'm going to factor top and bottom. Now, remember, how do we factor things like this? It's a quadratic, so I'm hoping to make it look like something that looks like this, all over something that looks like that. That's the hope. Now, in order to get an X^2 , what do I need? Well, I need an X and an X , so that when I distribute, I get X^2 . I need that to be true.
- 6.59** Now, because the coefficient of the X^2 is a 1, I then look at the negative 5 and the plus 6 to try and guess what the other terms are going to be. Now, think about it. Whatever goes here is going to be a number. Whatever goes here is going to be a number. When I take their product, this times this, that number has to equal positive 6.
- 6.60** You start racking your brain. You're like, what numbers multiply to give me 6? Well, a lot do. But the natural ones are like 6 and 1, or negative 6 and negative 1, or 2 and 3, or negative 2 and negative 3. I fuss around with those numbers, and I realized, OK, well, what's the property that those numbers have to satisfy?
- 6.61** Well, when I add them, I need to get negative 5. Clearly, 1 and 6 are out because 1 plus 6 is 7. Negative 1 and negative 6 are out because negative 1 plus negative 6 is negative 7. 2 and 3 are out because 2 plus 3 is 5, but negative 2 and negative 3 work. Negative 2 plus negative 3 is negative 5. Those are my factors. Minus 2, minus 3.

- 6.62 Then I can check, just as a sanity check, to make sure that I actually did it right. Similarly, down here, the coefficient is equal to a 1. So this has to be an X, this has to be an X. I'm looking for two numbers that multiply to give me positive 4 and add to give me negative 4, that's negative 2 and negative 2. I'm going to get a minus 2 and a minus 2.
- 6.63 Let me get rid of these arrows so that we can just view this expression by itself. That's the factored form. When I step back from this, I'm like, I'm really happy now. Because what I would like to do is just be like, well, look, I can cancel. How do fractions work? When I have a fraction that looks 5 over 6, I can write that as, multiply top and bottom by 10. I'm going to write as 2 times 5, or sorry, multiply by 2, 2 times 6. I can be like, oh, this is equal to that. Why? Because I can cancel that. Do we agree? I just cancel the 2's.
- 6.64 It's very tempting here to be like, oh, sweet. I have an X minus 2 here and X minus 2 here. Everything is multiplication, so I can just cancel the X minus 2's. That is exactly the right thing to do, but you are only allowed to do that because you are in a limit. You are only allowed to do that because you're in a limit. Let's finish this question, and then I'm going to belabor this point so that I can show you why I'm so picky on this. Because I'm in the limit, as X goes to 2, remember, limits don't care if you equal 2.
- 6.65 For example, this is like X is 2.1 or 2.01, or 1.9 or 1.99. If I plug those into this, X minus 2 isn't 0. X minus 2 isn't 0. If X is 2.1, 2.1 minus 2 is 0.1, not 0. If this is 2.1 minus 2, 2.1 minus 2, these are just numbers, no different than the 2 over 2 here that I can cancel. That's completely fine.
- 6.66 What I'm not allowed to do over here in the fraction world, though, is I'm not allowed to say, oh, OK, multiply top and bottom by 0. I get 0 times 5 all over 0 times 6, that's equal. That's garbage. That's nonsense. You're not allowed to multiply by 0 over 0 because you're not allowed to divide by 0, for goodness' sakes. That's not true. You have to be incredibly careful with what you're doing because look, X is close to 2. It's not equal to 2.
- 6.67 So I'm technically not multiplying by 0 over 0. I'm multiplying by something that's really close to 0, divided by something that's really close to 0, but not equal to 0. So I can cancel them. Because I can cancel them, I get the limit now, as X goes to 2, of what? X minus 3 all over X minus 2. And then I plug in. Maybe I can, maybe I can't. What happens? If I plug in the X is equal to 2 directly, I get what? Minus 1 over 0.
- 6.68 You're like, great. I'm not allowed to divide by 0. I had all this hope that this factorization was going to work nicely. You bait-and-switched me a little bit by the factorization worked nicely. I can cancel this, and suddenly I get a 0 in the denominator, and now I'm stuck again. It's like, well, yeah, that's how math works. You get stuck a lot. You need to figure your way out of it. All right. What can I do now? What can I do now? Well, let's see.

- 6.69** Remember, we had this notion that limits in general exist if the limit from the right and the limit from the left are equal. Now, I don't see how I could factor this any further. Maybe now I'm stuck. Now I have to consider the left-hand limit, the right-hand limit, and see what happens. We're going to take that off the table now and start computing things using a table. Again, this is the fact that I'm using right here.
- 6.70** The limit in equals L as X approaches A in general is equal to L if and only if the left- and right-hand limits are equal, too, to L . Let's consider one-sided limits. Let's look at the left-hand limit. I'm looking at the limit as X approaches 2 from the left, so 2 upper minus sign of X minus 3 over X minus 2. That's what I want to figure out. I hope and pray that that thing is some number that I can make sense of.
- 6.71** Then I hope and pray that when I look at the right-hand limit as X approaches 2 from the right-hand side, that that thing exists and that it's equal to the left-hand limit. That's what I'm hoping for. Maybe it's too hopeful. I don't know. Let's figure it out. Let's see what happens. As I approach 2 from the left-hand side, what's happening? Let's make a table. Let's make a table. I have X values. I have F of X values here. F of X is equal to my X minus 3 over X minus 2. And I'm approaching 2 from the left.
- 6.72** The way that I always remember this, this is probably a ridiculous way to remember it, in my opinion, but that's how my brain works. The way that I remember it is, I imagine myself walking on this one-dimensional line. I imagine myself as if I'm like some dot and I'm walking towards this direction. If I'm walking towards that direction and I look around myself, I say, what values am I?
- 6.73** It's like, well, when I'm over here, maybe I'm 0. Then once I get closer, I'm 1. Then I'm 1.5 and 1.6, and 1.7, 1.8, 1.9, 1.99, 1.9999, and so forth. That's how I think about limits. OK, I'm close. I'm 1.59's at this point. OK, great. Let's start plugging in values for that. OK, so X is equal to 1.9, 1.99, 1.999. Let's see what I get. Well, if I plug those into my function, what do I get? 1.9 minus 3. I don't know what that is. Frankly, I don't care what it is. Because 1.9 minus 3, is it positive or is it negative?
- 6.74** If I have 1.9 of something and I subtract away 3 of that something, I'm going to get negative. I'm going to get something less than 0. But what happens when I take 1.9 minus 2? I take 1.9 minus 2. What is this? This is negative, I'm just going to say negative upstairs, divided by what? This is going to be a minus 0.1. Is the whole fraction positive or negative? The numerator is negative, the denominator is negative. Is it positive or is it negative? It's going to be negative over negative, which is positive.
- 6.75** Negative 5 divided by negative 6 is positive five-sixths. Negative 2 divided by negative 3 is positive two-thirds. Negative divided by negative is a positive number. That's positive. What do I get when I plug in 1.99? I get 1.99 minus 3 over 1.99 minus 2. Again, I'm getting negative, divided by what? Minus 0.01. Negative over negative. It's still positive. What about 1.999? I get 1.999 minus 3. I'll have a 1.999 minus 2. Again, I'm getting negative over minus 0.001. OK. Still a positive number.

- 6.76** But the question is, what in the heck is actually going on here? What are these numbers? Let's think about this for a second. If I just gave you something to try and model this, if I gave you minus 2 divided by 0.1, what is that number? Taking negative 2 and I'm dividing by a tenth. What does that give me? Minus 2 divided by one-tenth. That's minus 2 times 10, which is minus 20. It got bigger. It got bigger.
- 6.77** If I try and model this and I said negative 2 divided by minus 0.1, that's now a minus here. This is a minus 10. This is a positive 20. Yes? What happens when I try and do something with minus 0.01? I get minus 2 over minus 0.01. That's minus 2 times minus 100, which is 200. It got bigger. The moral of the story is, if you divide by a smaller denominator, it makes the whole thing bigger. It makes the whole fraction bigger. If I did minus 2 over minus 0.001, this is minus 2 times minus 1,000, which is 2,000.
- 6.78** What do I notice? These values, look, the denominator keeps getting smaller and smaller and smaller. It's negative over negative, so the whole quantity is positive, and they're getting huge, like, plus infinity huge. It's going to positive infinity. What this limit is, is equal to plus infinity. It's blowing up to infinity. That's what it's doing.
- 6.79** That's what's happening as you approach 2 from the left-hand side. What's happening when you approach 2 from the right-hand side? What's happening there? Let's see. Once again, make our table. We're going to keep our table. Same function, just different X values. We're going to get rid of the left-hand limit now. We're going to look at the right-hand limit. So if I look at the right-hand limit, the right-hand limit says the following.
- 6.80** I have the limit of my function, $X - 3$ over $X - 2$ as X approaches 2. If it's a right-hand limit, what's the notation for it? I need an upper plus sign to say I'm approaching 2 from the right-hand side. Again, how does my brain reconcile all of this? It says, as X approaches 2 from the right, I imagine I'm a little dot over here and I'm just walking this direction. That's what I imagine. In particular, as I'm on my walk, I'm 3, I'm 2.5, 2.4, 2.3, 2.1, 2.01, 2.001, and so forth and so on. I'm getting closer and closer and closer to 2, but I'm bigger than 2.
- 6.81** What's my table now? I have 2.1, 2.01, 2.001, and so on. What happens when I plug it into my function? I get 2.1 minus 3 over 2.1 minus 2. That's negative over what? 0.1, negative over positive. If I give you a fraction which is negative in the numerator, positive in the denominator, is the overall thing positive or negative? If I gave you negative 5 divided by 6, that's negative five-sixths. If I give you negative 4 divided by 7, that's negative four-sevenths. So this whole quantity is negative.
- 6.82** What happens when I plug in 2.01? I get 2.01 minus 3 over 2.01 minus 2. That's negative again, divided by 0.01, and you can see the pattern. The denominator is always going to be positive. The numerator is always going to be negative, so the whole thing will be negative, but it's getting bigger. 2.001 minus 3 all over 2.001 minus 2. That's negative over 0.001.

- 6.83** So what's happening to these quantities? Well, look, if I gave you a negative 2 over a 0.1, we did this one earlier, minus 2 over one-tenth, this is negative 2 times 10, that's negative 20. If I give you negative 2 over 0.01, that's negative 2 times 100, that's negative 200. Negative 2 over 0.001, that's negative 2 times 1,000, that's negative 2,000. Where in the world are these things going? They're negative and they're going off to minus infinity. They're going off to minus infinity. So this thing is equal to minus infinity.
- 6.84** So what am I really saying? I'm saying, if I were to look at the graph of this function, X minus 3 over X minus 2, I would see this type of behavior. I would see something that, at X is equal to 2, if I approach 2 from the right-hand side, I see that I'm going down to minus infinity. I'm going to see some sort of behavior. As I'm close to 2, I'm going to see something like I'm going down like this. The values of my function are going down to minus infinity.
- 6.85** Whereas, if I approach 2 from the left-hand side, we said it was going to positive infinity. So I'm going to see some sort of behavior, something like that. So in particular, what did we just discover? We found that not only are limits valuable for what we're going to do with derivatives in calculus, we just found that limits are actually valuable for determining when we have a vertical asymptote to our graph.
- 6.86** When do I have a vertical line where my function is not allowed to pass that line and my function goes to infinity on one direction, minus infinity on the other, or both to infinity, both to minus infinity, things like that. Limits allow us to make sense of vertical asymptotes, which is pretty miraculous. That was a very long way of saying, does this limit exist? We said, look, the limit as I approach from the right-hand side is minus infinity.
- 6.87** The limit as I approach from the left-hand side is plus infinity. Those two things are about as diametrically opposite as possible. You have minus infinity and positive infinity. Those two things don't equal each other. Does this limit exist? The answer is no. So this limit does not exist. This limit does not exist.
- 6.88** One rationale for that is that the limit from the left does not equal the limit from the right. The fact that we're dealing with infinite limits as well, we also have to be a little bit careful as to, like, what do I mean by the limit existing? Go back to our intuitive notion of limits. Limits are, as X gets close to this value, the values of F get close to something else. What it mean to get close to infinity? That needs to be explored a little bit.
- 6.89** All of these things just come out of the woodwork when you start playing around with these little silly examples and trying to understand what exactly is happening. One indirect consequence of this example was, we now have a notion of limits telling us about behaviors of functions. In particular, vertical asymptotes. We want to

understand, OK, well, what does that mean? What am I actually capturing with these limits?

- 6.90** We made the observation. We made the following observation. The limit told me about the existence of a vertical asymptote. It told me I had a vertical asymptote because, as I got close to 2 from one on the sides, I went to plus infinity or minus infinity. So somehow, limits are telling me something about asymptotes, where the graph of my function doesn't exist.
- 6.91** But then we also had the example of the absolute value of X divided by X , where I had this behavior. So vertical asymptote had this behavior. It did something like this. That's like the vertical asymptote behavior. We also said that—that was one observation. The other observation was that the limit told me about jumps, jumps in a function. We had that example that was this example, the absolute value of X over X . We're like, well, that's odd. Somehow, limits have something to do about functions jumping at points as well.
- 6.92** What all of this is getting at, all of this is getting at the following thing, asymptotes, jumps, and holes. We had this one example that I motivated by saying, look, if I have a function that does something like this, this is roughly the graph that I wrote to justify that A doesn't actually need be in the domain of the function itself to actually make sense of limits, that limits can make sense for values at which those are not in the domain of your function.
- 6.93** Asymptotes, that's this case, jumps, that's this case, and holes, that's this case, are determined using limits. Those are determined using limits. A natural question would be to say, OK, question. Can we characterize this behavior for functions? Can we characterize what it means to have a hole like this picture over here, in a graph? Can we characterize what it means to have a jump, like we have over here? Can we characterize what it means to have an asymptote over here, simply by using limits? The is yes. All of that is covered, coming up in the notion of continuity of a function.

Lecture 7: Limits and Continuities

At a glance — Limits and continuity.

- 7.1** Now that we have limits, the sky is the limit, ultimately. Pardon the pun. However, derivatives and calculus and integrals and all of that stuff that you would cover in a calculus course all have their basis, if you will, in limits. We want to understand, what can we do with limits right now, and how do they get us back to the question of understanding instantaneous rates of change from average rates of change?
- 7.2** Remember, limits, we had the following idea. The limit of $f(x)$ as x approaches A , we said if that exists, then we give it a name and we call it L . We say that the limit as x approaches A of $f(x)$ is equal to L , means that as x gets close to A , the values of $f(x)$ are getting close to L . Then we did some examples.
- 7.3** Now, one of the important features that I want to be picky about is the following. This is more of a, let's call it a picky moment. Let's call it a picky moment. What I mean by this is, you have to be careful when you are simplifying fractions involving the variable x . For example, we had this example where we were computing the limit as x approaches 2 of $x^2 - 5x + 6$, divided by $x^2 - 4x + 4$.
- 7.4** This is an example that we worked out and we went through all the details and found out that the limit doesn't exist. The first step that we did was, we said, OK, well, let's just factor this into its individual pieces. I have x goes to 2. The top factor is $x - 2$ times $x - 3$. The bottom factor is $x - 2$ times $x - 2$. Then we said, well, I can just "cancel" the $x - 2$'s because I'm in a limit. As x approaches 2, I'm not equal to 2, so I'm like 2.1. If I'm 2.1, then I'm really not 0 over 0. I'm close to 0, divided by close to 0, and I can divide those numbers out.
- 7.5** We said that this was equal to the limit of, I can cancel these, of $x - 3$ over $x - 2$ as x approaches 2. The picky moment is this. Why can I cancel in a limit, but I can't just cancel function theoretically? What do I mean by that? If I were to give you this expression, $x - 2$ times $x - 3$ all over $x - 2$ times $x - 2$, and you wrote this, you give me an equal sign and you say, all right, that's equal to $x - 3$ over $x - 2$, that's equal to $x - 3$ over $x - 2$.
- 7.6** This is where I'm saying this is very picky. I'm being very picky because if you view these as functions on the left and on the right, this equality is wrong. This equality is false. Which seems weird because when we have fractions and we think of things like 3 times 5 over 3 times 2, you say, OK, cancel the 3, I get 5 over 2. That's totally fine. It seems like, well, that's, Robbie, that's what you're doing here. You're just canceling the $x - 2$'s, just like you're canceling the 3's. I would say, kind of.
- 7.7** I'm allowed to cancel the $x - 2$'s when x doesn't equal 2. I'm allowed to cancel the 3's always because 3 is never 0. But I have to be careful that if I cancel $x - 2$ from

top and bottom, I'm not somehow eliminating the possibility of X equaling 2. What am I really saying? Look, if I define a function which is F of X is $\frac{X - 2}{X - 3}$, all over $X - 2$ times $X - 2$, and I claim that this is equal to the function $X - 3$ over $X - 2$, which I give the name G of X , I give the function G of X , well, I have to be very careful here.

- 7.8** In this case, what do I notice? This is not the same, because of this issue of, if I'm getting rid of these terms, if I get rid of those terms, I'm losing information at 2. Now, you might say, well, that's OK because G isn't defined at 2 either. So F of 2 and G of 2 both don't exist, at X equals 2. But how could you change this example very simply? I could give you the function which is $X - 2$ times $X - 3$. Just change one of the denominators, $X - 2$ to $X + 2$. Just change this one to a plus.
- 7.9** If I said this now, let's give this a new function name, H of X . If I claimed that this was equal to $X - 3$ all over $X + 2$, and let's give that a name, let's call it K of X , why is this, why is this equality not true? Well, I can't plug in X equals 2 to the left because I get a 0 in the denominator. But now, magically, I can plug in 2 on the right. If functions are equal, they need to be equal at every single point. Any specific designation of X , they need to be able to equal each other.
- 7.10** It wouldn't make sense to say, H of X is equal to K of X , except for 20 different values of X . It's like, they're either equal or they're not. If I'm claiming that they're equal, K is allowed to be evaluated at 2, H is not allowed to be evaluated at 2, so what do I mean by they're equal? That's where I'm being very picky, saying, look, if I'm canceling terms, I have to be very careful with these things.
- 7.11** In a limit, it doesn't matter, because the limit doesn't care if you're equal to 2. It just cares that you're close to it. If I'm close to it, I'm not 0. Just be aware of that when you're computing limits. You're allowed to cancel terms. However, when you're doing it algebraically, you need to be very careful and very precise with what you're doing. That's a picky moment on that front. OK.
- 7.12** So, limits give us a way to understand continuity. We need to make sense of continuity. What do we mean by that? Well, intuitively, you would say that a function is continuous if its graph doesn't have any of these asymptotes or these jumps or these holes in it or any breaks or gaps or anything like that. We would say that a function is continuous at that point. Our definition is that we say F is continuous at, let's say, X is equal to A , if the following condition holds.
- 7.13** If, number one, the limit of F of X as X approaches A in general exists, if that limit exists, and I also need, number two, the value of my function at A is equal to that limit, as X approaches A . That needs to be true. In particular, if the limit exists, then so does the value of my function, because if F of A is equal to the limit and the limit exists, then F of A exists. That's fine. But I need these two principles to hold. Let's go back to what makes it not hold. This is continuity at a point.

- 7.14** What types of discontinuities? What types of discontinuities can I have? What I mean by that is, what types of points give me places where I'm not continuous, I'm discontinuous? We saw this when we talked about limits a little bit. We saw that we can have things that are called removable. That was the one that kind of looked like this. You have some hole there. This thing here, this is called a removable discontinuity.
- 7.15** Why is it removable? Well, because if I know what this limit is, let's say that this is L , if I know what the limit is, then I could really just patch this hole to some extent by just filling in the value of my function and declaring the value of my function to be equal to L at that point. It's like imagining, this is like some sidewalk, if you will, and then you have this minor tiny hole in the sidewalk, and then the sidewalk continues. It's like, what could you do? Well, you could just patch the sidewalk a little bit. You can just patch it up. That's not a big deal.
- 7.16** That's the notion of a removable discontinuity. Now, what are the other types of discontinuities? We can have one, or let's say, again, here's A . Maybe the function does something like, it comes up and then it comes over here. Maybe it's defined at that point. It doesn't have to be defined at that point, but maybe it is. But then it somehow jumps and then jumps down to here and then does its own thing from there. That's a possibility. That's a possibility.
- 7.17** We saw this in the example of taking the limit as X goes to 0 of the absolute value of X divided by X . We saw that that had value 1 to the right of 0 and it had value negative 1 to the left of 0. We have that. Those are good examples. Where do these things occur in nature, if you will, in the world? You could think about like, flipping on a light switch as an example of this. You come into a room, let's say, you come into a room and say T equals 1, let's say whatever T equals 1 is. That's maybe like one hour after you've woken up or something, I don't know, whatever T equals 1 is designating.
- 7.18** You say, OK, I go into a room and I flick the light switch. Well, the current flowing in the light switch was 0, up until you flipped the switch. You flipped the switch and suddenly what happens? You have current being 0, 0, 0, 0. You flip the switch, and suddenly current is now flowing. That's a jump. That's a jump discontinuity. You're going from nothing to something, instantly. That's a jump discontinuity. This is a jump.
- 7.19** Alternatively, we can have the behavior where I'm not finitely defined near A ; my function is approaching infinity or minus infinity at A . I can have an asymptote. Here's A . I could have something like this. Something like that. My function is still discontinuous there. I can have some asymptote behavior. Asymptote.
- 7.20** Sometimes these ones are just called infinite. They're called infinite discontinuities for the obvious reason that, on one side of it, I'm going to plus or minus infinity. On the other side, I'm going to plus or minus infinity. That's sometimes another way that you'll hear people talk about infinite discontinuities. Functions are continuous, essentially, if

this property holds. My function is defined at A , and the limit of the function as X approaches A is equal to the value of the function at that point. That's what we say.

- 7.21** What kinds of functions are continuous? We want to say, OK, what types of functions can we say are continuous? Continuous, let's say, continuous functions, and let's try and classify what those are. Things that we know are continuous. Things like polynomials. What do polynomials look like? Things like quadratic equations, cubic equations, and all of that. So things like, for example, things that look like $2X$ squared plus $3X$ minus 1, $3X$ to the sixth minus $4X$ to the third, plus X squared plus 4, and so forth and so on. Positive integer powers of X , basically, and then combinations of those. So polynomials are continuous. That's great.
- 7.22** The second thing would be things like our trigonometric functions, so sine and cosine. All right. So sine of X , cosine of X , those are continuous functions. Tangent is not continuous because remember, tangent is equal to sine divided by cosine. So wherever cosine is equal to 0, you're going to get vertical asymptotes. You're going to get vertical asymptotes. How do you see that? Again, you could graph these things. If I look at tangent, this is what I get. This is my π over 2 minus π over 2.
- 7.23** Let's say this is F of X is equal to tangent. Let's just illustrate how asymptotes come to be from a limit standpoint. Well, what do I notice? If I take the limit of tangent of X as X approaches π over 2 from the left, I'm approaching π over 2, and I'm approaching it from the left, it's like I'm walking this way on my axis. Where is my function going? My function is going to positive infinity. I can say that this is equal to plus infinity.
- 7.24** Similarly, I could say, well, look, if I approach negative π over 2 from the right, so if I take the limit of tangent of X as X approaches minus π over 2, but from the right-hand side, that should be minus infinity. So limits will diagnose for you when you have vertical asymptotes either going to plus infinity or down to minus infinity, simply by saying, the limit as I approach A directionally is a plus or minus infinity. That's how you characterize asymptotes.
- 7.25** So, sines and cosines are continuous. What else do we have? What else do we have? Well, logarithms are continuous, and so are exponential functions. Number three, exponentials. So how do I get those? Exponential functions. Things like B to the X , so things like that. Those are continuous. And then their inverses, which are logarithms, logarithmic functions. So, things like log base B of X . Those are also continuous. And that's great.
- 7.26** Now we have these families of continuous functions. So how do I characterize these types of discontinuities? We talked about this one, the infinite discontinuity. How do I know if a point is discontinuous or not, in terms of limits? This I'm drawing graphs, so I can look at it and say, OK, this is discontinuous because it has a hole. This is discontinuous because it has a jump. This is discontinuous because it has a vertical asymptote.

- 7.27** But what if I don't know what the graph looks like? How can I write down limit properties which actually diagnose discontinuities through limits without having to have a graph? That's the idea. What do I know here? This thing here satisfies the limit of F of X as X approaches A exists. It exists, but it's not equal to F of A . It's not equal to F of A . Maybe my function isn't defined here, or maybe it is defined, but it's, I don't know, defined maybe down here. The limit does not equal the value. But the limit does exist. That's a key thing.
- 7.28** What about the jump discontinuities? Jump discontinuities, I get that the limit of F of X as X approaches A , in general, does not exist. Because if I approach A from the left-hand side, I'm going to get something up here, versus as I approach A from the right-hand side, I'm going to get something down here. So the limit does not exist. But not only does it not exist, I get that the left and right limits are finite. I get that those are finite.
- 7.29** As I approach A from the left, I get something that's up here. That's a finite number. As I approach from the right, I get something here. That's a finite number. Over here, what do I get? I get that the limit of F of X as X approaches A , either from the left or the right, this is a plus or minus infinity. Or, so in this case, I'm approaching from the left, if I approach from the right, X goes to A from the right, that's a plus or minus infinity.
- 7.30** As I approach from the left, in this example, I go to infinity. As I approach from the right, I go down to minus infinity, either one of those. If that's true, then you say that A is an infinite discontinuity for F . That's our limit definition, if you will, way of diagnosing when a particular function is discontinuous at a point and what the discontinuity is.
- 7.31** The question we want to ask now is, well, OK, big deal. You have continuity, you have this idea. It's basically there's no gaps in your function. Fine. I could see maybe how that's useful in some ways. We can think about examples as to, OK, why would that be useful in the real world. The reason why it's useful in the real world is because of a big theorem. A big theorem, it's called the intermediate value theorem. You would explore this thoroughly and very heavily in a calculus course or in optimization as well.
- 7.32** What the intermediate value theorem says is exactly what our intuition tells us. The intermediate value theorem says the following thing. If I were to draw a line down the middle of this room, I was just to paint a line down the middle of this room, and I said, I'm going to draw a dot on this side of the room, and I'm going to draw a dot on this side of the room, so the line is in between them. I draw a dot over here, and I draw a dot over here.
- 7.33** I say, I'm going to take a marker, and I'm going to start at this dot, and I'm literally just going to keep the marker on the ground, and I'm just going to go over to this dot. I have to cross the line that I put on the ground. No matter how I do that, I have to cross a line. If I imagine just taking a line all the way, just infinitely long, I've got a point on this side, a point on this side. I want to connect them with some line. It doesn't matter how

squiggly I make it, how crazy I make it or anything. At some point, that line has to cross the one that I painted on the ground.

- 7.34** That's what the intermediate value theorem says. That's what it says. Let's talk about it. Why is the intermediate value theorem relevant? What does it have to do with anything, with continuity? How do I use it? Can I use it? Or is it just some high-level, I don't know, idea, not necessarily something that's practical, but just some thing like that? Or is there something I can actually do with it that's practical? It turns out, it's actually incredibly important for a lot of reasons.
- 7.35** Let's talk about the intermediate value theorem. Intermediate value theorem. Again, this would be covered very thoroughly in a single variable calculus course or in optimization, because you can extract an algorithm from this. But we're going to go through it from an intuitive standpoint right now. What does the intermediate value theorem say? It says, if F is continuous on the closed interval A, B , if F is continuous on the closed interval A, B , notice continuity immediately comes out of the gate. We have to make sure that our function is continuous. If the function is not continuous, this result is going to fail.
- 7.36** I'll give you an example about that. If F is continuous on A, B , and I look at the value of F at A , I look at the value of F at A , and I multiply the value of F at B , if I take those two numbers, whatever they are, F of A and F of B , they're numbers, and I multiply them, and if their result is negative, if that is negative, then there is some value of C , some C in A, B . I guess technically I can make this parentheses, in A, B , such that F of C is equal to 0. That's true.
- 7.37** Now, one thing I want to caution you about regarding mathematics in general, and I fell into this trap when I started learning mathematics, which is why I'm sharing this with you, is that I would read the chapter for the lesson or whatever it was. I'd read the book, I'd see a statement like this, and I would just memorize it. I'd say, OK, F is continuous on closed interval A, B , then if F of A times F of B is negative, then there's some point at which F is equal to 0.
- 7.38** I would kind of just, like, basically, like, regurgitate the statement. I'd say, oh, I "know" the statement because I can literally, just verbatim, give it back to you. I quickly realized that this was very, this was very poor, and this was not a good way to study because I didn't understand what the theorem was saying. I understood the words and where the words came and how to regurgitate where the words come and all of that stuff, but I couldn't actually apply the result.
- 7.39** What I would prefer, and what I do now when I learn, is, I say, OK, do I have examples to illustrate what this is even saying? Do I have a picture in mind illustrating what's actually being said? Do I have something that I can literally tell you that if I were to draw a line down the middle of the room, put a point over here, put a point over here,

and connect the two, then I have to cross the line? Do I have some sort of intuition for it? Because if I don't, I need to develop that in order to make the theorem relevant.

- 7.40** What is the picture here? Let's suppose, here's A. I don't care where they are. A and B, arbitrary points. It doesn't matter where they are. I know this result here, and this is just a really clever way, it's a very clever way of saying the following thing. Look, $f(A)$ is a number, $f(B)$ is a number, and I'm saying that the product of those two numbers is negative. What does that mean? Think about that. That means that $f(A)$ is either positive or negative, but then $f(B)$ has to be the opposite of whatever A is. If $f(A)$ is positive, $f(B)$ is negative. But if $f(A)$ is negative, $f(B)$ has to be positive. They have to be opposite signs.
- 7.41** If $f(A)$ is 2, then $f(B)$ has to be negative 5, versus if $f(A)$ is 5, then $f(B)$ has to be negative 2. They have to be opposite signs of each other. That's what this is really saying. I don't know what $f(A)$ is. I'm going to assume it's negative. It doesn't matter if it's positive or negative, but let's suppose it's down here. Here's $f(A)$. That's what it is. It's negative, which in this case, if $f(A)$ is negative, then $f(B)$ has to be positive. $f(B)$, so that's some point up here. So this is $f(B)$.
- 7.42** I've used this part of the statement, $f(A) \times f(B)$ is negative. I haven't used continuity yet. Continuity says what? It says that if I want to draw the graph of f , I can't break it. I can't have any holes, I can't have any jumps, I can't have any infinite asymptotes. I can't have anything like that. Basically, what I'm saying is, I have a line on the floor in the room. I have a point on one side of the line. I have a point on the other side of the line. I have to connect the two, and I have to cross the line somewhere.
- 7.43** Maybe I do it a couple of times. But at the very least, I have to cross it at least once. At least once. Do we agree? Maybe the graph looks like this. Maybe the graph looks like that, so it only crosses once. Maybe the graph looks like that. There's all sorts of ways that you can connect these points. Just like if I had a point over here and I had a point over here and a line down the middle, there's infinitely many ways that I could paint from that side to this side, but it doesn't matter how I do it. I have to cross the line in the middle of the room. I have to cross the X axis. Do we agree? OK.
- 7.44** Let's erase the dotted lines. Now we just have one function. We've got one function here. Now, what do I notice? This one I drew as if it crossed three times. That's great. But it doesn't have to cross three times. It could cross once. It could cross somehow like this. It could do that. But no matter what, I'm guaranteed that there has to be some C value such that $f(C)$ is equal to 0. That's the conclusion. The value of my function as C has to be equal to 0. That's what the intermediate value theorem says.
- 7.45** Now, technically, this is a special case of the intermediate value theorem. But again, like I said, you would do a deep dive on this in a single variable calculus course. So $f(C)$ is equal to 0. Why do I care? Why do I care? Let me show you why. All right, here we

go. We'll start small. Solve for X . We'll start small. First example, algebra course. $2X$ minus 3 is equal to minus X plus 4. Algebra course. Isolate the variable X .

- 7.46** How do I do it? I can add X on both sides. $2X$ plus X is a $3X$. I now have this. Move the 3 to the other side. Add 3, plus 3. I get $3X$ is equal to 7. Divide by 3, X is equal to seven-thirds. In what, one, two, three lines? We got the answer. X is equal to seven-thirds. OK, great.
- 7.47** Trigonometry. Let's figure that out. Sine of X is equal to one half. Sine of X is equal to one half. I have to use my unit circle. I use the unit circle. I say the sine is the Y variable, so I'm looking at sine of Y is one half. That's something like right here, Y equals one half. I'm looking for a corresponding angle associated to that. I'm calling my angle X , but I'm looking for the angle here. X is equal to, well, I could say, this is arc sine of one half, which an example of that is going to be π over 6.
- 7.48** Now, I could start going around in circles 2π and all that stuff. I could go around again 2π . I could go around again backwards 2π . I could do a lot of different things. I could say, oh, maybe I want to find this value over here, which is 5π over 6, and that's fine. But at the end of the day, I don't really care. I found one them. The question didn't specify. It didn't say solve for all of them. It just said solve for X . OK, fine. I found one. X is equal to π over 6.
- 7.49** All right, so we're looking pretty good. We look like we have a trigonometric function. Let's do exponentials and logarithms. Exponentials and logarithms. What if I said something like, 5 to the $2x$ minus 3 is equal to 7? Something like that. Solve for X . We need some method for being able to eliminate the 5 to the blank. I'm going to take the logarithm base 5. That was one of the properties that we had with logarithms. I take log base 5 of 5 to the $2X$ minus 3. That'll be log base 5 of 7. I end up getting that.
- 7.50** Properties of logarithms say that I then get log base 5 and 5. So what do I end up getting? $2X$ minus 3 is equal to log base 5 of 7. I add 3, I divide by 2, I get X is equal to 3 plus log base 5 of 7 all over 2. I have no clue what that number is. Frankly, I couldn't care less. Desmos would tell me what that number is. That's fine. Desmos can store that information. I don't want to store that information. But at the end of the day, we can solve for X . OK, great.
- 7.51** It looks like we're kind of, like, we kind of developed a sense of like, OK, we're untouchable to some extent. You give me an equation, I can solve it. That's great. This is a false sense of confidence on that front. Because what if I gave you the following question? Here's number four. What if I gave you this guy, number four? I'm going to say, what if I combine trigonometry and algebra? What if I say sine of X is equal to 2 minus X ? What can I say about that?
- 7.52** One, do those have a solution? If there is a solution, better question is, how do I find it? We can start messing around with this and we can be like, oh, OK, well, there's an X that's trapped in here, so I'm going to take arcsine of both sides. OK, great. If I take

arcsine of both sides, I get X is equal to arcsine of $2 - X$. Then I say, well, that wasn't super helpful, because the variable I'm interested in is now trapped in there.

- 7.53** I say, OK, fine. Well, if X is trapped inside of arcsine, then I just take sine on both sides. I'm going to get, OK, let's take sine on both sides. That's going to give me $2 - X$. I realize, OK, I just went back to where I started. I'm out. I don't know what to do. Algebraically, I'm completely stuck. I have no clue how to approach this question anymore, algebraically. I've literally exhausted my options. Take an arcsine to isolate this X , that then traps this X . Then undo arcsine by taking sine and that frees this X , but then traps this one. It's this never-ending game.
- 7.54** It's like you're trying to, it's like this frustration of, you have a rug that you're trying to fit into a room, but the rug is bigger than the room. You go over to one corner the room and you fuss with the rug a little bit and you get it to fit in the corner. Then you turn around and you look in that corner and it's bent over because it's too big. Then you run over that corner and then you try and fix that one and you get it fixed. Then you turn around and then it's messed up over here. It's like, the rug's too big. No matter what you do, it's never going to fit.
- 7.55** The same thing is happening here. No matter what you do, you're not going to be able to algebraically manipulate this to solve for X . Then the question is, well, is this just a hopeless question or does there exist a solution? If there does exist a solution, how would I find it? And so on. The answer is, of course, there's a solution. Look at the graphs. Look at the graphs. What does the graph of sine of X look like? Graph of sine of X looks something like this. I'm going to do my best. Desmos would do a far better job than I could ever do on this.
- 7.56** That's what the graph of that looks like. That's Y equals sine X . What does the graph of $2 - X$ look like? Well, that's a line. It looks like $MX + B$, where M is equal to minus 1 and B is equal to 2. So, I can say, oh, OK. I'm going to go up to 2, 0, 2. I'm going to down one, over one, down one, over one, down one, over. I'm going to get something like this.
- 7.57** Asking the question, is there a solution to this particular equation, is equivalent to saying, do these two things intersect each other? The answer is yes, they do intersect each other. They intersect each other right there. If I ask the question, is there a solution? The answer is yes, there is a solution here. At the very least, I can assert the existence of a solution. That's good. It exists.
- 7.58** But what if I need it? What if I need to know what it is? I need to know X equals blank. I actually need that number for some reason. How would I do that? The answer is intermediate value theorem. That's how you do it. Let's see how you do this. Yes, solution exists. That's great. These existential statements are one thing, actually being able to quantify them is another. It's one thing to be able to say some particular object

exists. It's another to say, look, here's said object. It's right here. Those are two very different questions.

7.59 OK, so sine of X is equal to 2 minus X . How do I do this? To use the intermediate value theorem, you need what? A continuous function. I need a continuous function, number one. What else do I need? I need a closed interval. And closed interval A, B , with F of A times F of B negative. That's what I need to be true. That's the definition. That's the requirement, the hypothesis, of the intermediate value theorem.

7.60 If that's the case, how do I generate a function? What could I say my function F of X is equal to? Well, a natural thing to do would be to say, look, what if I just move these over? What if I just move those over? I add X and subtract 2 . I add X and I subtract 2 . Then what would I get? I'd get sine of X plus X minus 2 . Do we agree? I'm just going to add X and subtract 2 . But what is this supposed to equal now? We want F of X equal to 0 . Because if F of X is equal to 0 , what do I get?

7.61 OK, well, add 2 , subtract X , I'm getting that sine X equals 2 minus X . We've turned the problem of saying, when does this thing, sine of X , equal this thing, 2 minus X , into saying, when is the function F zero? i.e., when does it cross the X axis? That's setting us up for the intermediate value theorem. This is a function.

7.62 Now I have to assert, is F continuous? Well, yes, F is continuous. So CTS is just continuous, because why? Well, sine is continuous. That's a trig function that's continuous. And X minus 2 is a polynomial. It's continuous everywhere. If I take a continuous function plus a continuous function, they're always continuous. F is continuous because it's a sum of continuous functions. It's a sum of continuous functions. That's good. I have this part for the intermediate value theorem. I have a continuous function. That's good.

7.63 Now I want to understand, how do I find A and B ? This is the art form here, finding A and B . This is the tricky bit. For me, this is how I approach it. I stand back and I look at the function, sine of X plus X minus 2 , and I ask myself the question, are there nice values of X that I can plug in? 0 is always a nice number, as long as it's in the domain. As long as X equals zero is in the domain, plugging in zero is a great number. It's fantastic.

7.64 What do I get at F of 0 ? F of 0 . I get sine of 0 plus 0 minus 2 . The sine of 0 . I have to go back to the unit circle. Unit circle never goes away, by the way. So don't just discount it and think it's this thing that you have to do in a trig course once. It never goes away. The unit circle is annoying that way. You always think you're rid of it until it pops up when you least expect it, which is annoying.

7.65 Sine of 0 is 0 . That's the Y coordinate. That's 0 , 0 plus 0 , 0 minus 2 . This is minus 2 . That's negative. That's negative. OK, if that's negative, what did we just find? We just found that when X is 0 , the function value is negative 2 . It's down here. What do I need to find? I need to find another point either over here or over here, where my function is positive, so that I can force it to cross the line.

- 7.66** I have to step back and say, OK, let's look at the function F of X again and say, what values of X could I plug in here to make it positive, for example? Well, what's going to make it negative? What do I know? Well, sine of X is stuck in between what? Sine of X is as low as negative 1. That's the lowest it goes, and it goes as high as 1. It's as low as negative 1, as high as 1. The worst case, the lowest this function can get, to some extent, between the sine of X and the minus 2, is negative 3.
- 7.67** Because sine of X , the lowest it gets is negative 1. Then if I tack on a negative 2, I'm going to get a negative 3. If I start looking at values of X that might offset a negative 3, I could think of, well, three, four, five, six, seven, things like that. But the problem with that is that, if I take X is equal to 5, for example, sure, X equals 5, 5 minus 2 is 3, that's a positive 3, and sine of 5, I don't know what it is, but I know that it's at least bigger than negative 1, then I could make some arguments about whether F is positive or negative there.
- 7.68** But what I would recommend is say, look, you're going to have to plug these values into a trigonometric function. Plug in things that you know. Plug in things you know. If I take, for example, what is F at π ? Because I know how to evaluate trig functions at π . If I take F of π , what do I get? This is a sine of π plus π minus 2. Sine of π , OK, unit circle again. π is over here, so this is π . Sine is the Y coordinate, so it's 0. So this is just equal to π minus 2.
- 7.69** But π is 3.14159, so forth. In particular, it's bigger than 3. It's bigger than 2, and this is positive. It's positive. So if I look at my zero here, π here, the value is something positive. It's got to cross somewhere in there by the intermediate value theorem. It says that there exists some C in open zero π , with F of C equal to 0. But run that through. What is F of C ? F of C is sine of C plus C minus 2 equals 0, i.e., sine of C is equal to 2 minus C .
- 7.70** So yes, there is a solution. Do I know what it is? No. But now I have a method for finding it. Now I have a method for finding it. How would I find it? Look, take this interval from 0 to π and cut it in half. Look in the middle. Let's say this is π over 2. Maybe this is π over 2. Evaluate your function F at π over 2. If it's positive, you can eliminate one half of your interval. If it's negative, you can eliminate the other half of your interval and apply intermediate value theorem to successively make these smaller and smaller intervals to trap your solution, C .
- 7.71** You can create an algorithm off the intermediate value theorem called the bisection method to actually numerically approximate solutions to this particular thing. Intermediate value theorem is incredibly powerful on that front. Let's look at other examples. Let's look at other examples. Maybe this is a fluke. Maybe it is just a fluke that trigonometric functions happen to satisfy this. I don't know. Or maybe the intermediate value theorem is more general than this example. Maybe it applies to more things.

- 7.72** What are other things that we could say? We could say, for example, what about the logarithm base 3 of X is equal to, let's say, sine of X ? We're going to completely abandon all algebra, and we're just simply going to go directly to trig and exponentials and logarithms, and start saying, what can I say about things like that? Now, again, we have graphs of these things, so I can say, all right, well, log base 3 looks like this. Here's 1, 0. So this is Y equals log base 3 of X . OK, fantastic.
- 7.73** And then I have this sine of X function. OK, so sine of X is doing something like that. What do I notice? They intersect. They intersect. The question is, is there a solution to this? It's like, yes, the solution exists. I know it exists. But the question is, well, what is it? No clue. This thing has to be my C value that's going to give me that corresponding solution. Again, rehash the intermediate value theorem. What do you need? You need a function which is continuous.
- 7.74** I'm going to choose to move this to the other side. You could move this to the other side. It doesn't matter. But I'm going to get log base 3 of X minus the sine of X . That's what I have. Is this a continuous function? Logarithm is continuous, trigonometric function sine X is continuous, their difference is continuous. It's continuous. Great.
- 7.75** What do I know about values of this? Well, I know that when I take F of 1, I'm not allowed to plug in X is equal to 0, remember? Because X is equal to 0 is not in the domain of our logarithm. Let's plug in something that I do know, F of 1. This is log base 3 of 1 minus sine of 1. I'm in trouble again. Sine of 1, I don't know the value for it, so I would need some Desmos tool or something. I'll show you how to get around that. Log base 3 of 1, this is equal to minus the sine of 1. OK. Minus the sine of 1.
- 7.76** I don't know what that is. Maybe it's negative, maybe it's positive. I don't know. But I need to know its sign. Pardon the pun. I need to know its sign, S-I-G-N. I need to know, is it positive or negative? What do I know? Well, look, on the unit circle, I know that the values of the circle, the Y values increase from here to here. They're getting bigger.
- 7.77** If I say theta is equal to 0 versus theta is equal to π over 2, π over 2 is what? π is 3.14, divided by 2, so what is that? 3.14, that's like 1.7ish, something like that, 1.7ish. In particular, a value of, say, theta is equal to 1 lives somewhere in here. Do we agree? If this is 0 and this is 1.7ish, then 1 has to live somewhere in between the two. Do we agree?
- 7.78** When I plug in sine of 1, what do I notice? I know the sine value is the Y value. I know that this quantity is positive. Therefore, since I'm multiplying it by negative, this whole quantity has to be negative. You don't need Desmos. You just need to know, what is the sine of the number, in both senses of the word, S-I-G-N and S-I-N-E. The sign of 1, trigonometric sign of 1 is positive, and then I multiply that by a negative, it makes the whole thing negative. OK, great.
- 7.79** What can I say about something like, I don't know, what is F of, say, I don't know, 9? Let's just say 9. For the sake of argument, what do I get? Well, this will be log base 3 of

9 minus the sine of 1. Log base 3 of 9 is log base 3 of 3 squared. That's why I chose 9, is I wanted a power of 3, so that I could use the logarithm properties to eliminate it. Log base 3 of 3 squared versus—sorry, this is sine of 9 up here. Sine of 9. Log base 3 of 3 squared is 2 minus the sine of 9.

- 7.80** But here's the point. What do I know? I have no clue what sine of 9 is. But I know that sine is in between negative 1 and 1. In particular, the largest sine can be is 1. This thing is 2 minus the largest thing is 1. It's positive. It's never going to get—sine of 9 can't be large enough to make this negative. So in particular, this has to be a positive number. So what does that mean? Well, that means that if I look at my corresponding graph, if I look at my corresponding graph here, I said, what, that X is equal to 1, so there's A equals 1.
- 7.81** Out here at B is equal to 9, I showed that, on the interval from 1 to 9, the intermediate value theorem says there is some C in 1 to 9 with F of C equal to zero. So i.e., F of C is this thing. This is log base 3 of C , minus the sine of C is equal to zero. So therefore, the log base 3 of C is equal to the sine of C . We actually have a solution to the equation that we're interested in.
- 7.82** Now, again, having the logarithm equal to the trigonometric function, that's great. We want to be able to use the algorithm of the bisection method to actually narrow down, to figure out what is C , so I can start approximating it, which is covered in optimization. But at the very least, the intermediate value theorem has now opened up a whole slew of different types of equations that we can now say, yeah, solutions do exist, and here's why.
- 7.83** Whereas before, we only had algebraic or trigonometric or exponential logarithmic techniques for trying to solve for X . We're not solving for X , but we are asserting that solutions do exist. Continuity is a very important property, and it all comes from limits. That was an interesting thing to get from limits. It was continuity. We miraculously got this deep result of the intermediate value theorem.
- 7.84** The question we kind of want to ask is, well, what can I do now? Can limits do something else? This is why I said limits are really just a starting point for all of the amazing things that you can do from a calculus standpoint. Everything has its fundamentals and limits. Let's go back to our average versus instantaneous, instantaneous rate of change. Average versus instantaneous rate of change.
- 7.85** Remember, how did we motivate this? We said, look, if I give you a graph that looks something like this, something like this, and I give you a point, let's say, A here, and I go up to my graph, and I look at the point which is A , F of A , and I move this point over to an A plus an H , where I get full control over H —you get to determine how far away you want A plus H to be— and I look at the point there, which is A plus H , F of A plus H , then I can look at the secant line between the two. That's my secant line.

- 7.86 And that secant line, by definition of it being a line, it has a slope. That has a slope. So the slope of the secant line is equal to the average value on the interval from A up to A plus H . So it's the average value of F on the interval from A to A plus H . We can write it down. We can say what this is. This is the F of A plus H minus F of A all over A plus H minus A . So A 's cancel, and I end up getting just an H . All right. So that's my secant slope. So maybe I'll call this M sub secant.
- 7.87 And we said, as I move A plus H closer to A , I'm going to start getting points here. I get a line there with a slope, I get a line here with a slope and so on. Eventually, I'm going to get the line which is tangent to this. So that's going to be M sub tangent. What am I really doing? Limits allow me to say that M sub tangent is equal to the limit as H goes to 0 of F of A plus H minus F of A all over H .
- 7.88 And that particular quantity, we give a name. We call this the F prime at A , and say that the M is giving me the instantaneous rate of change of the function F at the point A , and that is the notion of what is called the derivative of the function of F at A . That's the topic we're going to talk about next time.

Lecture 8: Introducing Derivatives

At a glance — Introducing derivatives.

- 8.1** We're finally here. We now get to talk about derivatives. All of this build to get to derivatives, all of this notion of limits and secant lines and average rates of change, all so that we could just get to instantaneous rates of change and the derivative. Let me remind you of the definition.
- 8.2** For derivatives, we have the following thing. The notation for it is going to be an F with an upper prime symbol. So F , upper little tick mark, this little apostrophe thing. At A , so F' in and of itself is a function. We're going to evaluate it at the point A . How is this going to be defined? This is defined as this limit as h goes to 0 of $F(A + h)$ minus $F(A)$ all divided by h . That's our definition of the derivative.
- 8.3** Now, what we want to try and do today is ultimately use this definition to find instantaneous rates of change. We're no longer talking about averages, we're talking about instantaneous things. So instantaneous rates of change. Then we want to try and understand, OK, well, what can we do with this object? Just like with limits, we figured out how we can compute with them, and then why are they useful?
- 8.4** Well, they tell us about continuity. They tell us about things like that. That's my definition, if you will, of the derivative. Let's just do a silly example. Let's just get our hands dirty a little bit. What if I just give you a quadratic function? I just give you something that looks like this. $F(x) = x^2 + 3x - 1$. Find F' at, say, 2. That's what I want to do.
- 8.5** Now, one thing that you notice is that our approach for limits is mocking you to some extent in this definition. Why? Because remember, with limits, what we said was, the first thing you do is you take h and you just plug it in and see if you get anything that makes sense. OK, let's do that. h is equal to 0. Fantastic. $F(A + 0)$, that's $F(A)$, minus $F(A)$, that's the numerator is 0 and I'm dividing by 0, I'm getting 0 over 0.
- 8.6** It's like, oh, that's fantastic. This is going to be one of those types of limits where I have to do something, do some manipulation, do some things along those lines. The answer is, unfortunately, yes, it is. But I can still use this definition. I can say, all right, how do I approach this? Well, F' at A here is what? A here is what I'm plugging in. So A is 2. So F' at 2 is equal to the limit as h goes to 0 of $F(2 + h)$ minus $F(2)$ all over h . That's what I have.
- 8.7** I have to step back for a second. Again, if I plug in h is equal to 0, I get nothing. h is 0 gives $F(2) - F(2)$. Great. That's 0 divided by 0. OK. What I really need to do is say, do I have a formula for F ? And now I'm going to have to plug in. And this is where everything just breaks loose, algebraically. This is where the statement of my calculus

teacher, back in the day, when he said, "You don't learn algebra until you learn calculus," and this is why that's true. So now all the algebra just rains down.

- 8.8** So the limit is H goes to 0. OK, what do I get? F of $2 + H$. So, remember the little rectangle example that we gave earlier. I just plug rectangles in for the X 's, and then I put the $2 + H$'s in. So what is this going to be? $2 + H$ squared plus 3 times $2 + H$ minus 1. What did we just write? That's F of $2 + H$. But what do I need to subtract off of that? I need to subtract F of 2. OK, so what is F of 2? That's going to be 2 squared plus 3 times 2 minus 1, and I have to take this whole thing and divide it out by H . All right?
- 8.9** And you realize very quickly, this is not a fun process. This is not enjoyable. All right? Because the algebra is terrible. Let's work through it. You got to go through this. You got to go through hard things to grow. It's just a fact of life. You don't learn to walk just by simply standing up and running a marathon. You stand up, take a step, fall on your butt. You stand up, take two steps, fall on your butt. You do that enough times, eventually you put it together.
- 8.10** It's the same idea with this. We have this really arduous definition of limit as H goes to 0 of this quotient. OK, that's great. There's got to be a better way. And there is. You'll find that in a first variable calculus course. But for now, we get the limit definition. That's the only tool we have. We get to stand and fall on our butts, and stand and fall on our butts. That's what we get to do.
- 8.11** All right, so what do I have? I have $2 + H$ squared. OK, $2 + H$ squared. Great. If I bring the 3 in, I get a 3 times 2 is 6 plus $3H$, so I'm going to get plus 6 plus $3H$. That's this term 6 and $3H$, and then a minus 1, minus 1, so that's nice. Then what do I get over here? Well, this was what? This is a $4 + 3$ times 2 is 6 minus 1. This is what? 10 minus 1 is 9. I'm going to get, remember, this is a minus. I'm going to get minus 9 here, all divided by—well, the denominator is nice. That's H . OK, that's not as scary as the numerator is.
- 8.12** What can I do here? This is equal to, the limit is H goes to 0. OK, so how do I factor out? How do I work out $2 + H$, the whole thing squared? Remember, $2 + H$ squared is equal to $2 + H$ times $2 + H$. OK, so what is that? 2 times 2 is 4, 2 times H and 2 times H , that's a $4H$, and then I get an H squared. I get a $4 + 4H + H$ squared. That's what goes here. I get a $4 + 4H + H$ squared. Plus, OK, what do I have? 6 minus 1 is a 5, so plus 5 plus $3H$ minus 9.
- 8.13** Goodness gracious. You quickly realize why this gets very frustrating very quickly. All right, so what do I have? What's amazing here is that a miracle happens at this stage. And what do I mean by that? Look, the $4 + 5$ gives me a 9. And then on the back end of that, I'm going to subtract a 9. So I get the 4 plus the 5 minus the 9. Those all cancel. OK, so it's like, OK, well, that's good. I get some cancelation.

- 8.14** And then I see a $4H$ and a $3H$ together. I can combine those to give a $7H$. This is the limit as H goes to 0 of what? I have an H squared plus a $7H$ all over H , and that's the miracle. Because now what I notice is, yes, even though I plug in H is equal to 0, I get 0 divided by 0. But notice that the numerator has an H that I can factor out. And because I'm in the limit, I can cancel that with the H in the denominator, and that springs it free for me.
- 8.15** So what am I saying? This is going to give me the limit of H times H plus 7 over H as H goes to 0 because I'm in the limit. Remember, this was that picky moment algebraically that I gave you guys earlier. H over H cancel. And so I'm saying, this is equal to the limit of H plus 7 as H goes to 0. Again, I reassess. This is a limit. What's the first thing that I do? Plug in, see if you get something that makes sense. If I plug in H is 0, I get 0 plus 7, which is 7. That's my answer, 7.
- 8.16** What that says is that, for this particular function, its instantaneous rate of change at A is equal to 2 is 7. You should congratulate yourself on this. You just computed your first derivative. That's a big deal. F' of 2, that's equal to 7. That's my instantaneous rate of change. That's fantastic. The question we want to ask, though, is like, all right, well, that was painful, algebraically, and I'd rather not try and do that again.
- 8.17** But so, hopefully there's some rules that make my life easier. It turns out that there are a lot of them. There's a lot of rules that make your life way easier on this front. You'll develop those in a first single variable calculus class. But I want to say, OK, what's the utility here? What's the utility? What can I actually do with this? What can I do with this? That's the question we want to ask.
- 8.18** One answer, which is very important, is the following. Here's the utility, if you will, utility of derivatives. Why are they useful? Why are they useful? Well, let's go back to the graph situation and where they came from. Let's go back to graphs. So if I give you something that looks like this, here's X , here's Y , and I give you the graph that we've been playing with this whole time, I give you this graph, and I give you, say, this point. So, let's say this is A , down here.
- 8.19** Remember, F' represented the slope of the tangent line to the graph. Remember, the tangent line was this nice line that just bounces off of the graph at that one point. The slope of this line, the slope is equal to F' at A . That's what it is. It's a slope. But that's just the slope. What's its equation? Well, I know what this point is. This point is A , F of A . I can say that the equation of the tangent line, so TGT for tangent line, at the point A , F of A , is given by what?
- 8.20** Well, I'm going to use the point slope form of a line. So point slope form. Point slope form says this. It's going to be Y minus the Y coordinate, F of A , is equal to the slope, F' at A times X minus A . That's what I get. That's what I get. That is the equation of this particular line. Any point on this line is obtained as an X and a Y , which satisfied this particular equation. That's the equation of the tangent line.

- 8.21** You might seem like, OK, well, I don't really care about tangent lines, and I would argue, neither do I. But I want to kind of understand, can I do anything with them? Can I do anything with tangent lines that's of use, that's of use? Well, let's try and come up with an example for where derivatives can be used in terms of these tangent lines to actually do something that we're interested in. Do something that we're interested in. Here we go.
- 8.22** The question I'm going to ask you is, I want to know, what is an approximation to the square root of 5? That's my question. Now, how would I do this? Really, I'd plug it into Desmos. I just plug in square root of 5. It'd give me some decimal expansion. I'd be happy with that. That's completely fine. But can I somehow utilize this in a precise way, that ties a lot of the stuff together that we've done?
- 8.23** All right, well, let's see. Let's see what I can do. Here's how we're going to approach it. This is going to tie most of what we have in this course together. So, solution is the following. Let's consider the following function. So recall that F of X equals the square root of X is the inverse function to X squared on the domain 0 to infinity. Remember when we were talking about functions and we were talking about inverses of these functions. When is a function invertible? It's got the horizontal line test. You have to satisfy that.
- 8.24** If I restrict my domain of the function, X goes to X squared to 0 to infinity, then square root of X is my inverse function to that. We have that. Well, I know that X squared looks like this, Y equals X squared, and I know that its inverse is mirrored about the line Y equals X . So Y equals X . So the inverse has to look something like this. This is Y is equal to the square root of X . It's got to be a mirror image of that. It's got to look something like that.
- 8.25** We've used functions and their inverses and their graphs. Now we want to try and understand, what does this have to do with the square root of 5? Well, it's got the following to do with the square root of 5. If this is what the square root of X looks like now, it looks like this, let me just draw it on 1 so that I don't have the X squared running around with it. If I come out here to X is equal to 5, I know what the exact value of the square root of 5 is by simply just going up to the graph, do we agree, and plugging it into this function.
- 8.26** The problem is that I don't know what that is. I have to have Desmos tell me what that is. I can't actually evaluate it. What I can do, though, is I can say, look, is there a number that's close to 5 that I do know the square root of, like 4? The answer is sure. If I go over here to 4 and I go up to the graph there, I know what that point is. It's 4, 2, the square root of 4. I know what that point is on the nose. That point right there is 4, 2. All right?
- 8.27** Here's what I'm going to do. I'm going to say, look, forget about 5 for a minute. Forget about 5. Let's just find the tangent line to Y equals square root of X at X is equal to 4.

What is that going to look like? That tangent line is going to look something like—see if I can pull this off. That tangent line is going to look something like that, whatever it is. I know what its formula is. Its formula is this. What I notice is, look, if I take that tangent line and I evaluate it at 5, I get this point right here. Do we agree? I get this point right above it.

- 8.28** That point is freakishly close to the one directly below it, which is the square root of 5, yes? Those points are very close to each other. If I were to somehow zoom this in, what am I going to look at? I've got this point here. I've got this point, which is 5, the square root of 5. I've got this line sitting above it, like this. This point up here will be 5, and then I don't know yet. We haven't figured that out. But this point here is very close. It would be a good approximation to the square root of 5. It would be a very good approximation to the square root of 5.
- 8.29** OK, so what I'm hearing is that tangent lines can somehow approximate values of functions that we otherwise don't know. The answer is, yeah, they can. Let's figure it out. Let's figure it out. Now, what bits of information do I need? I need F of 4. That's fine. F is going to be—so I know that F of 4 is equal to the square root of 4, that's 2. I have that information. I know this piece. I know what A is. A is 4. I know that piece. A is 4. Good. I know this one.
- 8.30** I need that. I need to know what the derivative of the square root function is evaluated at 4. Well, how do I do that? Well, I've got a formula for it. I've got this limit as H goes to 0 of F of A plus H minus F of A divided by H . OK, so let's run with that. Let me rewrite this formula down here so we know what we're going to be computing. We have Y minus F of A is equal to F prime at A times X minus A , so that I can erase this side, and we can do a computation that you're probably not going to like, but that's OK. That's OK.
- 8.31** Before you learn the nice skills and all that stuff, right, you have to learn the definitions associated to these things. OK. So let's find F prime at 4, where remember, F of X is equal to the square root of X . We're going to go step by step. OK. So what is my definition? F prime at 4 is equal to the limit as H goes to 0, F of 4 plus H minus F of 4 all over H . That's my definition. H goes to 0, limit F of A plus H minus F of A over H . That's my definition.
- 8.32** OK, so this becomes the limit as H goes to 0. What is F of 4 plus H ? Well, F was this. F of X is the square root of X . So 4 plus H , I take the square root of 4 plus H . Square root of 4 plus H minus, what's F of 4? We know that F of 4 is the square root of 4, that's 2. So minus 2 all over H .
- 8.33** Now, the number one mistake that people make, the number one mistake that people make, is that they think that square roots play well with addition and subtraction. What do I mean by that? People typically think that square roots—this is the most common

mistake I see. If you give me two numbers, A and B, and you say take the square root, people say this is the square root of A plus the square root of B.

- 8.34** That's what I mean by saying the square root, quote, "plays well" with addition. What I mean is that, what we want with our intuition is like, yeah, just take the square root of each piece and that'll do it. That's what I mean by playing well, and this is dead false. That's completely false. Let me prove that to you, why it's false. If this is true for all values of A and B, then I should be allowed to choose values of A and B to verify whether it's correct or not. Do we agree?
- 8.35** I'm going to choose A equals to B is equal to 1 in this example down here. What would I get on the left-hand side? I'd get the square root of 1 plus 1, that's 2. On the right-hand side, I'd get the square root of 1 plus the square root of 1. Well, that's 1 plus 1, that's 2. Except for the fact that this number is roughly 1.414 and so on. That is not equal to 2. This rule is not true in general. I can't simply look at the square root of 4 plus H and say, "Oh, that's the square root of 4 plus the square root of H." No, I'm not allowed to do that, which is a little annoying, but that's OK.
- 8.36** What do I do at this stage? What do I do at this stage? Well, again, it's a limit. Plug in H is equal to 0. What do I get? If H is 0, square root of 4 is 2, 2 minus 2, 0, divided by 0, I'm stuck. There's a trick to handle these things. There's some "math-a-magic", if you will, that happens with this. It's one of those things. It's like when somebody sends you a picture or something like that, and they say, "Do you see it?" And you look at it, and it's a picture of a birthday party or something. And then suddenly in the very background, you see somebody making some crazy face.
- 8.37** It's one of these things where you don't see it until you see it. This is what I mean by, it's kind of a "math-a-magic" trick. You wouldn't see this until you see it, until somebody says, "This is what you need to do." Here's the trick. I get the square root of 4 plus H minus 2 over H. Do you agree with me that I can multiply by 1 here and it's not going to change anything? If I multiply something by 1, it doesn't change a thing. 5 times 1, still 5. 12 times 1, still 12. Square root of stuff times 1, still square root of stuff.
- 8.38** But the quote, unquote, "1" that you multiply by is special. Here's what it is. It's not pretty, but here's what it is. Take the same numerator, but make this a plus instead. Square root 4 plus H plus 2 and divide out by, well, if you want to make it a 1, you need to make this quantity the same thing. So divide out by the square root of 4 plus H plus 2.
- 8.39** Now, if you're like me, I'd never see that in a million years, that that's the "right" thing to do. I would never see that in a million years unless somebody said, "Look, Robbie, this is what you do." You don't see it until you see it, kind of thing. All right? Let's clear some of this space out, because now algebra takes over.
- 8.40** All right, so I get the limit as H goes to 0. OK, what do I get? Well, I've got to do the distribution now. This term times this term. Square root of something times the same

square root. The square roots disappear. So I get 4 plus H. OK? Square root of 4 plus H times positive 2, I get plus 2 times the square root of 4 plus H. But then what happens? I take negative 2 times the square root of the same thing. I'm going to get minus 2 times the square root of 4 plus H. Then I'm going to get negative 2 times positive 2, which is minus 4.

- 8.41** This whole thing divided by what? The product of these two things here. I'm just going to write that as H times the square root of 4 plus H plus 2. That's what I get. It looks like I've made life a lot worse. Like, a lot worse. Because I've gone from something that looks like this, the square root of 4 plus H minus 2 over H, to something that looks like this, which is horrible.
- 8.42** But what I notice is, I say, well, look, 2 times the square root of 4 plus H minus 2 times the square root of 4 plus H, that cancels that. Those are done. I have 4 plus H minus 4, 4 cancels 4. There's a ton of cancelation that happens. That's kind of nice. OK, that's kind of nice. The denominator is what it is. This is just a general fact in math. It's kind of like physics. For every action, there's an equal and opposite reaction.
- 8.43** Mathematics is no different. It's like, well, what happened? I did this thing, and then what happened was, the numerator became really nice. It simply just became the single term of H. That's what the numerator is now, but the denominator is a nightmare. The denominator is now a nightmare. It's like, yeah, I made the numerator look all nice and pretty, but then the denominator is like, oh, that's not great. It's the same thing mathematically, that there's this equal and opposite reaction. I do one thing and I get something to look nice, but at the cost of what? Something else looks bad.
- 8.44** But maybe that's exactly what I need to solve the problem. We'll see. Let's rewrite this thing up here. So this is equal to the limit as H goes to 0. Remember, the numerator is now what? That's just H. That's a win. H divided by, what's the denominator? That's a huge win. I have an H times some stuff, square root of 4 plus H plus 2. Now, again, if I plug in H is 0, I get 0 over 0. But remember, because I'm in a limit, the H's can cancel each other, so those disappear.
- 8.45** If I cancel the H's, I'm looking at, this is equal to the limit as H goes to 0, 1 over the square root of 4 plus H plus 2. And now what happens when I plug in H is equal to 0 is, I get this is equal to 1 over the square root of 4 plus 2, which is 1 over 2 plus 2, which is 1 over 4, thankfully. That's a very long and drawn-out way of saying what? The slope of this line is one-fourth. It's a very long and drawn-out way of saying the slope of that line is equal to one-fourth.
- 8.46** Why is that useful? Well, now my tangent line at the point 4, 2, is what? Y minus 2, using my formula here, Y minus 2 is equal to slope, one-fourth, times X minus 4. That's what I have. I want to keep this. We can get rid of this stuff. This line has this equation. That's its defining equation. Let's simplify it a little bit. This is equal to what? Y is equal

to one-fourth X , so X divided by 4. Minus 1, so when this comes in, I get a minus 1, but I'm going to add 2, so minus 1 plus 2 is plus 1. That's what I get.

8.47 Now, what happens when I evaluate this thing at X is equal to 5? Remember, that was the whole goal. When X is equal to 5, I want to evaluate my tangent line. That's going to give me that point. And that's going to be a close approximation to what the square root of 5 was. So at X equals 5, what do I get? I get that Y is equal to 5 over 4 plus 1, which is 2.2, let's see, this is 2.25? Yes, 2.25. So this point up here on this line is 5, 2.25, two and a quarter.

8.48 But I notice that this line is above the graph of the square root of X . So really, this 2.25 is going to be bigger than the square root of 5, whatever the square root of 5 is. That's plausible. If you go into Desmos and you actually check what is the square root of 5, this is, roughly speaking, 2.236 and so forth. That's freakishly close. I can say that's a pretty darn good approximation, 2.25.

8.49 Tangent lines somehow allow us to take a problem that was otherwise completely hopeless, the square root of 5, and getting a value for it, and looking at this corresponding line, which I have complete control over, and then evaluating it at the corresponding point to get a decent approximation to what I wanted. That's amazing, if you think about it. It's amazing.

8.50 OK, what else can we do with derivatives? What else can we do? Remember that derivatives represent slope. So represent the slope of the tangent line. OK. Derivatives represent slopes of tangent lines. So why would this be useful? Well, what if I did the following setup? Remember what we talked about with the intermediate value theorem? The intermediate value theorem was useful, and we somehow took advantage of continuity to approximate solutions or show the existence of solutions of different equations. Do we agree?

8.51 We're showing the existence of solutions to equations from the intermediate value theorem. The question I want to come up with is, well, can I do better than that? Can I actually come up with another method for approximating roots or approximating solutions to equations? Let's take this square root of 5 example. Let's take this square root of 5 example and see what happens. What if I looked at the following thing?

8.52 Let's let X be equal to the square root of 5. Let X equal the square root of 5. What do I really want to try and do? I would like to say what the square root of 5 is equal to. I'd like to be able to say that it's actually 2.236, blah, blah, blah. I'd actually be able to write that down without having to use a calculator or anything like that. Well, then what I can do is, I can say, well, then X squared is equal to 5, if I square both sides, and if I subtract 5 over and X squared minus 5 is equal to 0.

8.53 So I'm really just saying the same thing in different fonts to some extent. I'm saying X is equal to square root of 5. That's the same thing as saying X squared is equal to 5. That's the same thing as saying X squared minus 5 is equal to 0, and that's great. What I'm

going to do is, I'm going to say, define a function F of X to be X squared minus 5. I'm going to define a function F of X equals X squared minus 5.

- 8.54** Now, why would I do that? Well, I'm trying to say, when X squared minus 5 is equal to 0, so this is setting me up for the intermediate value theorem side of things. When does a function equal 0? X squared minus 5 is a polynomial. When is that equal to 0? Why not call F of X equal to X squared minus 5? That seems reasonable.
- 8.55** But rather than using the intermediate value theorem, I'm going to ask the following question. Number one, if we were to graph this parabola, what does it look like? It looks like X squared just shifted down by negative 5. Just shifted down by negative 5. What does that look like? I'm down here at negative 5, let's say this is 0, negative 5, and then it comes up like this and comes up like this. OK, that's not too bad. All right, here's X , here's Y .
- 8.56** Now, where it intersects the X axis, this point right here is the square root of 5, 0. So that's what I'm trying to find. I'm trying to find, what is the X coordinate of that point? That's the square root of 5. All right. Here's what I'm going to do. It's kind of a cute example, actually. I'm going to say the following, just like I did over here, where I said, I'm interested in this point; what am I going to do? I'm going to choose a point close to it. I'm going to choose a point close to it.
- 8.57** Let's choose a point to the right, let's say. Let's say the point is 3. Let's give 3 a name. We're going to call it X_0 is equal to 3. That's this point right here. What am I going to do with that point? I'm going to go up to the graph, which is right up here. I'm going to draw the tangent line, and I'm going to recognize that that tangent line is going to intersect this X axis at some point. Do we agree? It's kind of a proof by pictures. It's like a proof of concept by pictures.
- 8.58** Here's what we're going to do. We're going to say, look at the graph X squared minus 5, of Y equals X squared minus 5. We're going to say, let's find the tangent line at X_0 equals 3. OK? So find the tangent line to Y equals X squared minus 5 at X_0 equals 3. OK? X_0 is playing the role of our A earlier. OK? It's playing the role of our A earlier. But we're going to have an X_0 , we're going to have an X_1 , and an X_2 , and an X_3 , and so forth, so I want to give them labels. I'm going to label it, rather than using A , I'm going to label it X_0 . It's just a name. That's it.
- 8.59** Find the tangent line to Y equals X squared minus 5 at X_0 is equal to 3. We know what that is. We know what that is. We know, let's call this thing F of X , instead of Y . This is another picky moment in mathematics, is that when you refer to the graph of a function, you should refer to it as Y equals something or variable equals something. When you refer to the function itself, you should refer to it as F of X equals something. You're kind of conflating two things by using F of X and Y . People understand what you mean.

- 8.60** But let's find the equation of the tangent line to the graph of F , which is Y equals X squared minus 5, where F of X is equal to X squared minus 5. What is the equation of the line? We know what that is. The equation is given by what? Y minus F at X_0 , that's equal to F prime at X_0 times X minus X_0 . We know what it is. OK?
- 8.61** Well, I know what F of X_0 is. If I plug in X_0 , that's 3. That's going to be 9 minus 5 is 4, so I know that much. I know this bit of information, that's a 4. I know X_0 is 3. I don't know F prime. I got to go over here and find out what is the corresponding slope. If I say, what is F prime at 3? This is my limit as H goes to 0, F of 3 plus H minus F of 3 all over H . This thing is the limit as H goes to 0.
- 8.62** OK, what happens when I get 3 plus H ? I plug it in for X . So what am I going to get? 3 plus H squared minus 5. That's my F of 3 plus H part. And then I subtract off F of 3. F of 3 is what? 3 squared is 9 minus 5, so that's a 4 all over H . OK, what do I get? Limit H goes to 0. If we work this out, again, the 3 plus H , 3 plus H , what do we get? 9 plus $3H$ and $3H$ is $6H$ plus H squared minus 5 minus 4 is minus 9 all over H .
- 8.63** The miracle happens again, and everything cancels except things with H 's. That seems like some standard thing. Maybe that's a fluke. Maybe that's something we could abstract later, potentially. It's true. It's called the local linearity property that you'd cover in an analysis-type course. This is equal to the limit as H goes to 0 of what? H times H plus 6 all over H . The H 's cancel, I evaluate at 0, and I get 6. Happy day. I get 6.
- 8.64** Now I know what the equation of my tangent line is. The equation of my tangent line is going to be what? Y minus F of X_0 , X_0 is 3, equals F prime at 3 times X minus 3. OK, well, we know all the information now. Let's plug it in. If X is 3, F of 3 is 9 minus 5, so that's 4. So this is Y minus 4 equals F prime of 3. Where's F prime of 3? We just did that. That's a 6. That's equal to 6 times X minus 3. So I get Y equals $6X$ minus 18 plus 4. That's Y equals $6X$ minus 14. OK, that's the equation of my line. Big deal.
- 8.65** What can I do with it? Well, remember, we're interested in finding the equation of the tangent line and finding out where the Y coordinate is equal to 0. Because this point here is when Y equals 0. Yes? Because it crosses the X axis. OK, so the question we now want to ask is, when is Y equal to 0? When is Y equal to 0? OK, well, let's set it equal to 0. Set Y equal to 0. That tells me 0 is equal to $6X$ minus 14.
- 8.66** That means that $6X$ is equal to 14, which means that X is equal to 14 over 6, which is what? 2 goes into this 7 times. 2 goes into that 3 times. That's 7 over 3. So what is that? That's like two and a third. So this is equal to 2.33 repeating. Do we agree? 3 goes into 7 twice with a remainder of 1, so I get two and one-third, so I get 2.333 and so on.
- 8.67** That gave me a new value that I could call maybe X_1 . That gave me a new value of X_1 , which I could then take back into my picture over here. Here's X_1 , plug it in. Look at the corresponding tangent line associated to that. Ask where that intersects the axis. That'll be an X_2 , and I can keep generating these things. We can keep iterating this.

- 8.68** OK, so what does that mean? What just happened? Well, we just found out that there's some sort of method for generating things like X_0, X_1, X_2, X_3 , and so on, that approximate when F of X is equal to 0. Yeah? What is this particular object? X_0, X_1, X_2, X_3 , that's a sequence. That's a sequence. This is a sequence. Again, the sequence is just a collection of points. It's a function such that when you give me an input, 0, 1, 2, 3, I give you an output, X_0, X_1, X_2, X_3 . It gives me a sequence.
- 8.69** The question I want to understand is, look, if I imagine just keeping this on the line, OK, over here is my square root of 5, here was X_0 , here was X_1 , here was X_2 , here's X_3 , and so on, I keep generating these things, I really hope that I could make sense of something such as saying the square root of 5 is equal to the limit of X_N as N goes to infinity. I'd really like to be able to say something like that.
- 8.70** What does that mean? I have this sequence, $X_N, X_0, X_1, X_2, X_3, X_4, X_5$, just like the arithmetic sequences that we talked about, just like the geometric sequences that we talked about. I would like to know as N , if N gets arbitrarily large, if N gets arbitrarily large, does X_N somehow go to the square root of 5? Does X_N somehow go to the square root of 5? The answer is yes, it does.
- 8.71** This is the notion that we talked about in the sequences and series section that said, what did we mean as N "gets large"? This is what we meant by that. As N gets large, as N goes to infinity, X_N converges to the square root of 5. What do I mean by that? I mean that, as N gets arbitrarily large, let's say N is equal to a billion. By the time you get out to a billion, whatever values you're generating for X_N , they'll be indistinguishable from the number square root of 5. They'll give you an approximation to the square root of 5.
- 8.72** Maybe this has value. Maybe this thing that we just started goofing around with, this idea of, maybe if I just look at a point and look at a tangent line and find where it intersects, maybe I can generate something out of this. Maybe this actually has some merit mathematically, in terms of this. What we've been talking about is what's called Newton's method. This entire thing that we were doing is called Newton's method.
- 8.73** Newton's method is arguably one of the most powerful mathematical techniques of all time, and it's used everywhere. It's used everywhere in terms of optimization, in terms of things like that. So, derivatives are powerful for some reason. For some reason, they have a lot to do with kind of, like, everything. We have slopes, which are rates of change.
- 8.74** We have average rates of change, which we found by taking this difference in the F 's, divided by the difference in their inputs. That was our average speed that we talked about in terms of cycling as our examples or in our cars. But then we said our cars give us the speedometer, which is an instantaneous speed. We want to understand, what in the world is instantaneous speed? We said, well, that's a limit of averages.

- 8.75** In order to make averages instantaneous, we had to pass to this notion of a limit. We had to pass to this notion of a limit. Now, when we have limits, we saw that continuity came out from limits, and we could diagnose when a function was continuous or not, or whether it was discontinuous. We stumbled upon the intermediate value theorem, which then gave us an algorithm for being able to determine when certain equations have solutions or not. That's fantastic.
- 8.76** But then we said, well, those limits are nice. They gave us this notion of the derivative, which is the slope of the tangent line to the graph. Somehow these tangent lines are incredibly important because they generate one of the most important mathematical techniques in Newton's method. That's beautiful, actually, how all of that is connected. It's beautiful how all of that connected. Newton's method is incredibly powerful. It's used in optimization.
- 8.77** Now, optimization is what? If Newton's method is tied to tangent lines somehow, and tangent lines have derivatives running around, then it stands to reason that if Newton's method is all about optimization, then derivatives should have something to do with optimization, presumably. Let's think about this for a second. Why would this be relevant? Let's suppose I want to motivate the following thing.
- 8.78** Let's suppose you have a particular function which models something of interest to you. Whatever that is, that's up to you. That's up to you. Whether that's, you're a business owner, and it models your revenue or your profits or your costs or whatever it is, if I give you a function, let's say it does something like this. Something along these lines. Let's suppose I give you this.
- 8.79** Now, again, this function is supposed to be relevant for you. Whatever that is, whatever it is. Pick your favorite, most important thing, whatever that is, whether that's how much money you can generate from a business standpoint, whether that's some sort of drive times, whether that's athletic performance, whatever this is, let's suppose this is the graph underlying all of that.
- 8.80** What are some important features of this graph? For the sake of this discussion, I'm going to say that, suppose this is the revenue in a business. Suppose this is the revenue in a business, where X is saying something, maybe X is time as time goes on, and the vertical axis is the revenue that we're generating in our business. Something like that.
- 8.81** Well, what are important features of this graph then? For me, when I step back and I say, what are important features, I say, well, if this is revenue, I'm really interested in that point, because that's saying I've made a lot of money and I'm really happy with that one. I'm saying I'm a little concerned about this one, and I'm concerned about this one, and I'm really freaking out about this one, if that's the revenue, because something bad is going on. Those three are pretty bad.
- 8.82** This one's kind of nice. This is a nice little one. And this one is, too. It's a little minor one, but that's nice, too. Can I say something about characteristics that every single

one of these points shares? Think derivatives. Think tangent lines. If I draw a tangent line here, it's horizontal. If I draw a tangent line here, it's horizontal. Horizontal, horizontal, horizontal, horizontal. Horizontal lines don't have slope. They have slope equal to zero. That's the feature to notice. Horizontal tangent line has slope equal to zero.

- 8.83** OK, so what does that mean? That means that if I can go and find this function, and I can find F' as a function in and of itself, I want to know, if F' is representing the slope of my tangent line, I want to know, when is F' equal to zero? When is that equal to zero? Why do I care about that? Because F' is the slope of the tangent line. The slope of the tangent line tells me whether or not I get to eat each month or not.
- 8.84** Because it's telling me whether or not my revenue is increasing, decreasing, giving me a max or giving me a minimum. That's relevant. That's relevant. So trying to find out when F' of X is equal to zero gives me locations for where these things are occurring. So this point here, this point here, here, here, here, and here. It's giving me locations for those.
- 8.85** All right, well, that's all fine. Horizontal tangent lines have slope equals zero. We already have methods for being able to try and find this. We say, well, if F' is a continuous function, then the intermediate value theorem, intermediate value theorem gives me a method for algorithmically computing the zeros to this thing. I also have Newton's method now. I've got Newton's method. Do I have others? The answer is, yeah, there's a lot of others.
- 8.86** What's one of the most popular ones? One of the most popular ones, people throw this word around all the time. Well, this phrase around, I should say, is what's called gradient descent. Gradient descent. It sounds like this big technical, scary word. What is gradient? That's weird. I don't use that word daily. Descent, that sounds sophisticated. I don't know. What does it mean? It's embarrassingly simple, actually.
- 8.87** It says, if you want to find where you bottom out, for example, because we're talking about descent—we could talk about ascent if you want as well, but we're going to talk about descent. If I want to figure out where I'm bottoming out because I want to figure out where my revenue is lowest and I want to try and hedge against those opportunities, I want to try and increase those from being as low as they are. I want to find out where they bottom out, gradient descent literally says, point yourself downhill and walk. That's literally all it says.
- 8.88** What does it mean by point yourself downhill? It means look at the derivative of your function and then go in the direction of opposite your derivative because that's going to point in the direction of greatest decrease. It's saying the following. If I start with a point here, let's call it X_0 , and I go up to my graph, it's going to tell me walk downhill. You want to find the bottom of this hill. It's going to tell me walk in this direction.

- 8.89** I start going downhill, maybe I walk for a period of time, and I end up over here, X_1 . Here's what I get there. It's going to tell me, go back, walk back this way until you get a point here and here and here, until you eventually get close down to the bottom, meaning the limit of your sequence converges to a point at the bottom.
- 8.90** You might ask the question, well, that's not the lowest one. I would say, I agree. This is the lowest one. How do you find it? Why did you choose X_0 here? Choose X_0 over here, over here, over here. Choose a bunch of X_0 s, run them, find whichever one gives you the lowest point.
- 8.91** This is the fundamentals of mathematics and precalculus and gives you the tools necessary to get to a place of optimization, which is really how the world works. Everything we do is an optimization problem. Now, through this course and others, you've got the fundamental tools to start your journey to optimization in higher level mathematics.